6CCSAALA: Advanced Logics for Artificial Intelligence

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1 Introduction to Non-monotonic Logics

Learning Objectives

By the end of this section you should be able to:

- ☐ Define what it means for a logic to be monotonic/non-monotonic;
- ☐ Apply Sequent rules to construct proofs of logical consequences in (classical) propositional logic;
- ☐ Apply the Gabbay–Makinson rules to construct proofs of rational consequences in propositional non-monotonic logics.

1.1 Prelude: What are (Non-classical) Logics?

Before we get into the details of *"what are non-classical logics"*, the first question we must ask is

"What are logics"?

At first this may seem to be an odd question to ask, for how can there be more than one logic. Surely logic is just logic. And while you will recall that we have seen both two logics already — propositional logic and predicate logic — predicate logic is essentially just an extension of propositional logic. Indeed, we could consider propositional variables to simply be predicate symbols with arity zero; in other words their value is determined to either be true or false in a model irrespective of any arguments.

So let's first describe what we mean by a 'logic', which typically starts by fixing some formal language.

Definition 1.1: Well-formed Formulas

Given a set of **propositional variables** $\mathsf{Prop} = \{p_1, \ldots, p_n\}$ we define the set *Fml* of all **(well-formed) formulas** inductively as follows:

- If $p \in \text{Prop}$ then $p \in \mathit{Fml}$ is a well-formed formula;
- If $F, G \in \mathit{Fml}$ are well-formed formulas then so are:

$$\neg F, \qquad (F \wedge G), \qquad (F \vee G), \qquad (F \to G)$$

Logics can be defined in many ways, broadly breaking down into two categories:

Semantics

Defined in terms of models of how the world "really" is and what it means for a statement to be "true" in that world.

Syntax

Defined in terms of mechanical rules to "derive" conclusions from premises via a finite sequence of steps.

In the land of **semantics** we define some notion of a **model** that assigns **truth values** to the atomic components of our formal language, and define how those truth values can be extended to more complex formulas of our language. In propositional logics, we capture these models with truth tables, while in predicate logic, we needed a much richer definition of a model that told us not only what are the individual objects we wanted to model, but how to **interpret** each of the symbols from our formal language in terms of the objects from that domain.

Definition 1.2: Models in Propositional Logic

We define a **(propositional) model** or **valuation** to be a function $V : \text{Prop} \to \{0, 1\}$ that maps each propositional variable $p \in \text{Prop}$ to a Boolean value: 0 ('false') or 1 ('true').

We extend each propositional valuation V to the domain of all well-formed formulas *Fml* inductively on the structure of the formula:

$$\begin{aligned} V(\neg\varphi) &= 1 - V(\varphi) \\ V(\varphi \wedge \psi) &= \min(V(\varphi), V(\psi)) \\ V(\varphi \vee \psi) &= \max(V(\varphi), V(\psi)) \\ V(\varphi \to \psi) &= \max(1 - V(\varphi), V(\psi)) \end{aligned}$$

We say that V is a **model of** φ if $V(\varphi) = 1$, and denote the set of all models of φ to be

$$\mathsf{mod}(\varphi) \;=\; \{V : \mathsf{Prop} \to \{0,1\} : V(\varphi) = 1\}.$$

It will sometimes be convenient to associate each model $V : \mathsf{Prop} \to \{0,1\}$ with a formula $F_V \in$ *Fml* such that $V'(F_V) = 1$ if and only if $V' = V$:

$$F_V \;=\; \bigwedge_{p \in \mathsf{Prop}} p^{V(p)}$$

where $p^1 = p$ and $p^0 = \neg p$, for $p \in \mathsf{Prop}$.

Meanwhile in the land of **syntax**, we have formal systems of proof, such as **Natural Deduction**, where we have some formal rules for how to transform formulas or collections of formulas into new formulas based only on their structure.

Definition 1.3: Natural Deduction

A **(natural deduction) rule** is of the form:

$$\frac{F_1 \quad F_2 \quad \ldots \quad F_n}{G}$$

where $F_1, \ldots, F_n \in$ *Fml* denote the **premises** of the rule and $G \in$ *Fml* denotes the **conclusion** of the rule.

A **(natural deduction) proof system** is a finite set $\mathcal{ND}$ of natural deduction rules. A **proof** in a system $\mathcal{ND}$ is a sequence of formulas in which each formula is either an assumption / input data or is the conclusion of some instance of rule of $\mathcal{ND}$ where the premises are all formulas that have appeared previously in the sequence[a].

[a] The caveat here is the behaviour of the implication introduction rule, which has a slightly different formulation, but we are not here to re-learn natural deduction so we will ignore this for now.

In both the land of semantics and the land of syntax we have some **consequence** or **provability relation** saying that, given some set of formulas that we take to be true we are able to conclude that some other formula must also hold.

But regardless of how we formulate our logic, either semantically or syntactically, and with whatever models or proof systems we may dream up, ultimately we obtain some notion of a consequence relation — we can therefore identify a 'logic' with a consequence relation over some formal language.

Definition 1.4: What is a logic?

Given a set of formulas *Fml* generated by some grammar, we identify a **logic** with a binary relation, called a **consequence relation**

$$\models \ \subseteq \ \mathscr{P}(\mathit{Fml}) \times \mathit{Fml}$$

between sets of formulas from $\mathscr{P}(\mathit{Fml})$ and formulas from *Fml*, often satisfying some desirable properties (to be determined), where $\mathscr{P}(\mathit{Fml})$ denotes the **powerset** of all *subsets* of formulas from *Fml*.

It may look strange at first to see $\models \ \subseteq \ \mathscr{P}(\mathit{Fml}) \times \mathit{Fml}$ since we are accustomed to seeing $\models$ written in infix notation, for example, $P, P \to Q \models Q$. However, we can

think of this as just a convenient way to write down that the pair $(\{P, P \to Q\}, Q) \in \models$ belongs to the consequence relation that we denote by $\models$[1].

Logics can be defined either **syntactically**, by some **deductive system/proof theory**, or **semantically**, by some definition of **models** or what it means for a formula to be 'true'.

Typically, to distinguish the two, we use $\models$ to denote a consequence relation that is defined semantically, and $\vdash$ to denote a consequence relation that is defined syntactically (where we may better describe the relation as a **provability relation**).

Definition 1.5

We say that a formula $\varphi \in$ *Fml* is a **logical consequence** of a set of formulas $\Gamma \subseteq$ *Fml* if every model that is a model of Γ is also a model of φ:

$$\Gamma \models \varphi \quad \iff \quad \mathsf{mod}(\Gamma) \subseteq \mathsf{mod}(\varphi)$$

We say that φ is **provable/derivable** from Γ if there is some **proof** of φ from some finite subset of formulas in Γ:

$$\Gamma \vdash \varphi \quad \iff \quad \text{There is a proof of } \varphi \text{ from } \Gamma.$$

Note that the two definitions of logical consequence, defined semantically, and of provability, defined syntactically, have very different 'flavours' — logical consequence is a **universal statement** that requires that 'all' models that satisfy all the formulas of Γ must also satisfy φ. Whereas provability is an **existential statement** that requires only that there 'exists' some proof of φ from 'some' subset of data from Γ.

It is therefore quite remarkable that, at least for propositional and predicate logic, we are able to make these two definitions agree on the same consequence relation.

[1] Similar to how we define the binary relation $<$ on integers as the set of all pairs $(x, y) \in \mathbb{Z} \times \mathbb{Z}$ where x is less than y, but write $x < y$ rather than $(x, y) \in <$.

Theorem 1.6: The Completeness Theorem of Propositional Logic

Let $\Gamma \subseteq$ *Fml* and $F \in$ *Fml* then

$$\Gamma \models F \quad \Longleftrightarrow \quad \Gamma \vdash F$$

That is to say that $\models \; = \; \vdash$.

The completeness theorem says that although the two definitions of logical consequence and provability differ considerably, they ultimately define the *same consequence relation* on *Fml*.

Propositional and predicate logic are well-suited to describe and model the world of mathematics, but straying outside of that Platonic realm of abstract ideals, we quickly encounter situations where such 'classical' logics break down, or at best become unnecessarily cumbersome to reason with, and clearly not aligned with how we typically reason. If we wish to design automated systems to reason about the real world, and interact with human reasoners such as ourselves, then we had better design them to be able to reason **non-classically**!

1.2 Motivation for Non-Monotonicity

Consider the following argument:

Premise 1)	Tweety is a bird;
Premise 2)	Birds can fly;
Conclusion)	Tweety can fly.

This seems reasonable enough, and certainly something that we could formalise in predicate logic with predicates for isBird, canFly and constant *tweety*, or even

in propositional logic with propositions *"I am Tweety"* (t), *"I am a bird"* (b), and *"I can fly"* (f):

$$t \to b,\ b \to f \ \models\ t \to f$$

However, if we were also to learn two additional premises

Premise 3)	Tweety is a penguin ($t \to p$);
Premise 4)	Penguins can't fly ($p \to \neg f$);

then we may well not be quite so committed to our initial conclusion that *"Tweety can fly"*. And yet, classical logics (both propositional and predicate) are **monotonic**!

Definition 1.7

We say that a logic is **monotonic** if, for all $\Gamma, \Delta \subseteq$ *Fml* and $F \in$ *Fml*

$$\Gamma \models F \qquad \text{implies} \qquad \Gamma \cup \Delta \models F$$

In other words, adding additional premises to our argument *cannot* invalidate any conclusions that have already been drawn[2] And so, if we reason about the world monotonically, or develop systems that reason about the world monotonically, then we or they would be unable to retract the conclusion that *"Tweety can fly"* even in the face of additional data signalling to the contrary.

Worse still is the fact that by introducing these additional premises into our knowledge base now ends up poisoning the whole system, since we can now derive the contradiction that *"Tweety can and cannot fly"*, from which we can then derive *any* statement we like such as that *"Tyrannosaurs make excellent accountants (a)"*

$$t \to b,\ b \to f,\ t \to p,\ p \to \neg f,\ t \ \models\ a$$

This phenomenon is known as the **principle of explosion**.

[2]This is clear when we consider that any natural deduction would remain valid even if we introduced additional premises.

So what went wrong!? Well, with the benefit of hindsight, we can perhaps identify that the problematic premise was that *"Birds can fly"*, which is now revealed to have been false. It is tempting then to simply remove this premise entirely on the basis that we do not want any false statements in our knowledge base. However, this would mean throwing out a rule that in most usual circumstances is very useful. If we learned that *Perry the pigeon is a bird* then we would perhaps want to, at least initially, infer that Perry can fly[3] Instead, what we should have said was that *"Birds can usually fly"*, or rather, *"If I am a bird then usually I can fly"*.

Non-monotonic logics allow us to deal with potentially conflicting information and reason with so-called **defeasible inferences** that *usually* hold but may fail in exceptional cases. To emphasise the non-monotonicity, we use the symbol $\mid\!\approx_{\vec{S}}$ to denote the consequence relation of a non-monotonic logic, with the interpretation that

$$\Gamma \mid\!\approx_{\vec{S}} F \qquad \Longleftrightarrow \qquad F \textit{ usually } \text{follows from } \Gamma$$

for all $\Gamma \subseteq$ *Fml* and $F \in$ *Fml*.

[3]We may again wish to retract this conclusion if we later learn than Perry was injured in a tragic mix-up while attending a clay pigeon shooting competition.

Exercise

Formulate the following statements in terms of a rational consequence relation $\mid\!\sim$:

(i) *"Birds usually fly"*;

$$b \mid\!\sim f$$

(ii) *"Penguins usually don't fly"*;

$$p \mid\!\sim \neg f$$

(iii) *"Penguins are birds"*.

We could use the classical $p \models b$, or reframed in the language of rational consequence we could say that *"Penguins that are not birds are usually impossible"*

$$p \wedge \neg b \mid\!\sim \bot$$

But how do we define such a consequence relation that would satisfy our implicit intuitions of what it means for a conclusion to *usually* follow from some given set of premises?

In contrast to propositional and predicate logic where we first defined a suitable **semantics** and then sought a proof system that was sound and complete with respect to the semantics; for non-monotonic logics where we don't have an immediate notion of what a non-monotonic 'model' would look like, it is more reasonable to start by defining a **proof system** that captures how we would like our consequence relation to behave, and then seek a corresponding semantics.

1.3 Interlude: Sequent Calculi

When it comes to proof systems, natural deduction is not the only player in town. In the land of syntax, there also live completely different genres of proof systems such as **Hilbert systems**, favoured by logician David Hilbert, where proofs are simply sequences of formulas, each of which is either an axiom or an application of one of a very small number of logical rules[4]. Or **Sequent Calculi** — sometimes known as **Gentzen systems** after logician Gerhard Gentzen who first proposed such systems in 1935. Unlike natural deductions which operate on formulas, the rules of Sequent calculi operate on objects known as **Sequents**.

Definition 1.8: Sequents

A **sequent** is a pair of **finite** sets of formulas from *Fml*, written:

$$A_1, \ldots, A_n \;\Rightarrow\; B_1, \ldots, B_m$$

where $A_1, \ldots, A_n, B_1, \ldots, B_m \in$ *Fml* are formulas.

Although, formally a sequent is just a finite sequence of formulas separated into two parts with a $\Rightarrow$ symbol, we can intuitively interpret a sequent as saying

"If **everything** on the left is true, then **something** on the right is true"

(This is, of course, purely a design choice which then motivates the kinds of rules we introduce as part of our sequent calculus).

We can think of a Sequent Calculus as a kind of *'meta-proof'* that operates directly on the consequence relation between finite sets of formulas, in contrast to Natural Deduction or Hilbert systems that operate at the level of individual formulas.

[4]For (classical) propositional logic, typically the only rule we have is **modus ponens** $F, F \to G/G$, whereas for (classical) predicate logic, we have just **modus ponens** and **universal generalisation** $F(x)/\forall x\, F(x)$. However, the axioms required to obtain soundness and completeness results are considerably more opaque.

Rules of our Sequent calculi will take some number of Sequents as input, written on the top, with exactly one Sequent as output, written on the bottom.

Definition 1.9: Sequent Calculi

A **sequent rule** is a rule that takes as input zero or more sequents as **premises** and returns a single sequent as its **conclusion**

$$\frac{\Gamma_1 \;\Rightarrow\; \Delta_1 \quad \ldots \quad \Gamma_n \;\Rightarrow\; \Delta_n}{\Gamma \;\Rightarrow\; \Delta}$$

A rule may have one or more **'side conditions'** that stipulate additional criteria under which the rule may be applied. A sequent rule with zero premises is referred to as an **axiom schema**. A **sequent calculus** is any finite set $\mathcal{S}$ of sequent rules.

For example, the following is a sequent calculus for (classical) propositional logic

A Sequent Calculus for Propositional Logic

The following set of rules $\mathcal{S}$ is adapted from Gentzen [3]. There is a single **Reflexivity axiom (REF)**, and a *left* and *right* rule for each of the Boolean connectives $\wedge$, $\vee$, $\neg$, $\rightarrow$:

$$\frac{\varnothing}{F \;\Rightarrow\; F}\ \textbf{(REF)}$$

$$\frac{\Gamma,\ F,\ G \;\Rightarrow\; \Delta}{\Gamma,\ (F \wedge G) \;\Rightarrow\; \Delta}\ \textbf{(L}\wedge\textbf{)} \qquad \frac{\Gamma \;\Rightarrow\; \Delta, F \quad \Gamma \;\Rightarrow\; \Delta, G}{\Gamma \;\Rightarrow\; \Delta,\ (F \wedge G)}\ \textbf{(R}\wedge\textbf{)}$$

$$\frac{\Gamma,\ F \;\Rightarrow\; \Delta \quad \Gamma,\ G \;\Rightarrow\; \Delta}{\Gamma,\ (F \vee G) \;\Rightarrow\; \Delta}\ \textbf{(L}\vee\textbf{)} \qquad \frac{\Gamma \;\Rightarrow\; \Delta,\ F,\ G}{\Gamma \;\Rightarrow\; \Delta,\ (F \vee G)}\ \textbf{(R}\vee\textbf{)}$$

$$\frac{\Gamma \;\Rightarrow\; \Delta,\ F}{\Gamma,\ \neg F \;\Rightarrow\; \Delta}\ \textbf{(L}\neg\textbf{)} \qquad \frac{\Gamma,\ F \;\Rightarrow\; \Delta}{\Gamma \;\Rightarrow\; \Delta,\ \neg F}\ \textbf{(R}\neg\textbf{)}$$

$$\frac{\Gamma \;\Rightarrow\; \Delta,\ F \quad \Gamma,\ G \;\Rightarrow\; \Delta}{\Gamma,\ (F \rightarrow G) \;\Rightarrow\; \Delta}\ \textbf{(L}\rightarrow\textbf{)} \qquad \frac{\Gamma,\ F \;\Rightarrow\; \Delta,\ G}{\Gamma \;\Rightarrow\; \Delta,\ (F \rightarrow G)}\ \textbf{(R}\rightarrow\textbf{)}$$

together with structural rules:

$$\frac{\Gamma \;\Rightarrow\; \Delta}{\Gamma,\ F \;\Rightarrow\; \Delta}\ \textbf{(MON)} \qquad \frac{\Gamma_1 \;\Rightarrow\; \Delta_1, \Theta \quad \Gamma_2, \Theta \;\Rightarrow\; \Delta_2}{\Gamma_1, \Gamma_2 \;\Rightarrow\; \Delta_1, \Delta_2}\ \textbf{(CUT)}$$

where $F, G \in$ *Fml* are arbitrary formulas, and Γ and Δ stand for finite sets (or sequences) of formulas that are not **active** in the application of the rule.

As with natural deduction, each rule acts as a **template** that can be **instantiated** with arbitrary formulas that fit the same pattern — we call such instantiations **instances** of the rule.

Notes

Write down three instances of the rule:

$$\frac{\Gamma, F \;\Rightarrow\; \Delta}{\Gamma \;\Rightarrow\; \Delta, \neg F}\ \textbf{(R}\neg\textbf{)}$$

(i)

$$\frac{p, q \;\Rightarrow\; r}{p \;\Rightarrow\; r, \neg q}$$

(ii)

$$\frac{(p \to q) \;\Rightarrow\; \varnothing}{\varnothing \;\Rightarrow\; \neg(p \to q)}$$

(iii)

$$\frac{\neg(p \wedge q), p \;\Rightarrow\; \neg q}{\neg(p \wedge q) \;\Rightarrow\; \neg p, \neg q}$$

A proof in our Sequent calculi is a finite tree in which each internal node of our tree is labelled by an instance of a rule, where the inputs to each rule are the outputs of the children nodes.

Definition 1.10: Proofs

A **proof** (or **proof tree**) in a sequent calculus $\mathcal{S}$ is a (directed) tree $T = (V, E)$ in which each vertex of the tree is labelled by an instance of a sequent rule, such that if a vertex $v \in V$ is labelled by an instance of a rule

$$\frac{\Gamma_1 \;\Rightarrow\; \Delta_1 \quad \ldots \quad \Gamma_n \;\Rightarrow\; \Delta_n}{\Gamma \;\Rightarrow\; \Delta}$$

with n premises, then v has n children $u_1, \ldots, u_n$ with $(v, u_i) \in E$, for $i = 1, \ldots, n$, such that u_i is labelled by an instance of a rule with conclusion $\Gamma_i \;\Rightarrow\; \Delta_i$.

We say that a formula $F \in$ *Fml* is **derivable** or **provable** in $\mathcal{S}$ from some set of formulas $\Gamma \subseteq$ *Fml* if there is a proof in $\mathcal{S}$ of the sequent $\Gamma' \;\Rightarrow\; F$ for some *finite* subset $\Gamma' \subseteq \Gamma$.

In which case, we write that $\Gamma \vdash^{\mathcal{S}} F$, or simply $\Gamma \vdash F$ if it is understood that we are working within the system $\mathcal{S}$.

Exercise

Prove the following consequences are derivable in $\mathcal{S}$:

(i) $\neg(p \wedge q) \vdash (\neg p \vee \neg q)$

$$
\begin{array}{c}
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\varnothing}{p, q \;\Rightarrow\; p}\,\textbf{(REF)} \qquad \dfrac{\varnothing}{p, q \;\Rightarrow\; q}\,\textbf{(REF)}}{p, q \;\Rightarrow\; (p \wedge q)}\,\textbf{(R}\wedge\textbf{)}}{\neg(p \wedge q), p, q \;\Rightarrow\; \varnothing}\,\textbf{(L}\neg\textbf{)}}{\neg(p \wedge q), p \;\Rightarrow\; \neg q}\,\textbf{(R}\neg\textbf{)}}{\neg(p \wedge q) \;\Rightarrow\; \neg p, \neg q}\,\textbf{(R}\neg\textbf{)}}{\neg(p \wedge q) \;\Rightarrow\; (\neg p \vee \neg q)}\,\textbf{(R}\vee\textbf{)}
\end{array}
$$

(ii) $\neg(p \vee q) \vdash (\neg p \wedge \neg q)$

$$
\begin{array}{c}
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\varnothing}{p \;\Rightarrow\; p, q}\,\textbf{(REF)}}{p \;\Rightarrow\; (p \vee q)}\,\textbf{(R}\vee\textbf{)}}{\neg(p \vee q), p \;\Rightarrow\; \varnothing}\,\textbf{(L}\neg\textbf{)}}{\neg(p \vee q) \;\Rightarrow\; \neg p}\,\textbf{(R}\neg\textbf{)} \qquad \dfrac{\dfrac{\dfrac{\dfrac{\varnothing}{q \;\Rightarrow\; p, q}\,\textbf{(REF)}}{q \;\Rightarrow\; (p \vee q)}\,\textbf{(R}\vee\textbf{)}}{\neg(p \vee q), q \;\Rightarrow\; \varnothing}\,\textbf{(L}\neg\textbf{)}}{\neg(p \vee q) \;\Rightarrow\; \neg q}\,\textbf{(R}\neg\textbf{)}}{\neg(p \vee q) \;\Rightarrow\; (\neg p \wedge \neg q)}\,\textbf{(R}\wedge\textbf{)}
\end{array}
$$

As with natural deduction, we can show that the above sequent calculus $\mathcal{S}$ is sound and complete with respect to the usual semantics for propositional logic.

Theorem 1.11: Completeness Theorem

Let $\Gamma \subseteq$ *Fml* and $F \in$ *Fml* then

$$\Gamma \models F \quad \Longleftrightarrow \quad \Gamma \vdash^{\mathcal{S}} F$$

where $\models$ is defined by the usual truth-table semantics of Definition 1.1.

1.4 Gabbay–Makinson Rules

In contrast to classical (monotonic) propositional logic, where we first defined a **semantics** (truth tables) for what it means for a formula to be true and, by extension, defined our logical consequence relation in terms of the models that makes those formulas true; for non-monotonic logics, we start by defining the **syntactic rules** that we expect that any reasonable non-monotonic logic should follow.

One such set of rules are those proposed by Dov Gabbay and David Makinson [2, 6]. In order to distinguish the sequents $\Gamma \Rightarrow \Delta$ used in the sequent calculus $\mathcal{S}$ for propositional logic, we will make a stylistic choice to denote the sequents of the Gabbay–Makinson system $\mathcal{GM}$ with $F \mid\!\sim G$.

Gabbay–Makinson Rules

The Gabbay–Makinson proof system $\mathcal{GM}$ comprises the following rules:

$$\frac{\varnothing}{F \mid\!\sim F}\ \textbf{(REF)}$$

$$\frac{F \mid\!\sim G \quad F \mid\!\sim H}{F \mid\!\sim (G \wedge H)}\ \textbf{(R}\wedge\textbf{)} \qquad \frac{F \mid\!\sim H \quad F \mid\!\sim H}{(F \vee G) \mid\!\sim H}\ \textbf{(L}\vee\textbf{)}$$

$$\frac{F \mid\!\sim H \quad F \equiv G}{G \mid\!\sim H}\ \textbf{(LLE)} \qquad \frac{F \mid\!\sim G \quad G \models H}{F \mid\!\sim H}\ \textbf{(RWE)}$$

$$\frac{F \mid\!\sim G \quad F \mid\!\sim H}{(F \wedge G) \mid\!\sim H}\ \textbf{(CMON)} \qquad \frac{F \mid\!\sim H \quad F \not\mid\!\sim \neg G}{(F \wedge G) \mid\!\sim H}\ \textbf{(RMON)}$$

where $F, G, H \in$ *Fml* are arbitrary formulas.

Before proceeding, let us examine each of the rules one at a time, and hopefully convince ourselves that they are 'reasonable' rules that we would like to hold for our intuitive notion of "usually".

Reflexivity Axiom As with the sequent calculus $\mathcal{S}$ for (classical) propositional logic, we have a single **axiom schema (REF)** that says that we are always allowed to infer that *"F usually follows from F"*, for all formulas $F \in$ *Fml*

$$\frac{\varnothing}{F \;\mid\!\sim\; F} \qquad \textbf{(REF)}$$

Left Logical Equivalence We have a rule **(LLE)** that says that if H usually follows from F, and we know that F and G are (classically) logically equivalent formulas[5], then it should also be the case that H usually follows from G.

$$\frac{F \;\mid\!\sim\; H \quad F \equiv G}{G \;\mid\!\sim\; H} \qquad \textbf{(LLE)}$$

Right Weakening Similar to **(LLE)**, we have a rule that **(RWE)** that says that if G usually follows from F and H is a logical *consequence* of G then H usually follows from F.

$$\frac{F \;\mid\!\sim\; G \quad G \models H}{F \;\mid\!\sim\; H} \qquad \textbf{(RWE)}$$

Right Conjunction The **(R$\wedge$)** rule says that if G and H both usually follow from F, then it should be the case that their conjunction $(G \wedge H)$ usually follows from F.

$$\frac{F \;\mid\!\sim\; G \quad F \;\mid\!\sim\; H}{F \;\mid\!\sim\; (G \wedge H)} \qquad \textbf{(R}\wedge\textbf{)}$$

[5]Strictly speaking, to avoid mixing classical logic and non-monotonic logic, the condition that $F \equiv G$ should be considered as a *side-condition* under which the rule is applicable, but it is notationally convenient to write this side condition as a premise of the rule.

Left Disjunction Conversely, the **(L$\vee$)** rule says that if H usually follows from F and also usually follows from G, then it should not matter which of F or G is true for H to usually follow.

$$\frac{F \mid\!\sim H \quad G \mid\!\sim H}{(F \vee G) \mid\!\sim H} \qquad \textbf{(L}\vee\textbf{)}$$

Cautious Monotonicity [6] The **(CMON)** rule says that if we know that H usually follows from F then we can conclude that H also follows from $(F \wedge G)$ *provided that* G usually follows from F.

$$\frac{F \mid\!\sim G \quad F \mid\!\sim H}{(F \wedge G) \mid\!\sim H} \qquad \textbf{(CMON)}$$

Rational Monotonicity The **(RMON)** rule, on the other hand, says that if we know that H usually follows from F then we can conclude that H also follows from $(F \wedge G)$ *provided that* the negation $\neg G$ doesn't usually follow from F.

$$\frac{F \mid\!\sim H \quad F \not\mid\!\sim \neg G}{(F \wedge G) \mid\!\sim H} \qquad \textbf{(RMON)}$$

Note that rational monotonicity differs from the other rules in that as one of its premises, it requires that we first prove that something *doesn't* usually follow.

We define proofs/proof trees in $\mathcal{GM}$ as in Definition 1.3, and say that $G \in$ *Fml* is **provable** or **derivable** from $F \in$ *Fml* if there is some proof in $\mathcal{GM}$ of the sequent $F \mid\!\sim G$. In which case we write that $F \mid\!\sim^{\mathcal{GM}} G$, or simply $F \mid\!\sim G$, if it is understood that we are working with the full set of Gabbay–Makinson rules $\mathcal{GM}$.

[6] This is what Gabbay refers to as *Restricted* Monotonicity.

1.5 Some Derived Rules of $\mathcal{GM}$

As with Natural Deduction, it is often convenient to work not only with the basic rules of the system $\mathcal{GM}$, but also with some helpful **derived rules** that can be derived from the basic rules.

Some derived rules of $\mathcal{GM}$

The following **derived rules** can each be derived from the basic rules of $\mathcal{GM}$:

$$\frac{F \models G}{F \mid\!\sim G}\ \textbf{(SC)} \qquad \frac{(F \wedge G) \mid\!\sim H}{F \mid\!\sim (G \to H)}\ \textbf{(CON)}$$

$$\frac{F \mid\!\sim G \quad (F \wedge G) \mid\!\sim H}{F \mid\!\sim H}\ \textbf{(CC)} \qquad \frac{F \mid\!\sim H \quad (F \wedge \neg G) \not\mid\!\sim H}{(F \wedge G) \mid\!\sim H}\ \textbf{(NRAT)}$$

where $F, G, H \in$ *Fml* are arbitrary formulas.

Supraclassicality The **(SC)** rule says that if G always follows from F — that is to say that $F \models G$ is a classical consequence — then G *usually* follows from F.

$$\frac{F \models G}{F \mid\!\sim G} \qquad \textbf{(SC)}$$

Conditionalisation The **(CON)** rule is similar to the **(R→)** rule of $\mathcal{S}$ and says that if H usually follows when F and G are both true, then the implication $(G \to H)$ usually follows from F on its own.

$$\frac{(F \wedge G) \mid\!\sim H}{F \mid\!\sim (G \to H)} \qquad \textbf{(CON)}$$

Cautious Cut The **(CC)** rule says that if H usually follows when F and G are both true, and G is usually true whenever F is true, then G is somewhat redundant and we can conclude that H usually follows from F alone.

$$\frac{F \mid\!\sim G \quad (F \wedge G) \mid\!\sim H}{F \mid\!\sim H} \qquad \textbf{(CC)}$$

Negation Rationality The **(NRAT)** rule says that if H usually follows when F, but doesn't usually follow from F together with $\neg G$, then it should be the case that H usually follows from F together with G.

$$\frac{F \mid\!\sim H \quad (F \wedge \neg G) \not\mid\!\sim H}{(F \wedge G) \mid\!\sim H} \qquad \textbf{(NRAT)}$$

This rule is sometimes also written in terms of its contrapositive as:

$$\frac{(F \wedge G) \not\mid\!\sim H \quad (F \wedge \neg G) \not\mid\!\sim H}{F \not\mid\!\sim H} \qquad \textbf{(NRAT)'}$$

which is to say that if H doesn't usually follow from either F together with G or from F together from $\neg G$, then there must be something inherent about F that means that H doesn't usually follow from F.

Exercise

Use the Gabbay–Makinson Rules (and featured derived rules) to prove that the following statements follow rationally from the premises that *"Birds usually fly"*, *"Penguins usually don't fly"* and *Penguins are birds*".

(i) *"Birds that are penguins usually don't fly"*;

$$\frac{p \mid\!\sim \neg f \quad \dfrac{p \models b}{p \mid\!\sim b}\,\textbf{(SC)}}{(p \wedge b) \mid\!\sim \neg f}\,\textbf{(CMON)}$$

(ii) *"Birds are usually not penguins"*.

$$\frac{\dfrac{\dfrac{\dfrac{(\text{i})}{(b \wedge p) \mid\!\sim \neg f}}{b \mid\!\sim (p \to \neg f)}\,\textbf{(CON)} \quad b \mid\!\sim f}{b \mid\!\sim (p \to \neg f) \wedge f}\,\textbf{(R}\wedge\textbf{)} \quad (p \to \neg f) \wedge f \models \neg p}{b \mid\!\sim \neg p}\,\textbf{(RWE)}$$

2 Semantics for Rational Consequence

Learning Objectives

By the end of this section you should be able to:

- ☐ Define what is a **ranked preferential model**;
- ☐ Determine whether $F \mid\!\approx_{\vec{S}} G$ for a given ranked preferential model $\vec{S}$;
- ☐ Show that $\mid\!\approx_{\vec{S}}$ is a **rational consequence relation** for all ranked models $\vec{S}$;
- ☐ Apply to Representation Theorem to show that certain properties/rules hold for all rational consequence relations.

2.1 Ranked Preferential Models

So far, we have defined rational consequence relations purely syntactically with a set of rules — the Gabbay–Makinson rules — that we felt captured our intuitions of how we would like our notion of "usually" to behave. However, as with (classical) propositional logic, we would also like a semantic definition that is sound and complete with respect to the proof theory we have defined. One such semantics is that of **ranked preferential models** [4].

Definition 2.1: Ranked Preferential Models

A **ranked preferential model** is an ordered sequence $\vec{S} = \langle S_1, S_2, \ldots, S_k \rangle$ of length k, for some $k \geq 1$, where each S_i is a set of propositional valuations.

Intuitively, we can think of $\vec{S}$ as an ordering over the **"most usual"** worlds, with S_1 being considered the most usual, and S_k being considered the most exceptional case[7]. We can define an associated consequence relation so that $F \mid\!\approx_{\vec{S}} G$ by

[7]Note that the definition does not require that every model appears in some S_i — indeed, there may be some models that we deem, not only to be exceptional, but impossible!

checking that G is true in all the most usual worlds in which F is true, but allowing that there may be some less usual worlds in which F is true but G is false.

Definition 2.2

Given a ranked preferential model $\vec{S} = \langle S_1, \ldots, S_k \rangle$ we can define an associated consequence relation $\mathrel{|\!\!\approx}_{\vec{S}}$ by taking

$$F \mathrel{|\!\!\approx}_{\vec{S}} G \quad \iff \quad \begin{cases} \mathsf{mod}(F) \cap S_i = \varnothing & \text{for all } 1 \leq i \leq k, \\ \mathsf{mod}(F) \cap S_j \subseteq \mathsf{mod}(G) & \text{where } j \text{ is minimal such that } \mathsf{mod}(F) \cap S_j \neq \varnothing, \end{cases}$$

for all $F, G \in$ *Fml*.

In other words, to determine whether $F \mathrel{|\!\!\approx}_{\vec{S}} G$ holds, we successively check the sets S_i until we find some S_j that contains a model satisfying F (*i.e.* when $\mathsf{mod}(F) \cap S_j \neq \varnothing$). We then disregard all other sets except S_j, and check whether G is satisfied in all valuations in S_j that satisfy F. This is effectively the same criterion that we check when deciding whether $F \models G$ (namely, whether $\mathsf{mod}(F) \subseteq \mathsf{mod}(G)$), only this time we ignore all valuations that do not belong to S_j.

Exercise

Consider the following ranked preferential model $\vec{S} = \langle S_1, S_2, S_3 \rangle$ where

$$\begin{aligned} S_1 &= \{\ (p \wedge \neg q \wedge \neg r)\ \}, \\ S_2 &= \{\ (\neg p \wedge q \wedge r),\ (p \wedge q \wedge \neg r)\ \}, \\ S_3 &= \{\ (p \wedge \neg q \wedge r)\ \}. \end{aligned}$$

Determine which of the following rational consequences hold:

(i) $p \mathrel{|\!\!\approx}_{\vec{S}} \neg q$

True. Since S_1 is minimal such that $\text{mod}(p) \cap S_1 = \{(p \wedge \neg q \wedge \neg r)\} \neq \varnothing$ and $\text{mod}(p) \cap S_1 \subseteq \text{mod}(\neg q)$

(ii) $(p \wedge r) \mathrel{|\!\!\approx}_{\vec{S}} q$

False. Since S_3 is minimal such that $\text{mod}(p \wedge r) \cap S_3 = \{(p \wedge \neg q \wedge r)\} \neq \varnothing$ yet $(p \wedge \neg q \wedge r) \notin \text{mod}(q)$

(iii) $q \mathrel{|\!\!\approx}_{\vec{S}} (p \to r)$

True. Since S_2 is minimal such that $\text{mod}(q) \cap S_2 = \{(\neg p \wedge q \wedge r),\ (p \wedge q \wedge \neg r)\} \neq \varnothing$ and $\text{mod}(q) \cap S_2 \subseteq \text{mod}(p \to r) = \text{mod}(\neg p) \cup \text{mod}(r)$

2.2 The Representation Theorem: Part I

With ranked preferential models we certainly have a definition of a consequence relation, but is this a *rational* consequence relation? We now show that the consequence relation $\mid\!\approx_{\vec{S}}$ associated with an arbitrary ranked model is indeed a rational consequence relation; that is to say that $\mid\!\approx_{\vec{S}}$ satisfies all of the Gabbay–Makinson rules.

Exercise

Prove the following set relationships:

(i) $\text{mod}(F \wedge G) = \text{mod}(F) \cap \text{mod}(G)$

$$\begin{aligned} V \in \text{mod}(F \wedge G) &\iff V(F \wedge G) = 1 \\ &\iff V(F) = 1 \text{ and } V(G) = 1 \\ &\iff V \in \text{mod}(F) \text{ and } V \in \text{mod}(G) \\ &\iff V \in \text{mod}(F) \cap \text{mod}(G) \end{aligned}$$

(ii) $\text{mod}(F \vee G) = \text{mod}(F) \cup \text{mod}(G)$

$$\begin{aligned} V \in \text{mod}(F \vee G) &\iff V(F \vee G) = 1 \\ &\iff V(F) = 1 \text{ or } V(G) = 1 \\ &\iff V \in \text{mod}(F) \text{ or } V \in \text{mod}(G) \\ &\iff V \in \text{mod}(F) \cup \text{mod}(G) \end{aligned}$$

(iii) $\text{mod}(\neg F) = \overline{\text{mod}(F)}$

$$\begin{aligned} V \in \text{mod}(\neg F) &\iff V(\neg F) = 1 \\ &\iff V(F) = 0 \\ &\iff V \notin \text{mod}(F) \\ &\iff V \in \overline{\text{mod}(F)} \end{aligned}$$

Theorem 2.3

Let $\vec{S} = \langle S_1, \ldots, S_k \rangle$ be any ranked preferential model. Then $\mid\!\approx_{\vec{S}}$ is a **rational consequence relation**.

Proof: Suppose that $\vec{S} = \langle S_1, \ldots, S_k \rangle$ is a ranked preferential model. It is sufficient to show that $\mid\!\approx_{\vec{S}}$ satisfies each of the Gabbay–Makinson rules in turn.

(REF) If $\mathsf{mod}(F) \cap S_i = \varnothing$ for all i then $F \mid\!\approx_{\vec{S}} F$ holds automatically. Otherwise, let j be minimal such that $\mathsf{mod}(F) \cap S_j \neq \varnothing$. Then clearly $\mathsf{mod}(F) \cap S_j \subseteq \mathsf{mod}(F)$, as required.

(R∧) Suppose that $F \mid\!\approx_{\vec{S}} G$ and $F \mid\!\approx_{\vec{S}} H$. If $\mathsf{mod}(F) \cap S_i = \varnothing$ for all i then we are done. Otherwise, let j be minimal such that $\mathsf{mod}(F) \cap S_j \neq \varnothing$ then $\mathsf{mod}(F) \cap S_j \subseteq \mathsf{mod}(G)$ and $\mathsf{mod}(F) \cap S_j \subseteq \mathsf{mod}(H)$. And so it follows that

$$\mathsf{mod}(F) \cap S_j \subseteq \mathsf{mod}(G) \cap \mathsf{mod}(H) = \mathsf{mod}(G \wedge H)$$

as required.

(L∨) Suppose that $F \mid\!\approx_{\vec{S}} H$ and $G \mid\!\approx_{\vec{S}} H$ If $\mathsf{mod}(F \vee G) \cap S_i = \varnothing$ for all i then we are done. Otherwise, let j be minimal such that $\mathsf{mod}(F \vee G) \cap S_j \neq \varnothing$. Since $\mathsf{mod}(F \vee G) = \mathsf{mod}(F) \cup \mathsf{mod}(G)$, we have that

$$\begin{aligned} \mathsf{mod}(F \vee G) \cap S_j &= (\mathsf{mod}(F) \cup \mathsf{mod}(G)) \cap S_j \\ &= (\mathsf{mod}(F) \cap S_j) \cup (\mathsf{mod}(G) \cap S_j) \neq \varnothing \end{aligned}$$

It follows that if $\mathsf{mod}(F) \cap S_j \neq \varnothing$ then j must be minimal such that $\mathsf{mod}(F) \cap S_j \neq \varnothing$, otherwise j would not be minimal with $\mathsf{mod}(F \vee G) \cap S_j \neq \varnothing$. And so $\mathsf{mod}(F) \cap S_j \subseteq \mathsf{mod}(H)$[a].

Similarly, if $\mathsf{mod}(G) \cap S_j \neq \varnothing$ then j must also be minimal such that $\mathsf{mod}(G) \cap S_j \neq \varnothing$. And so $\mathsf{mod}(G) \cap S_j \subseteq \mathsf{mod}(H)$.

Hence we have that $\mathsf{mod}(F \vee G) \cap S_j = (\mathsf{mod}(F) \cap S_j) \cup (\mathsf{mod}(G) \cap S_j) \subseteq \mathsf{mod}(H)$, as required.

Notes

Sketch a Venn diagram illustrating the above set inclusions between $\text{mod}(F)$, $\text{mod}(G)$, S_j and $\text{mod}(H)$ for **(R$\wedge$)** and **(L$\vee$)**.

(LLE) Suppose that $F \mathrel{|\!\!\approx}_{\vec{S}} H$ and $F \equiv G$, i.e. $\text{mod}(F) = \text{mod}(G)$. If $\text{mod}(G) \cap S_i = \varnothing$ for all i then we are done. Otherwise, let j be minimal such that $\text{mod}(G) \cap S_j \neq \varnothing$ then $\text{mod}(G) \cap S_j = \text{mod}(F) \cap S_j \subseteq \text{mod}(H)$, as required.

(RWE) Suppose that $F \mathrel{|\!\!\approx}_{\vec{S}} G$ and $G \models H$, i.e. $\text{mod}(G) \subseteq \text{mod}(H)$. If $\text{mod}(F) \cap S_i = \varnothing$ for all i then we are done. Otherwise, let j be minimal such that $\text{mod}(F) \cap S_j \neq \varnothing$ then $\text{mod}(F) \cap S_j \subseteq \text{mod}(G) \subseteq \text{mod}(H)$, as required.

Notes

Sketch a Venn diagram illustrating the above set inclusions between $\text{mod}(F)$, $\text{mod}(G)$, S_j and $\text{mod}(H)$ for **(LLE)** and **(RWE)**

(CMON) Suppose that $F \mathrel{|\!\!\approx}_{\vec{S}} G$ and $F \mathrel{|\!\!\approx}_{\vec{S}} H$. If $\mathsf{mod}(F \wedge G) \cap S_i = \varnothing$ for all i then we are done. Otherwise, let j be minimal such that $\mathsf{mod}(F \wedge G) \cap S_j \neq \varnothing$ then

$$\mathsf{mod}(F \wedge G) \cap S_j = (\mathsf{mod}(F) \cap \mathsf{mod}(G)) \cap S_j \subseteq \mathsf{mod}(F) \cap S_j \subseteq \mathsf{mod}(H)$$

as required.

(RMON) Suppose that $F \mathrel{|\!\!\approx}_{\vec{S}} H$ and $F \not\mathrel{|\!\!\approx}_{\vec{S}} \neg G$. If $\mathsf{mod}(F \wedge G) \cap S_i = \varnothing$ for all i then we are done. Otherwise, let j be minimal such that $\mathsf{mod}(F \wedge G) \cap S_j \neq \varnothing$. It follows that $\mathsf{mod}(F) \cap S_j \neq \varnothing$ since

$$\mathsf{mod}(F \wedge G) \cap S_j = (\mathsf{mod}(F) \cap \mathsf{mod}(G)) \cap S_j \subseteq \mathsf{mod}(F) \cap S_j \tag{1}$$

Now let $i < j$ then we have that $\mathsf{mod}(F \wedge G) \cap S_i = \mathsf{mod}(F) \cap \mathsf{mod}(G) \cap S_i = \varnothing$. Hence, it follows that

$$\mathsf{mod}(F) \cap S_i \subseteq \overline{\mathsf{mod}(G)} \;=\; \mathsf{mod}(\neg G) \tag{2}$$

since $X \cap Y = \varnothing$ iff $X \subseteq \overline{Y}$.

It follows that j must be minimal with $\mathsf{mod}(F) \cap S_j = \varnothing$, else by (2) we would have that $F \mathrel{|\!\!\approx}_{\vec{S}} \neg G$, contrary to the assumption.

But then, it follows from (1) and $F \mathrel{|\!\!\approx}_{\vec{S}} H$ that

$$\mathsf{mod}(F \wedge G) \cap S_j \subseteq \mathsf{mod}(F) \cap S_j \subseteq \mathsf{mod}(H)$$

as required.

Notes

Sketch a Venn diagram illustrating the above set inclusions between $\mathsf{mod}(F)$, $\mathsf{mod}(G)$, S_j and $\mathsf{mod}(H)$ for **(CMON)** and **(RMON)**

One application of this result is that it allows us to show that certain conclusions *cannot* be rationally inferred from some set of premises — similar to how in (classical) propositional logic, the easiest way to show that some consequence is *not* provable, is to construct a counter-model where the premises are true but where the conclusion fails.

Notes

Consider the following premises:

Premise 1) Birds usually fly ($b \mid\!\sim f$);

Premise 2) Penguins usually don't fly $p \mid\!\sim \neg f$;

Premise 3) Penguins that are not birds are usually impossible ($p \wedge \neg b \mid\!\sim \bot$)

Show that we cannot conclude from these premises that *"non-flying birds are usually penguins"* ($\neg f \mid\!\sim p$).

Consider the rational consequence relation $\mid\!\approx_{\vec{S}}$ generated by the model $\vec{S} = \langle S_1, S_2, S_3 \rangle$ where

$$S_1 = \{b \wedge \neg p \wedge f\}, \qquad S_2 = \{b \wedge \neg p \wedge \neg f\} \quad \text{and} \quad S_3 = \{b \wedge p \wedge \neg f\}$$

- $b \mid\!\approx_{\vec{S}} f$ since S_1 is minimal such that $\text{mod}(b) \cap S_1 \neq \varnothing$ and $\text{mod}(b) \cap S_1 = \{b \wedge \neg p \wedge f\} \subseteq \text{mod}(f)$.
- $p \mid\!\approx_{\vec{S}} \neg f$ since S_3 is minimal such that $\text{mod}(p) \cap S_3 \neq \varnothing$ and $\text{mod}(p) \cap S_3 = \{b \wedge p \wedge \neg f\} \subseteq \text{mod}(\neg f)$.
- $p \wedge \neg b \mid\!\approx_{\vec{S}} \bot$ since $\text{mod}(p \wedge \neg b) \cap S_i = \varnothing$ for all $i = 1, 2, 3$.
- However, $\neg f \not\mid\!\sim p$ since S_2 is minimal such that $\text{mod}(\neg f) \cap S_2 \neq \varnothing$, yet $\text{mod}(\neg f) \cap S_2 \{b \wedge \neg p \wedge \neg f\} \not\subseteq \text{mod}(p)$.

[a] Of course this also holds if $\text{mod}(F) \cap S_j = \varnothing$, since trivially $\varnothing \subseteq \text{mod}(H)$.

2.2.1 Historical Note

You may have asked yourself *"If this is what a ranked preferential model is, is there such a thing as a non-ranked preferential model?"*, and indeed there is! We can think of our ranked preferential models $\vec{S} = \langle S_1, \ldots, S_k \rangle$ not as a sequence of but as an order $(\{S_1, \ldots, S_k\}, \prec)$ where $\prec$ is a **total linear ordering** on $\{S_1, \ldots, S_k\}$ — this is just an alternative formulation of the same definition with $S_i \prec S_j$ if and only if $i < j$.

But now, under this formulation, it is natural to consider alternative types of ordering on $\{S_1, \ldots, S_k\}$, such as only requiring $\prec$ to be a **partial order** (*i.e.* one that is reflexive, transitive, and anti-symmetric). This defines a **preferential model** which, with a suitable tweak to the definition of the relation $\mathrel{|\!\approx}_{\vec{S}}$, gives a slightly weaker consequence relation; one that satisfies all of the Gabbay–Makinson rules *except* for Rational Monotonicity **(RMON)** [4]. If we felt strongly that rational monotonicity was not a "reasonable" rule for how we would like to model our notion of "usually", then we could consider preferential models instead of ranked preferential models.

2.3 The Representation Theorem: Part II

In Theorem 2.2 we saw that the consequence relation $\mid\!\approx_{\vec{S}}$ generated by any ranked preferential model is indeed a rational consequence relation. But can *every* rational consequence relation be represented by some suitably chosen ranked preferential mode? It turns out that the answer to this question is: yes!

Theorem 2.4: The Representation Theorem

Let $\mid\!\sim$ be any **rational consequence relation**. Then there is some **ranked preferential model** $\vec{S}$ such that $\mid\!\approx_{\vec{S}} = \mid\!\sim$.

Proof: Suppose that $\mid\!\sim \;=\; \mid\!\approx_{\vec{S}}$ for some ranked preferential model $\vec{S} = \langle S_1, \ldots, S_k \rangle$, for some $k > 0$.

Step 1) We can deduce S_1 by looking at all the formulas that are **usually true** — those that **usually follow from a tautology** $\top \mid\!\sim F$, since by definition we would have

$$S_1 \;=\; S_1 \cap \mathsf{mod}(\top) \;\subseteq\; \mathsf{mod}(F)$$

Hence, if $V \in S_1$ then it must be in $\mathsf{mod}(F)$ for every $F \in \mathit{Fml}$ such that $\top \models F$. So by taking the common intersection of all such $\mathsf{mod}(F)$, we get the largest set of valuations that are models of all formulas that are usually true:

$$S_1 \;=\; \bigcap \{\, \mathsf{mod}(F) \;:\; \top \mid\!\sim F \,\}$$

Step 2) To deduce S_2 we look for all the formulas that are usually true in the **remaining** models. To do this, we first need a formula Θ_1 that is false in all valuations in S_1 and true in all valuation $V \notin S_1$

Notes

Construct a formula Θ_1 such that $V(\Theta_1) = 1$ if and only if $V \notin S_1$, where $S_1 = \{(p \wedge \neg q \wedge r), (\neg p \wedge q \wedge r)\}$.

We can read the DNF of Θ_1 directly form the valuations that do not below to S_1:

$$\Theta_1 \;=\; (\neg p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \\ \vee (p \wedge \neg q \wedge \neg r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r)$$

We can then define S_2 as the largest set of valuations that are models of all formulas that **usually follow from** Θ_1 $\Theta_1 \mid\!\sim F$:

$$S_2 \;=\; \bigcap \{\, \mathsf{mod}(F) \;:\; \Theta_1 \mid\!\sim F \,\}$$

Step 3) In general, we can deduce S_{i+1} by looking at the formulas that are usually true in all of the remaining models that are not among the $S_1, \ldots, S_i$ already deduced. So let $\Theta_i \in$ *Fml* be such that

$$\mathsf{mod}(\Theta_i) = \overline{S_1 \cup \cdots \cup S_i} \tag{3}$$

and take

$$S_{i+1} \;=\; \bigcap \{\, \mathsf{mod}(F) \;:\; \Theta_i \mid\!\sim F \,\} \tag{4}$$

It follows that we can recursively use (4) to generate an infinite sequence of sets $S_1, S_2, S_3, \ldots$. All that remains is to show that this sequence eventually stabilises with $S_{k+1} = S_{k+2} = \cdots = \varnothing$, for some finite $k \geq 1$.

Exercise

(i) Show that if $S_i = \varnothing$ then $S_{i+1} = \varnothing$.

(ii) Use the observation that $S_j \subseteq \mathsf{mod}(\Theta_j)$ to show that $S_i \cap S_j = \varnothing$, for $i < j$.

(i) If $S_i = \varnothing$ then we would choose $\Theta_{i+1} \in \mathit{Fml}$ such that

$$\mathsf{mod}(\Theta_i) = \overline{S_1 \cup \cdots \cup S_{i-1} \cup S_i} = \overline{S_1 \cup \cdots \cup S_{i-1}} = \mathsf{mod}(\Theta_{i-1})$$

and so $\Theta_i \equiv \Theta_{i-1}$. Hence by **(LLE)**

$$\begin{aligned} S_{i+1} &= \bigcap\{\mathsf{mod}(F) : \Theta_i \mid\!\sim F\} \\ &= \bigcap\{\mathsf{mod}(F) : \Theta_{i-1} \mid\!\sim F\} = S_i = \varnothing \end{aligned}$$

(ii) Let $i < j$, then

$$\begin{aligned} S_i \cap S_j &\subseteq S_i \cap \mathsf{mod}(\Theta_j) \\ &= S_i \cap \overline{S_1 \cup \cdots \cup S_j} \\ &\subseteq S_i \cap \overline{S_1} \cap \cdots \cap \overline{S_i} \cap \ldots \overline{S_j} \subseteq S_i \cap \overline{S_i} = \varnothing \end{aligned}$$

Since there are only finitely many possible valuations, by the pigeonhole principle, there must be some finite k after which this definition ceases to generate any further non-empty sets using. Hence we can take $\vec{S} = \langle S_1, \ldots, S_k \rangle$ to be the resulting (finite) ranked preferential model.

Of course, this construction started with the bold assumption that $\mid\!\sim\, = \,\mid\!\approx_{\vec{S}}$ for some ranked preferential model $\vec{S}$, which seems notoriously circular since this was precisely what we were trying to prove in the first place! Fortunately, now that we have a definition of what we expect $\vec{S}$ should be if $\mid\!\sim\, = \,\mid\!\approx_{\vec{S}}$, we can proceed to show that it really does satisfy the property that $F \mid\!\sim G$ if and only if $F \mid\!\approx_{\vec{S}} G$. Thus eliminating any trace of circularity.

Proof:

($\Rightarrow$) Suppose that $F \mid\!\sim G$, then if $\mathsf{mod}(F) \cap S_i = \varnothing$ for all $i = 1, \ldots, k$ then we are done. Otherwise, suppose that j is minimal such that $\mathsf{mod}(F) \cap S_j \neq \varnothing$, which is to say that $\mathsf{mod}(F) \cap S_i = \varnothing$ for all $i = 1, \ldots, (j-1)$.

It then follows that

$$\mathsf{mod}(F) \cap (S_1 \cup \cdots \cup S_{j-1}) \;=\; \varnothing \qquad \text{since } j \text{ is minimal}$$

Hence, by (3), we have that

$$\mathsf{mod}(F) \;\subseteq\; \overline{S_1 \cup \cdots \cup S_{j-1}} \;=\; \mathsf{mod}(\Theta_{j-1})$$

which is to say that $F \models \Theta_{j-1}$.

Notes

Use **(SC)**, **(CMON)**, and **(CON)** to show that $S_j \subseteq \mathsf{mod}(F \to G)$

We then have that

$$\dfrac{\dfrac{\dfrac{F \models \Theta_{j-1}}{F \mid\!\sim \Theta_{j-1}}\,\textbf{(SC)} \qquad F \mid\!\sim G}{F \wedge \Theta_{j-1} \mid\!\sim G}\,\textbf{(CMON)}}{\Theta_{j-1} \mid\!\sim F \to G}\,\textbf{(CON)}$$

From which it follows from the definition of S_j that $S_j \subseteq \mathsf{mod}(F \to G)$.

From $S_j \subseteq \mathsf{mod}(F \to G)$ it follows that

$$\begin{aligned}
\mathsf{mod}(F) \cap S_j &\subseteq \mathsf{mod}(F) \cap \mathsf{mod}(F \to G) \\
&= \mathsf{mod}(F \wedge (F \to G)) \\
&\subseteq \mathsf{mod}(G) \qquad \text{since } F \wedge (F \to G) \models G
\end{aligned}$$

Hence we have that $F \mid\!\approx_{\vec{S}} G$, as required.

($\Leftarrow$) Conversely, suppose that $F \mid\!\approx_{\vec{S}} G$.

Exercise

Show that the set $R = \{\text{mod}(F) : \Theta_i \mid\!\sim F\}$ is closed under intersections. *i.e.* If $X, Y \in R$ both belong to the set then $(X \cap Y) \in R$.

Suppose that $X, Y \in R$ then by definition $X = \text{mod}(F)$ and $Y = \text{mod}(G)$ for some F and G with $\Theta_i \mid\!\sim F$ and $\Theta_i \mid\!\sim G$.

By **(R$\wedge$)** we have that $\Theta_i \mid\!\sim (F \wedge G)$ and hence $\text{mod}(F \wedge G) = \text{mod}(F) \cap \text{mod}(G) = (X \cap Y) \in M$.

Since $\{\text{mod}(F) : \Theta_i \mid\!\sim F\}$ is finite and closed under intersections, we must have that it contains its common intersection S_i. Let Ψ_i be such that $\text{mod}(\Psi_i) = S_i$ then this means that

$$\Theta_{i-1} \mid\!\sim \Psi_i \tag{5}$$

Exercise

Show that $\Theta_i \mid\!\sim \bot$ iff $S_i = \varnothing$.

($\Rightarrow$) Suppose that $\Theta_i \mid\!\sim \bot$ then $S_i \subseteq \text{mod}(\bot) = \varnothing$, as required.

($\Leftarrow$) If $S_i = \varnothing$ then $\text{mod}(\Psi_i) = S_i = \varnothing = \text{mod}(\bot)$ and so $\Psi_i \models \bot$. Hence by **(RWE)** we have

$$\frac{\Theta_i \mid\!\sim \Psi_i \quad \Psi_i \models \bot}{\Theta_i \mid\!\sim \bot}\ \textbf{(RWE)}$$

We have two cases to consider:

Case 1) If $\mathsf{mod}(F) \cap S_i = \varnothing$ for all $i = 1, \ldots, k$ then it follows that

$$\mathsf{mod}(F) \cap (S_1 \cup \cdots \cup S_k) = \varnothing$$

Then, by (3), we have that

$$\mathsf{mod}(F) \subseteq \overline{S_1 \cup \cdots \cup S_k} = \mathsf{mod}(\Theta_k)$$

Hence $F \models \Theta_k$ and so

$$F \wedge \Theta_k \equiv F \tag{6}$$

Since $S_{k+1} = \varnothing$ we have that $\Theta_k \mid\!\sim \bot$. Hence we have that

Notes

Use **(RWE)**, **(CMON)** and **(LLE)** to show that $F \mid\!\sim G$.

$$\dfrac{\dfrac{\dfrac{\Theta_k \mid\!\sim \bot \quad \bot \models F}{\Theta_k \mid\!\sim F}\,\textbf{(RWE)} \qquad \dfrac{\Theta_k \mid\!\sim \bot \quad \bot \models G}{\Theta_k \mid\!\sim G}\,\textbf{(RWE)}}{\Theta_k \wedge F \mid\!\sim G}\,\textbf{(CMON)} \qquad \Theta_k \wedge F \equiv F}{F \mid\!\sim G}\,\textbf{(LLE)}$$

Hence in this case we have that $F \mid\!\sim G$, as required.

Case 2) If $\mathsf{mod}(F) \cap S_i \neq \varnothing$ for all $i = 1, \ldots, k$ then let j be minimal such that $\mathsf{mod}(F) \cap S_j \neq \varnothing$. From which it follows that

$$S_j \not\subseteq \overline{\mathsf{mod}(F)} = \mathsf{mod}(\neg F)$$

It then follows from (4) that we must have

$$\Theta_{j-1} \not\mid\!\sim \neg F \tag{7}$$

Furthermore, it also follows that

$$\mathsf{mod}(F) \cap (S_1, \ldots, S_{j-1}) = \varnothing$$

since j is minimal. And so by (3), we get that

$$\mathsf{mod}(F) \subseteq \overline{S_1 \cup \cdots \cup S_{j-1}} = \mathsf{mod}(\Theta_{j-1})$$

and so $F \models \Theta_{j-1}$. An immediate consequence of this that will be useful later is that

$$F \;\equiv\; F \wedge \Theta_{j-1} \tag{8}$$

Since $F \mid\!\approx_{\vec{S}} G$ we must have that $\mathsf{mod}(F) \cap S_j \subseteq \mathsf{mod}(G)$. Hence by (5),

$$\mathsf{mod}(F) \cap \mathsf{mod}(\Psi_j) \quad \subseteq \quad \mathsf{mod}(G)$$

which is to say that $F \wedge \Psi_j \models G$, which means that

$$\Psi_j \models (F \to G) \tag{9}$$

Combining (7), (5), (9) and (8) we can show that $F \mid\!\sim (F \to G)$.

Notes

Use **(RWE)**, **(RMON)**, and **(LLE)** together with (7),(5), (9) and (8) to show that $F \mid\!\sim (F \to G)$

$$\dfrac{\dfrac{\Theta_{j-1} \not\mid\!\sim \neg F \qquad \dfrac{\Theta_{j-1} \mid\!\sim \Psi_j \quad \Psi_j \models (F \to G)}{\Theta_{j-1} \mid\!\sim (F \to G)}\,\textbf{(RWE)}}{F \wedge \Theta_{j-1} \mid\!\sim (F \to G)}\,\textbf{(RMON)} \qquad F \equiv F \wedge \Theta_{j-1}}{F \mid\!\sim (F \to G)}\,\textbf{(LLE)}$$

Finally, we can show that from $F \mid\!\sim (F \to G)$ we can rationally deduce that $F \mid\!\sim G$.

Notes

Use **(REF)**, **(R∧)**, and **(RWE)** together with $F \mid\!\sim (F \to G)$ to show that $F \mid\!\sim G$.

$$\dfrac{\dfrac{\dfrac{\varnothing}{F \mid\!\sim F}\,\textbf{(REF)} \qquad F \mid\!\sim (F \to G)}{F \mid\!\sim F \wedge (F \to G)}\,\textbf{(R}\wedge\textbf{)} \qquad F \wedge (F \to G) \models G}{F \mid\!\sim G}\,\textbf{(RWE)}$$

Hence in this case we also have that $F \mid\!\sim G$, as required.

Exercise

Use the Representation Theorem to show that the following derived rule of Negation Rationality holds for all rational consequence relations:

$$\frac{(F \wedge G) \not\mid\!\sim H \quad (F \wedge \neg G) \not\mid\!\sim H}{F \not\mid\!\sim H} \textbf{ (NRAT)}$$

By the Representation Theorem, every rational consequence relation is of the form $\mid\!\approx_{\vec{S}}$ for some ranked preferential model $\vec{S} = \langle S_1, \ldots, S_k \rangle$. And so it suffices to show that $F \not\mid\!\approx_{\vec{S}} H$ whenever $(F \wedge G) \not\mid\!\approx_{\vec{S}} H$ and $F \mid\!\approx_{\vec{S}} H$, for an arbitrary $\vec{S}$. Suppose that $(F \wedge G) \not\mid\!\approx_{\vec{S}} H$ and $(F \wedge \neg G) \not\mid\!\approx_{\vec{S}} H$, then there is some minimal j_1, j_2 such that $\text{mod}(F \wedge \neg G) \cap S_{j_1} \not\subseteq \text{mod}(H)$ and $\text{mod}(F \wedge \neg G) \cap S_{j_2} \not\subseteq \text{mod}(H)$. It follows that $\text{mod}(F) \cap S_{j_1} \not\subseteq \text{mod}(H)$ and $\text{mod}(F) \cap S_{j_2} \not\subseteq \text{mod}(H)$
Note that since $\text{mod}(F) = \text{mod}(F \wedge G) \cup \text{mod}(F \wedge \neg G)$ we have that for $i < \min(j_1, j_2)$

$$\text{mod}(F) \cap S_i \subseteq (\text{mod}(F \wedge G) \cap S_i) \cup (\text{mod}(F \wedge \neg G) \cap S_i) = \varnothing$$

And hence $j = \min(j_1, j_2)$ must be minimal such that $\text{mod}(F) \cap S_j \neq \varnothing$. However, we then have that

$$\text{mod}(F) \cap S_j \subseteq (\text{mod}(F \wedge G) \cap S_j) \cup (\text{mod}(F \wedge \neg G) \cap S_j) \not\subseteq \text{mod}(H)$$

which is to say that $F \not\mid\!\approx_{\vec{S}} H$, as required.

3 Introduction to Fuzzy Logics

Learning Objectives

By the end of this section you should be able to:

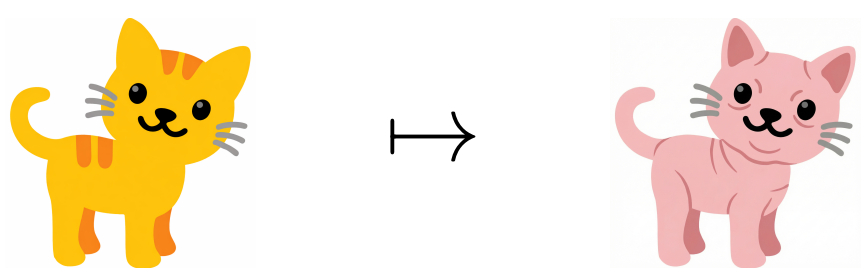
- ☐ Evaluate arbitrary propositional formulas in each of the Fuzzy logics $\mathbb{F}^1$, $\mathbb{F}^2$, and $\mathbb{F}^3$;
- ☐ Demonstrate Trillas' Theorem by constructing an appropriate isomorphism for any given 'reasonable' function for negation;

3.1 Motivation and Definitions

Consider the following argument:

Premise 1)	Mr Whiskers is not bald;
Premise 2)	If Mr Whiskers is not bald and we pluck one hair from Mr Whiskers, then Mr Whiskers would still remain not bald;
Conclusion)	Mr Whiskers still remains not bald.

On the face of it this seems like a perfectly natural argument that we might formulate in propositional logic[8]. However, if we were to continue painstaking plucking hairs from Mr Whiskers — in what is doubtless a far more painful experience for him than for us —then we would continue to maintain that 'Mr Whiskers remains not bald' is true even once Mr Whiskers looks like this:



[8] or if we were concerned about the temporal aspect of this, we could formulate this in predicate logic in terms of Mr Whiskers is not bald at time t

How, then, are we to make sense of what is clearly an undesirable conclusion?

Well, upon closer inspection, we may question the validity of a proposition such as Premise 2, and perhaps even if it makes sense to assign a boolean value to a proposition such as 'Mr Whiskers is (not) bald' — surely the issue here is that 'baldness' is not a binary switch where one is either 'bald' or 'not bald' but a sliding scale[9].

Definition 3.1: Boolean Truth Functions

We define a **Boolean truth function** to be a function on n-variables to be a function $F : \{0,1\}^n \to \{0,1\}$ that takes as input an n-tuple of values that are either 0 (false) or 1 (true) and returns a boolean output from $\{0,1\}$.

Each of our Boolean connectives $\neg$, $\wedge$, $\vee$, $\to$, therefore correspond to an equivalent truth function $F_\neg$, $F_\wedge$, $F_\vee$, $F_\to$, of the appropriate arity.

$$\begin{aligned} F_\neg(x) &= 1 - x \\ F_\wedge(x,y) &= \text{min}(x,y) \\ F_\vee(x,y) &= \text{max}(x,y) \\ F_\to(x,y) &= \text{max}(1-x,y) \end{aligned}$$

We can also associate each propositional formula F, with n-variables, with a corresponding truth function $\overline{F} : \{0,1\}^n \to \{0,1\}$ by composing the above functions

[9] If we object that 'baldness' is indeed a binary switch and possessing even a single hair is sufficient to be 'not bald', then swap this example for 'grains of sand' in a 'pile', where surely we would contest that a single gain of sand or even half a dozen grains of sand formed a 'pile of sand'

according to the structure of F. For example, if $F = (p \wedge \neg q) \to r$ then we would have that

$$\overline{((p \wedge \neg q) \to r)}(x_1, x_2, x_3) \;=\; F_{\to}(F_{\wedge}(x_1, F_{\neg}(x_2)), x_3),$$

In general we have that

$$V(F) = 1 \qquad \Longleftrightarrow \qquad \overline{F}\big(V(p_1), \ldots, V(p_n)\big) = 1$$

Effectively, this is just a reformulation of the semantics of classical propositional logic in terms of composition of truth functions. However, with this new formulation, we have the opportunity to relax the condition that our truth functions only take on Boolean values from $\{0, 1\}$, and instead consider truth values over the whole **unit interval** $[0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$.

Definition 3.2: Fuzzy Truth Functions

We define a **Fuzzy** or **real-valued truth function** on n-variables to be a function $F : [0, 1]^n \to [0, 1]$.

A **model** of **fuzzy logic** is a real-valued function $V : \mathsf{Prop} \to [0, 1]$ that assigns a real-value $V(p) \in [0, 1]$ to each propositional variable $p \in \mathsf{Prop}$.

Unlike Boolean truth functions — of which there are at most 2^{2^n} such functions on n-variables — there is seemingly no limit to how creative we could be with our choice of real-valued truth functions on n-variables. However, in practice, when people speak of "Fuzzy logic" they are usually speaking of one of the following semantics:

Min-Max/Zadeh Logic

$$\begin{aligned} F_{\neg}(x) &= 1 - x \\ F_{\wedge}(x, y) &= \mathsf{min}(x, y) \\ F_{\vee}(x, y) &= \mathsf{max}(x, y) \\ F_{\to}(x, y) &= \mathsf{min}(1, 1 - x + y) \end{aligned}$$

Product/Gödel Logic $\mathbb{F}^2$

$$
\begin{aligned}
F_{\neg}(x) &= 1 - x \\
F_{\wedge}(x, y) &= xy \\
F_{\vee}(x, y) &= x + y - xy \\
F_{\rightarrow}(x, y) &= \begin{cases} 1 & \text{if } x \leq y \\ y/x & \text{otherwise} \end{cases}
\end{aligned}
$$

Łukasiewicz Logic $\mathbb{F}^3$

$$
\begin{aligned}
F_{\neg}(x) &= 1 - x \\
F_{\wedge}(x, y) &= \mathsf{max}(0, x + y - 1) \\
F_{\vee}(x, y) &= \mathsf{min}(1, x + y) \\
F_{\rightarrow}(x, y) &= \mathsf{min}(1, 1 - x + y)
\end{aligned}
$$

Exercise

Evaluate the following formulas in each of the fuzzy logics $\mathbb{F}^1$, $\mathbb{F}^2$ and $\mathbb{F}^3$ under the valuation $V(p) = 0.3$ and $V(q) = 0.8$:

Formula	**Min–Max** $\mathbb{F}^1$	**Product** $\mathbb{F}^2$	**Łukasiewicz** $\mathbb{F}^3$
$\neg p$	0.7	0.7	0.7
$p \wedge q$	0.3	0.24	0.1
$p \vee q$	0.8	0.86	1
$p \rightarrow q$	1	1	1
$q \rightarrow p$	0.5	0.375	0.5
$(p \wedge q) \rightarrow p$	1	1	1
$p \wedge p$	0.3	0.09	0
$p \vee \neg p$	0.7	0.79	1
$p \wedge \neg p$	0.3	0.21	0
$(p \rightarrow q) \vee (q \rightarrow p)$	1	1	1

Definition 3.3

Given an n-variable function $F(p_1, \ldots, p_n)$ with variables $p_1, \ldots, p_n$. We say that a model $V : \text{Prop} \to [0,1]$ is a **model of** F **with respect to a fuzzy logic** $\mathbb{F}$ if $\overline{F}(V(p_1), \ldots, V(p_n)) = 1$.

We say that F is **valid with respect to** $\mathbb{F}$ or a **theorem of** $\mathbb{F}$ if $\overline{F}(x_1, \ldots, x_n) = 1$ for all $x_1, \ldots, x_n \in [0,1]$. In which case we write

$$\models^{\mathbb{F}} F \qquad \Longleftrightarrow \qquad \overline{F}(x_1, \ldots, x_n) = 1$$

3.2 Interlude 1: The Intermediate Value Theorem

Many of the results in the following sections depend on an important (and almost obvious) property of continuous real-valued functions, called the **Intermediate Value Theorem (I.V.T.)**.

Definition 3.4

A function $f : [a, b] \to \mathbb{R}$ is said to be **continuous at** $c \in [a, b]$ if $\lim_{x \to c} f(x) = f(c)$, *i.e.*

$$\forall \varepsilon > 0;\ \exists \delta > 0;\ |x - c| < \delta \implies |f(x) - f(c)| < \varepsilon$$

We say that f is **continuous on** $[a, b]$ (or simply **continuous**) if f is continuous at c for all $c \in [a, b]$.

Theorem 3.5: The Intermediate Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be any continuous real valued function defined on the interval $[a, b]$, and let $N \in \mathbb{R}$ be some value between $f(a)$ and $f(b)$. Then there is some $c \in [a, b]$ such that $f(c) = N$.

Proof: Suppose that $f(a) < N < f(b)$ and let $c = \sup(H)$ where

$$H = \{x \in [a, b] : f(x) \leq N\},$$

which is to say that c is the **least upper bound** of all $x \in [a, b]$ such that $f(x) \leq N$.

Notes

Show that $H \neq \varnothing$ and that $\sup(H)$ is well-defined.

We have that $a \in H$ since $a \in [a, b]$ and $f(a) \leq N$. We have that H has an upper bound since $b \notin H$ since $N < f(b)$, and $x \leq b$ for all $x \in H$ since $H \subseteq [a, b]$.

Note that $c \in (a, b)$ since $f(a) < N < f(b)$, by supposition.

Case 1) Suppose that $f(c) < N$ and let $\varepsilon = N - f(c)$, so by continuity there is some $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < \varepsilon$. Let $d = c + \frac{\delta}{2}$ then $|f(d) - f(c)| < N - f(c)$ from which we have that $f(d) < N$. Hence $d \in H$ and so $c \geq d = c + \frac{\delta}{2}$ since c is an **upper bound** on H, which is a contradiction.

Notes

Sketch a figure illustrating Case 1.

Case 2) Similarly, if $f(c) > N$ then let $\varepsilon = f(c) - N$, so by continuity there is some $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < \varepsilon$. Let $d = c + \frac{\delta}{2}$ then for all $x \in [d, c]$ it follows that $|f(x) - f(c)| < f(c) - N$, from which we get that $N - f(c) < f(x) - f(c)$, so $f(x) > N$. Which is to say that if $x \in [d, c]$ then $x \notin H$.

Hence $H \subseteq [a, d)$ and so d is an upper bound of H, which is a contradiction since $c > d$ is the **least** upper bound.

Notes

Sketch a figure illustrating Case 2.

Since $<$ is a **total linear order** on $[0, 1]$, we must have that $f(c) = N$, as required.

Exercise

Use the Intermediate Value Theorem to show that there are always two antipodal points on Earth (points on opposite sides of the Earth) that have the same temperature.

(You may assume that the Earth is a perfect sphere, and that temperature changes continuously as we navigate around the surface on the Earth.)

- Choose any great circle that divides the Earth into two hemispheres — for example the equator.
- Define a (continuous) function $t : [0, 360] \to \mathbb{R}$ by taking

$$t(x) \;=\; \text{temp}(0, x) - \text{temp}(0, x + 180)$$

to be the difference in temperature between the antipodal points at coordinates $(0°N, x°E)$ and $(0°N, (x + 180)°E)$.

Let $a = 0$ and $b = 180$ then we note that

$$\begin{aligned} t(b) &= \text{temp}(0, 180) - \text{temp}(0, 180 + 180) \\ &= \text{temp}(0, 180) - \text{temp}(0, 0) \qquad \text{since } 360° = 0° \\ &= -t(a) \end{aligned}$$

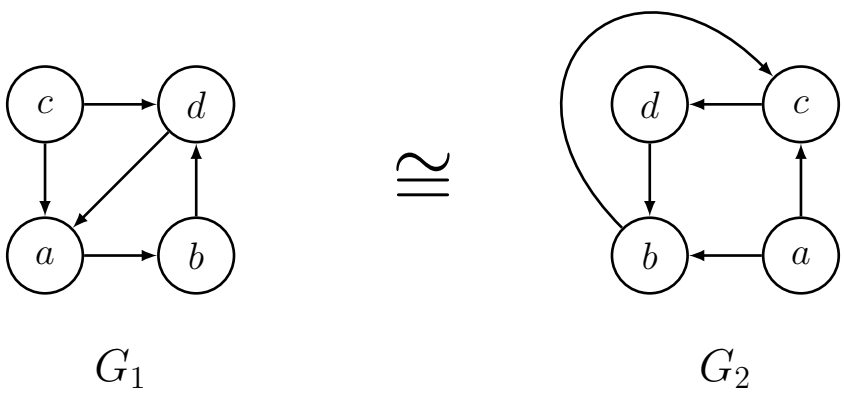
If $t(a) > 0$ then $N = 0$ is an intermediate value between $t(a)$ and $t(b)$, and hence by the I.V.T. there is some $a < c < b$ such that $f(c) = 0$, which is to say that

$$\text{temp}(0, 0) - \text{temp}(0, 180) \;=\; 0 \qquad \Longleftrightarrow \qquad \text{temp}(0, 0) = \text{temp}(0, 180)$$

as required.

3.3 Interlude 2: Isomorphism

We say that two 'structures' are **isomorphic** if they are fundamentally the same up to some relabelling of objects. For example, we have that the following two graphs are isomorphic since we can relabel and rearrange the vertices and edges so that G_1 "looks like" G_2 using the relabelling function that maps $f(a) = c$, $f(b) = d$, $f(c) = a$, $f(d) = b$:



Definition 3.6: Graph Isomorphism

Given two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, we say that G_1 is **isomorphic** to G_2 (denoted $G_1 \cong G_2$) if there exists some bijection $\rho : V_1 \to V_2$ such that

$$(x, y) \in E_1 \quad \iff \quad \big(\rho(x), \rho(y)\big) \in E_2$$

In which case we call ρ an **isomorphism** between G_1 and G_2.

However, we can also define isomorphisms between arbitrary structures; of interest to us will be isomorphisms between **ordered structures**[10] of the form $\langle I, <, f_1, \ldots, f_n, < \rangle$ where I is some interval of $\mathbb{R}$, $<$ is a **total linear ordering** on I and $f_1, \ldots, f_n$ are functions defined on the domain I.

[10] In general, isomorphisms can be defined over arbitrary first-order structures with arbitrarily many predicate symbols, function symbols, and constant symbols.

Definition 3.7: Isomorphism of Ordered Structures

We say that two ordered structures $M_1 = \langle I, <, f \rangle$ and $M_2 = \langle I', <, g \rangle$ are **isomorphic** if there exists some bijection $\rho : I \to I'$ such that:

- **Order is preserved** For all $x, y \in I$ we have that

$$x < y \qquad \Longleftrightarrow \qquad \rho(x) < \rho(y)$$

- **Functions are preserved** For all $x_1, \ldots, x_n, y \in I$, we have that

$$f(x_1, \ldots, x_n) \;=\; y \qquad \Longleftrightarrow \qquad g\big(\rho(x_1), \ldots, \rho(x_n)\big) \;=\; \rho(y)$$

In which case we call ρ an **isomorphism** between M_1 and M_2.

Exercise

Show that the following pairs of ordered structures are isomorphic by constructing an appropriate isomorphism:

- $\langle [0,1], <, x^2 \rangle$ and $\langle [0,2], <, x^2/2 \rangle$

  We can take $\rho(x) = 2x$ since

$$\rho(x^2) = 2x^2 = \frac{4x^2}{2} = \frac{(2x)^2}{2} = \frac{\rho(x)^2}{2}$$

- $\langle \mathbb{R}^{>0}, <, \times \rangle$ and $\langle \mathbb{R}, <, + \rangle$

  We can take $\rho(x) = \log_2(x)$ since $x < y$ implies $\log_2(x) < \log_2(y)$ since $\log_2(\cdot)$ is increasing and

$$\rho(xy) = \log_x(x, y) = \log_2(x) + \log_2(y) = \rho(x) + \rho(y)$$

- $\langle [0,1], <, \min \rangle$ and $\langle [-1,1], >, \max \rangle$ (note the change of ordering from $<$ to $>$)

  We can take $\rho(x) = 1 - 2x$ since $x < y$ implies $\rho(x) = 1 - 2x > 1 - 2y = \rho(y)$, and if $x \leq y$ then

$$\rho(\min(x,y)) = 1 - 2\min(x,y) = 1 - 2x = \max(1-2x, 1-2y) = \max(\rho(x), \rho(y))$$

  whereas if $x > y$

$$\rho(\min(x,y)) = 1 - 2\min(x,y) = 1 - 2y = \max(1-2x, 1-2y) = \max(\rho(x), \rho(y))$$

3.4 Trillas' Theorem

One of the first things we notice when looking at the three logics $\mathbb{F}_1$, $\mathbb{F}_2$, and $\mathbb{F}_3$, is that despite the variety of truth functions for $F_\wedge$, $F_\vee$ and $F_\rightarrow$, all three logics are oddly in agreement that $F_\neg(x) = 1 - x$. To answer this question we can start by first introducing some desirable properties that we would like $F_\neg$ to have:

(N1) Boolean: $\neg 0 = 1$ and $\neg 1 = 0$;

(N2) Involutive: $\neg\neg x = x$;

(N3) Continuous: $\neg$ is *continuous*, *i.e.* For all $\varepsilon > 0$ there is a $\delta > 0$ such that $|\neg x - \neg y| < \varepsilon$, whenever $|x - y| < \delta$;

(N4) Decreasing: $F_\neg$ is *decreasing*. *i.e.* If $x < y$ then $\neg x \geq \neg y$;

The first **(N1)** says that negation behaves as we would expect on Boolean inputs. The second **(N2)** says that $F_\neg$ is an **involution**, which also aligns with our expectations, as we would ideally like that double negations cancel each other out — *"It is not the case that Mr Whiskers is not bald"* should be assigned the same truth value as *"Mr Whiskers is bald"*. Continuity **(N3)**, says that if we change the degree to which we assign a truth value to some formula φ changes by some small amount, we should expect that the degree to which we assign a truth value to the negation $\neg\varphi$ also changes by some small amount. And lastly, **(N4)** says that as the degree to which we assign truth to a formula φ increases, then the degree to which we assign truth to its negation $\neg\varphi$ decreases.[11].

[11]Note that **(N4)** only says that $\neg$ is not *increasing*; however, it can be shown that any function satisfying **(N1)–(N4)** must be *strictly* decreasing.

Exercise

Show that if $\neg : [0,1] \to [0,1]$ is a function satisfying **(N2)**, then $\neg$ is a **bijection**.

- **One-to-One (Injective):**

  Suppose that $\neg(x) = \neg(y)$. Then by (**(N2)**)

  $$x \;=\; \neg\neg x \;=\; \neg\neg y \;=\; y$$

  which is to say that if $x \neq y$ then $\neg x \neq \neg y$, as required.

- **Onto (Surjective):**

  For any $y \in [0,1]$ then we take $x = \neg y$ so that $\neg x = \neg\neg y = y$, by (**(N2)**), as required.

So is $f(x) = 1 - x$ the only function that has these desirable properties? Surely not!?

Exercise

Write down a function, other than $f(x) = 1 - x$, that satisfies **(N1)**–**(N4)**.

$$f(x) = \sqrt{1 - x^2}$$

So what gives!?

Well, the following theorem of Trillas (/tɾi.ʝas/) shows that, although there is nothing unique about the function $f(x) = 1 - x$, any function satisfying **(N1)–(N4)** is in, some sense, "equivalent" to $f(x)$ [10].

Theorem 3.8: Trillas' Theorem [10]

If $\neg$ satisfies **(N1)–(N4)**, then $\langle [0,1], <, \neg \rangle$ is isomorphic to $\langle [0,1], <, 1-x \rangle$. That is to say that $\neg x = 1 - x$, up to isomorphism.

Proof:

- First define a function $f(x) = \neg x - x$, and note that, by **(N1)**, $f(0) = \neg 0 - 0 = 1$ and $f(1) = \neg 1 - 1 = -1$. Hence by the **Intermediate Value Theorem**, there is some $c \in [0,1]$ such that $f(c) = \neg c - c = 0$, which is to say that $\neg c = c$.

- Define new function $\rho : [0,1] \to [0,1]$ by taking

$$\rho(x) = \begin{cases} \dfrac{x}{2c} & \text{if } x \leq c \\ 1 - \dfrac{\neg x}{2c} & \text{if } x \geq c \end{cases}$$

- Note that $f(0) = 0$ and $f(1) = 1$

- We claim that $\rho(\neg x) = 1 - \rho(x)$.

  - If $x \leq c$ then $\neg x \geq \neg c$, by **(N4)**, and so by **(N2)** we have that

$$\rho(\neg x) = 1 - \frac{\neg(\neg x)}{2c} = 1 - \frac{x}{2c} = 1 - \rho(x)$$

  - Similarly, if $x > c$ then $\neg x < \neg c$, by **(N4)**, and so

$$\rho(\neg x) = \frac{\neg x}{2c} = 1 - \left(1 - \frac{\neg x}{2c}\right) = 1 - \rho(x)$$

One way to interpret Trillas' Theorem is that although we may disagree on the function $F_\neg$, we can explain these differences as simply a disagreement in the **scale**; that is how we subjectively assign a numerical real-value in $[0,1]$ to our propositions. Of all such reasonable functions satisfying **(N1)–(N4)**, the function $F_\neg(x) = 1 - x$ is the simplest, and so it is the natural choice of truth function for negation that is shared across all fuzzy semantics $\mathbb{F}^1, \mathbb{F}^2, \mathbb{F}^3$

Exercise

Consider the function $\neg : [0,1] \to [0,1]$ with

$$\neg x \;=\; \frac{1-x}{1+x}$$

Show that $\langle [0,1], <, \neg \rangle$ is isomorphic to $\langle [0,1], <, 1-x \rangle$ by constructing an order-preserving isomorphism $\rho : [0,1] \to [0,1]$ such that

$$\rho(\neg x) \;=\; 1 - \rho(x)$$

4 The Mostert–Shields Theorems

Learning Objectives

By the end of this section you should be able to:

- ☐ Demonstrate the Mostert–Shield Theorem by constructing an appropriate isomorphism for any given 'reasonable' function for conjunction.
- ☐ Demonstrate the Mostert–Shield Theorem by constructing an appropriate isomorphism for any given 'reasonable' function for disjunction.

4.1 The Mostert–Shields Theorems

4.1.1 ...for Conjunction

In contrast to negation, it would appear that there are a variety of options for reasonable functions for conjunction and disjunction that are not simply isomorphic versions of one another. For instance, it is clear that $F^1_\wedge = \min(x, y)$ and $F^2_\wedge(x, y) = xy$ are not isomorphic since $F^1_\wedge(x, x) = x$ whereas $F^2_\wedge(x, x) = x^2$ which is strictly less than x, whenever $0 < x < 1$, which cannot happen if there were an order-preserving isomorphism between $F^1_\wedge$ and $F^2_\wedge$.

In what follows it will be convenient to define the notation:

$$B_\delta(a, b) \;=\; \{(x, y) \in [0, 1]^2 \;:\; (x - a)^2 + (y - b)^2 < \delta^2\}$$

to be the set of all points $(x, y) \in [0, 1]^2$ that lie within a disc[12] of radius δ, centred at the point $(a, b) \in [0, 1]^2$.

As with Trillas' Theorem, we start by identifying some desirable properties that we expect any 'reasonable' interpretation of conjunction to follow. As before, we take the convention of denoting $F_\wedge(x, y)$ as simply $(x \wedge y)$, for $x, y \in [0, 1]$.

[12]topologist consider a disc to simply be a **ball** of dimension 2, hence the notation B

The following properties **(C1)–(C4)** are some properties that all each of our functions $F^1_\wedge$, $F^2_\wedge$, and $F^3_\wedge$:

(C1) Boolean: $0 \wedge 0 = 1 \wedge 0 = 0 \wedge 1 = 0$, and $1 \wedge 1 = 1$;

(C2) Increasing: $\wedge$ is *increasing* in each component. *i.e.* If $z < z'$ then $(x \wedge z) \leq (x \wedge z')$ and $(z \wedge y) \leq (z' \wedge y)$, for all $x, y \in [0, 1]$;

(C3) Continuous: $\wedge$ is *continuous*, *i.e.* For all $(a, b) \in [0, 1]^2$ and $\varepsilon > 0$ there is a $\delta > 0$ such that $|(x \wedge y) - (a \wedge b)| < \varepsilon$, whenever $(x, y) \in B_\delta(a, b)$;

(C4) Associative: $\wedge$ is associative. *i.e.* $((x \wedge y) \wedge z) = (x \wedge (y \wedge z))$

Any binary function satisfying the above properties **(C1)–(C4)** is called a **(continuous) triangular norm** or, more commonly, a **(continuous) t-norm**.

As a direct consequence of **(C1)–(C4)**, we get the following natural properties that also seem to align with our expectations of how conjunction should behave, which gives us more confidence in the 'reasonableness' of these properties as desirable properties for conjunction.

Exercise

Show that, for all $0 \leq x \leq 1$, if $\wedge$ is any binary function satisfying **(C1)–(C4)**, then:

(i) $x \wedge 0 = 0 \wedge x = 0$

(ii) $x \wedge 1 = 1 \wedge x = x$

(iii) $x \wedge x \leq x$

(i) By **(C2)** and **(C1)** $x \wedge 0 \leq 1 \wedge 0 = 0$ since $x \leq 1$

(ii) By **(C1)** We have that $0 \wedge 1 \leq x \leq 1 \wedge 1$. Hence, by the IVT, there is some $t \in [0, 1]$ such that $x = t \wedge 1$. Then $x \wedge 1 = (t \wedge 1) \wedge 1 = t \wedge (1 \wedge 1) = t \wedge 1 = x$, by **(C4)**

(iii) By **(C2)**, $x \leq 1$ and so $x \wedge x \leq x \wedge 1 = x$ by (ii).

We would like to show that any 'reasonable' function satisfying each of the properties **(C1)–(C4)** is, if not identical to one of the t-norms $F^1_\wedge$, $F^2_\wedge$, or $F^3_\wedge$, then is at least — as with Trillas' Theorem — isomorphic to one of these t-norms.

Unfortunately, this is not quite the case! But rather, any function satisfying **(C1)–(C4)** is a sort of **chimera** that behaves like one of these t-norms (that is to say is isomorphic to one of these t-norms) over certain intervals of the domain [8].

Theorem 4.1: Mostert–Shields Theorem for $\wedge$ [8]

Suppose that $\wedge$ satisfies **(C1)–(C4)**, and that $A = \{z \in [0,1] : z \wedge z = z\}$. Then for all $[a,b] \subseteq [0,1]$ such that $a, b \in A$ and $(a,b) \cap A = \varnothing$:

- If $z \wedge z \neq a$ for **all** $z \in (a,b)$ then

$$\langle [a,b], <, \wedge \rangle \;\cong\; \langle [0,1], <, F^2_\wedge \rangle$$

- If $z \wedge z = a$ for **some** $z \in (a,b)$ then

$$\langle [a,b], <, \wedge \rangle \;\cong\; \langle [0,1], <, F^3_\wedge \rangle$$

otherwise $(x \wedge y) = F^1_\wedge(x,y) = \min(x,y)$.

If we picture the domain $[0,1]^2$ of $F_\wedge$ as a unit square comprising all pairs of points (x,y) with $0 \leq x, y, \leq 1$, then the Mostert–Shields Theorem says that there are portions of this domain $[a,b]^2 \subseteq [0,1]^2$ arranged along the main diagonal, over which $F_\wedge$ either behaves like the $F_\wedge$ of product logic $\mathbb{F}^2$ or behaves like the $F^3_\wedge$ of Łukasiewicz logic $\mathbb{F}^3$. And that outside of these square subset $[a,b]^2$, the function behaves exactly like the $F^1_\wedge$ minimum of the $\mathbb{F}^1$ logic.

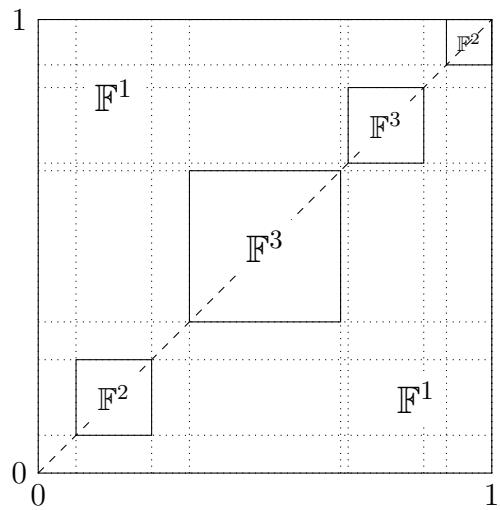


Figure 1: Illustration of the domain of a function for $\wedge$ satisfying **(C1)–(C4)**.

Case 1: Portions that behave like $F^1_\wedge$ Of the three t-norms $F^1_\wedge$, $F^2_\wedge$ and $F^3_\wedge$, the first $F^1_\wedge(x, y) = \min(x, y)$ stands alone as the only t-norm for which $(z \wedge z) = z$, for all $z \in [0, 1]$; the other t-norms are *strictly* increasing and so $(z \wedge z) < (z \wedge 1) = z$, whenever $0 < z < 1$.

The following lemma shows that whenever we have a point $z \in [0, 1]$ such that $(z \wedge z) = z$, for some t-norm satisfying **(C1)–(C4)**, then $\wedge$ behaves like $F^1_\wedge(x, y) = \min(x, y)$, whenever z lies **between** x and y.

Lemma 4.2

Let $z \in [0, 1]$ be such that $(z \wedge z) = z$, then

(i) For all $x \leq z \leq y$ we have that $(x \wedge y) = x = \min(x, y)$;

(ii) For all $y \leq z \leq x$ we have that $(x \wedge y) = y = \min(x, y)$.

Proof:

- Suppose that $x \leq z \leq y$ where $z \wedge z = z$.
- It follows that $0 = (z \wedge 0) \leq x \leq (z \wedge 1) = z$. Hence by the I.V.T. there is some $t \in [0, 1]$ such that $x = (z \wedge t)$

- We then have that

$$\begin{aligned} x &= z \wedge t \\ &= (z \wedge z) \wedge t \quad \text{since } z \wedge z = z \\ &= z \wedge (z \wedge t) \quad \text{by } \textbf{(C4)} \\ &\leq y \wedge (z \wedge t) \quad \text{by } \textbf{(C2)}\text{, since } z \leq y \\ &= y \wedge x \quad \text{by definition} \end{aligned}$$

- We also have that

$$y \wedge x \;\leq\; 1 \wedge x \;=\; x$$

  by **(C2)**, since $y \leq 1$

- Which is to say that $x \wedge y = x = \min(x, y)$, as required.

- The case where $y \leq z \leq x$ is proved analogously.

A consequence of Lemma 4.1.1 is that we can carve up the domain $[0,1]^2$ of $\wedge$ into intervals where $(z \wedge z) = z$ and where $(z \wedge z) \neq z$, which then dictate portions of domain that are forced to behave like $F^1_\wedge$. Specifically, let

$$A \;=\; \{z \in [0,1] : (z \wedge z) = z\} \tag{10}$$

We can then identify sub-intervals $[a_i, b_i] \subseteq [0,1]$, such that $a_i, b_i \in A$ and $(a_i, b_i) \cap A = \varnothing$, over which the behaviour of $\wedge$ remains to be determined — Outside of these sub-intervals we conclude that $\wedge$ behaves like the $F^1_\wedge$.

Notes

Sketch an illustration of the the set A and the intervals $[a_i, b_i]$.

Case 2: Portions that behaves like $F^2_\wedge$ Of the intervals $[a_i, b_i] \subseteq [0, 1]$, such that $a_i, b_i \in A$ and $(a_i, b_i) \cap A = \varnothing$, there are two distinct 'types'; those where $(z \wedge z) = a_i$ for some $z \in (a_i, b_i)$ **(type 3)**, and those where $(z \wedge z) \neq a_i$ for any $z \in (a_i, b_i)$ **(type 2)**.

The next lemma shows that those intervals of type 2 behave like the t-norm $F^2_\wedge(x, y) = xy$ of $\mathbb{F}^2$, over the intervals $[a_i, b_i]$.

Lemma 4.3

Let $[a, b] \subseteq [0, 1]$ be an interval of type 2, such that $a, b \in A$ and $(a, b) \cap A = \varnothing$, and $z \wedge z \neq a$ for any $z \in (a, b)$. Then $\langle [a, b], <, \wedge \rangle$ is isomorphic to $\langle [0, 1], <, \times \rangle$.

Proof:

- Suppose that $z \wedge z \neq a$ for any $z \in [a, b]$, and fix some value $c \in [a, b]$ arbitrarily — for example, we could take $c = (a + b)/2$ to be the midpoint of a and b.
- We will define a form of $\wedge$**-'exponentiation'** on $[a, b]$ similar to how x^y is defined in terms of multiplication:
  - For each $n = 1, 2, \ldots$, we can define $x^{\wedge n}$, by taking:

$$x^{\wedge n} = \underbrace{x \wedge x \wedge \cdots \wedge x}_{n \text{ times}}$$

- We next define a notion of **fractional exponents** $x^{\wedge\frac{1}{n}}$ for $n = 1, 2, \ldots$. We note that since $a \wedge a = a$ and $b \wedge b = b$, (by definition), we have that, for all $x \in [a, b]$, $a = a^n \leq x \leq b^n = b$. Hence, by the I.V.T. there is some $\beta \in [a, b]$ such that $y^n = x$.

Exercise

Show that the choice of y is unique by showing that if $y^{\wedge n} = z^{\wedge n} = x$ then $y = z$.

---

* Suppose WLOG that $y < z$. We now prove by induction on n that $y^{\wedge n} < z^{\wedge 2}$.
* The base case where $n = 1$ follows immediately from the definition.
* So suppose that $y^n < z^n$. Then since $\wedge$ is strictly increasing, we have that

$$y^{\wedge(n+1)} \;=\; (y^{\wedge n} \wedge y) \;<\; (y^{\wedge n} \wedge z) \;<\; (z^{\wedge n} \wedge z) \;=\; z^{\wedge(n+1)}$$

  as required.
* It follows, by contraposition, that if $y^{\wedge n} = z^{\wedge n}$ then $y = z$.

Hence we can define $x^{\wedge\frac{1}{n}} = y$.

- We can then define $x^{\wedge\frac{n}{m}}$ for all $n, m = 1, 2, \ldots$ by taking

$$x^{\wedge\frac{n}{m}} \;=\; (x^{\wedge n})^{\wedge\frac{1}{m}}$$

Exercise

Show that $x^{\wedge\frac{nk}{mk}} = x^{\wedge\frac{n}{m}}$.

---

* Suppose $y = x^{\wedge\frac{nk}{mk}}$ which is to say that $y^{\wedge mk} = x^{\wedge nk}$
* And so $y^{\wedge m} = x^{\wedge n}$ using the same reasoning as above and the fact that $\wedge$ is strictly increasing.
* Hence $y = x^{\wedge\frac{n}{m}}$, as required.

- It remains to define x^r for $r \in \mathbb{R}^{>0}$. Let $r_0, r_1, \ldots, \in \mathbb{Q}$ be (any) sequence of rational numbers such that $\lim_{n\to\infty} r_n = r$, *i.e.* for all $\varepsilon > 0$ there is some $N_0 \geq 0$ such that $|r_n - r| < \varepsilon$ for all $n \geq N_0$.

  Then we define:

$$x^{\wedge r} = \lim_{n\to\infty} x^{\wedge r_n} \tag{11}$$

  to be the limit of the sequence of values x^{r_n} for $n \geq 0$.

Exercise

Show that $(x^{\wedge r} \wedge x^{\wedge s}) = x^{\wedge(r+s)}$ for any real $r, s \in \mathbb{R}^{>0}$.

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* Let $r_0, r_1, \ldots$ and $s_0, s_1, \ldots$ be two sequences of rational numbers such that $\lim_{n\to\infty} r_n = r$ and $\lim_{n\to\infty} s_n = s$ then we have that:

$$\begin{aligned}
\alpha^{\wedge r} \wedge \alpha^{\wedge s} &= \left(\lim_{n\to\infty} \alpha^{\wedge r_n}\right) \wedge \left(\lim_{m\to\infty} \alpha^{\wedge s_m}\right) \\
&= \lim_{m\to\infty} \left[\left(\lim_{n\to\infty} \alpha^{\wedge r_n}\right) \wedge \alpha^{\wedge s_m}\right] \\
&= \lim_{m\to\infty} \lim_{n\to\infty} (\alpha^{\wedge r_n} \wedge \alpha^{\wedge s_m}) \\
&= \lim_{n\to\infty} (\alpha^{\wedge r_n} \wedge \alpha^{\wedge s_n}) \\
&= \lim_{n\to\infty} \alpha^{\wedge(r_n+s_n)} \\
&= \alpha^{\wedge(r+s)}
\end{aligned}$$

Exercise

Show that if $r < s$ then $x^{\wedge r} \geq x^{\wedge s}$, for $r, s \in \mathbb{R}^{>0}$

* Let $s = r + t$ for some $t \in \mathbb{R}^{>0}$ then we have that:

$$x^{\wedge s} = x^{\wedge(r+t)} = (x^{\wedge r} \wedge x^{\wedge t})$$

* However, by **(C2)**, we have that:

$$(x^{\wedge r} \wedge x^{\wedge t}) \leq (x^{\wedge r} \wedge 1) = x^{\wedge r}$$

since $x^{\wedge t} \leq 1$

We can now use this to define an isomorphism $\rho : [0,1] \to [a,b]$ from $\langle [0,1], \times, < \rangle$ to $\langle [a,b], \wedge, < \rangle$ by taking

$$\rho(x) = \begin{cases} a & \text{if } x = 0 \\ b & \text{if } x = 1 \\ c^{\wedge r} & \text{if } 0 < x < 1 \text{ where } r = -\log_2(x) \end{cases}$$

Exercise

Show that ρ is an isomorphism

- First we need to show that $\rho : [0,1] \to [a,b]$ is a bijection, which can be done by showing that ρ is invertible:

$$\rho^{-1}(x) = \begin{cases} 0 & \text{if } x = a \\ 1 & \text{if } x = b \\ (\frac{1}{2})^x & \text{if } a < x < b \end{cases}$$

  from which we can check that $\rho(\rho^{-1}(x)) = \rho^{-1}(\rho(x)) = x$ for all x.

- We also have that

$$\begin{aligned} \rho(x \times y) &= c^{\wedge(-\log_2(xy))} \\ &= c^{\wedge(-\log_2(x) - \log_2(y))} \\ &= c^{\wedge(-\log_2(x))} \wedge c^{\wedge(-\log_2(y))} \\ &= \rho(x) \wedge \rho(y) \end{aligned}$$

- And for $x \leq y$ we have that

$$\begin{aligned} x \leq y &\implies \log_2(x) \leq \log_2(y) \\ &\implies -\log_2(x) > -\log_2(y) \\ &\implies c^{\wedge(-\log_2(x))} \leq c^{\wedge(-\log_2(y))} \\ &\implies \rho(x) \leq \rho(y) \end{aligned}$$

  with $\rho(x) = \rho(y)$ only when $x = y$, since ρ is injective.

Case 3: Portions that behave like $F^3_\wedge$ For intervals $[a_i, b_i] \subseteq [0,1]$ of type 3, where there is some $z \in (a,b)$ such that $z \wedge z = a_i$, the next lemma shows that these intervals behave like the t-norm $F^3_\wedge(x,y) = \max(0, x+y-1)$.

Lemma 4.4

Let $[a,b] \subseteq [0,1]$ be an interval of type 3, such that $a, b \in A$ and $(a,b) \cap A = \varnothing$, and there is some $z \in [a,b]$ such that $z \wedge z = a$. Then $\langle [a,b], <, \wedge \rangle$ is isomorphic to $\langle [0,1], \max(0, x+y-1), < \rangle$.

Proof:

- Let $[a,b] \subseteq [0,1]$ be an interval of type 3 and fix $c \in [a,b]$ to be the **largest** value such that $c \wedge c = a$, which exists by definition.
- We define $x^{\wedge r}$ as in the proof of Lemma 4.1.1. However, we observe that $c^{\wedge r} = a$, for all $r \geq 2$, since $c \wedge c = a$, by choice of c.

Exercise

Show that $c^{\wedge r} = a$, for all $r \geq 2$.

- Suppose that $r \geq 2$ and let $r = 2 + s$ so that

$$c^{\wedge r} = c^{\wedge(2+s)} = c^{\wedge 2} \wedge c^{\wedge s} = a_i \wedge c^{\wedge s}$$

- However, by **(C2)**

$$a_i = (a_i \wedge a_i) \leq (a_i \wedge c^{\wedge s}) \leq (a_i \wedge 1) = a_i$$

since $a_i \leq c^{\wedge s} \leq 1$, and $a_i \wedge a_i = a_i$ since $a_i \in A$.

- We can, this time, define a different isomorphism $\rho : [0,2] \to [a,b]$ from $\langle [0,2], >, \min(2, x+y) \rangle$ to $\langle [a,b], <, \wedge \rangle$ by taking

$$\rho(x) \;=\; \begin{cases} a & \text{if } x = 2 \\ b & \text{if } x = 0 \\ c^{\wedge x} & \text{for } 0 < x < 2 \end{cases}$$

Note that this is not an order-preserving isomorphism, but rather an **order-reversing** isomorphism, since $\rho(x) < \rho(y)$ if and only if $x > y$.

Exercise

Show that ρ is an order-reversing isomorphism.

- First we need to show that $\rho : [0,1] \to [a,b]$ is a bijection, which can be done by showing that ρ is invertible:

$$\rho^{-1}(x) \;=\; \begin{cases} 0 & \text{if } x = a \\ 1 & \text{if } x = b \\ (\frac{1}{2})^x & \text{if } a < x < b \end{cases}$$

  from which we can check that $\rho(\rho^{-1}(x)) = \rho^{-1}(\rho(x)) = x$ for all x.

- We also have that if $x + y \leq 2$ then

$$\rho(\min(x+y, 2)) = \rho(x+y) = c^{\wedge(x+y)} = c^{\wedge x} \wedge c^{\wedge y} = \rho(x) \wedge \rho(y)$$

  and if $x + y > 2$ then $\rho(\min(x+y, 2)) = \rho(2) = c^{\wedge 2} = a$, while

$$a = a \wedge a = \rho(2) \wedge \rho(2) \leq \rho(x) \wedge \rho(y) = c^{\wedge x} \wedge c^{\wedge y} = c^{\wedge(x+y)} \leq c^{\wedge 2} = a$$

  since $x, y \leq 2$ and ρ is order-reversing. Hence $\rho(\min(x+y, 2)) = \rho(x) \wedge \rho(y) = a$, as required.

- This is not quite the desired result. However we can get the required result by noticing the following order-reversing isomorphism $\sigma : [0,1] \to [0,2]$ from $\langle [0,1], <, \text{max}(0, x+y-1)\rangle$ to $\langle [0,2], >, \text{min}(x+y, 2)\rangle$

$$\sigma(x) \;=\; 2(1-x)$$

- Hence $(\rho \circ \sigma) : [0,1] \to [a,b]$ is a **order-preserving** isomorphism from $\langle [0,1], <, \text{max}(0, x+y-1)\rangle$ to $\langle [a,b], <, \wedge\rangle$, as required.

Piecing it together Having done most of the heavy-lifting, we are now in a position to prove the Mostert–Shield Theorem.

Theorem 4.5: Mostert–Shields Theorem for $\wedge$ (restated)

Suppose that $\wedge$ satisfies **(C1)–(C4)**, and that $A = \{z \in [0,1] : z \wedge z = z\}$. Then for all $[a,b] \subseteq [0,1]$ such that $a, b \in A$ and $(a,b) \cap A = \varnothing$:

- If $z \wedge z \neq a$ for **all** $z \in (a,b)$ then

$$\langle [a,b], <, \wedge\rangle \;\cong\; \langle [0,1], <, F^2_\wedge\rangle$$

- If $z \wedge z = a$ for **some** $z \in (a,b)$ then

$$\langle [a,b], <, \wedge\rangle \;\cong\; \langle [0,1], <, F^3_\wedge\rangle$$

otherwise $(x \wedge y) = F^1_\wedge(x,y) = \text{min}(x,y)$.

Proof:

- Suppose that $\wedge$ is a function satisfying **(C1)–(C4)** and let $[a,b] \subseteq [0,1]$ be any interval such that such that $a, b \in A$ and $(a,b) \cap A = \varnothing$.

- If $z \wedge z \neq a$ for all $z \in (a, b)$ then by Lemma 4.1.1, we have that $\langle [a, b], <, \wedge \rangle \cong \langle [0, 1], <, F^2_\wedge \rangle$.
- If $z \wedge z = a$ for some $z \in (a, b)$ then by Lemma 4.1.1, we have that $\langle [a, b], <, \wedge \rangle \cong \langle [0, 1], <, F^3_\wedge \rangle$.
- Otherwise, if x, y do not belong to the same interval $[a, b]$ with $a, b \in A$ and $(a, b) \cap A = \varnothing$, then there must be some $z \in A$ such that $x \leq z \leq y$ or $y \leq z \leq x$. Hence by Lemma 4.1.1, we have that $(x \wedge y) = \min(x, y) = F^1_\wedge(x, y)$, as required.

4.1.2 ...for Disjunction

As with the Mostert–Shields Theorem for conjunction, we can also state some desirable properties for disjunction, with the hope of arriving at some similar characterisation of all binary functions that satisfy them. For instance, we may decide that the following properties **(D1)–(D4)** are some properties that we expect all reasonable functions for $\vee$ to possess. And indeed, are properties that are held by each of our functions $F^1_\vee$, $F^2_\vee$, and $F^3_\vee$, which again gives us some reassurance that they are reasonable:

(D1) **Boolean:** $0 \vee 0 = 0$, and $1 \vee 0 = 0 \vee 1 = 1 \vee 1 = 1$;

(D2) **Increasing:** $\vee$ is *increasing* in each component. *i.e.* If $z < z'$ then $(x \vee z) \leq (x \vee z')$ and $(z \vee y) \leq (z' \vee y)$, for all $x, y \in [0, 1]$;

(D3) **Continuous:** $\vee$ is *continuous*, *i.e.* For all $(a, b) \in [0, 1]^2$ and $\varepsilon > 0$ there is a $\delta > 0$ such that $|(x \vee y) - (a \vee b)| < \varepsilon$, whenever $(x, y) \in B_\delta(a, b)$;

(D4) **Associative:** $\vee$ is associative. *i.e.* $((x \vee y) \vee z) = (x \vee (y \vee z))$

Any binary function satisfying the above properties **(D1)–(D4)** is called a **(continuous) triangular conorm** or, more commonly, a **(continuous) t-conorm**.

The Mostert–Shields Theorem for disjunction says that any binary functions satisfying **(D1)–(D4)** must be a chimera of the t-conorms $F^1_\vee$, $F^2_\vee$, and $F^3_\vee$.

Theorem 4.6: Mostert–Shields Theorem for $\vee$ [8]

Suppose that $\vee$ satisfies **(D1)–(D4)**, and that $A = \{z \in [0,1] : z \vee z = z\}$. Then for all $[a,b] \subseteq [0,1]$ such that $a, b \in A$ and $(a,b) \cap A = \varnothing$:

- If $z \vee z \neq b$ for **all** $z \in (a,b)$ then

$$\langle [a,b], <, \vee \rangle \;\cong\; \langle [0,1], <, F^2_\vee \rangle$$

- If $z \vee z = b$ for **some** $z \in (a,b)$ then

$$\langle [a,b], <, \vee \rangle \;\cong\; \langle [0,1], <, F^3_\vee \rangle$$

otherwise $(x \vee y) = F^1_\vee(x,y) = \max(x,y)$.

We could prove this result in much the same way as we did the version for conjunction. However, an alternative approach utilises the fact that, for every binary function $\vee$ satisfying **(D1)–(D4)**, we can **define** a binary function $\wedge$ by taking

$$(x \wedge y) = 1 - \big((1-x) \vee (1-y)\big)$$

which we can readily check satisfies the properties **(C1)–(C4)**. Hence, we can utilise Theorem 4.1.1 to show that $\wedge$ is a chimera of $F^1_\wedge$, $F^2_\wedge$, and $F^3_\wedge$. Moreover, we have that

$$\begin{aligned} F^1_\wedge(x,y) &= 1 - F^1_\vee(1-x, 1-y) \\ F^2_\wedge(x,y) &= 1 - F^2_\vee(1-x, 1-y) \\ F^3_\wedge(x,y) &= 1 - F^3_\vee(1-x, 1-y) \end{aligned}$$

And so if $\langle [a,b], <, \wedge \rangle$ is isomorphic to $\langle [0,1], <, F^k_\wedge \rangle$, for $k = 2, 3$, then $\langle [a,b], <, \vee \rangle$ is isomorphic to $\langle [0,1], <, F^k_\vee \rangle$.

5 Łukasiewicz Logic and McNaughton's Theorem

Learning Objectives

By the end of this section you should be able to:

- ☐ Show that the truth-function of every propositional formula in 1 variable satisfies McNaughton's Theorem
- ☐ Show that every (1 variable) function satisfying McNaughton's Theorem can be represented as the truth function of some propositional formula in 1 variable.

5.1 An Argument for Łukasiewicz's Logic

Of the three logics $\mathbb{F}^1$, $\mathbb{F}^2$ and $\mathbb{F}^3$, Łukasiewicz logic $\mathbb{F}^3$ is most well-studied.

$\mathbb{F}^3$		
$F_\neg(x)$	$=$	$1 - x$
$F_\wedge(x, y)$	$=$	$\mathsf{max}(0, x + y - 1)$
$F_\vee(x, y)$	$=$	$\mathsf{min}(1, x + y)$
$F_\rightarrow(x, y)$	$=$	$\mathsf{min}(1, 1 - x + y)$

One argument against $\mathbb{F}^1$ and $\mathbb{F}^2$ is that both fail to validate some formulas that we would otherwise expect to be true such as $F = p \vee \neg p$. This is often codified in the so-called **Law of Excluded Middle**.

(LEM) Law of Excluded Middle: $x \vee \neg x = 1$, for all $x \in [0, 1]$;

Exercise

Show that $\mathbb{F}^1$ and $\mathbb{F}^2$ fail to satisfy **(LEM)**.

- For $\mathbb{F}^1$:
  - If $0 < x \leq \frac{1}{2}$ then $x \leq 1-x < 1$ and so $x \vee \neg x = \max(x, 1-x) = 1-x < 1$.
  - Otherwise $\frac{1}{2} < x < 1$ then $x \geq 1-x$ and so $x \vee \neg x = \max(x, 1-x) = x < 1$.
- For $\mathbb{F}^2$:

$$\begin{aligned} x \vee \neg x = 1 \quad &\iff \quad x + (1-x) - x(1-x) = 1 \\ &\iff \quad 1 - x + x^2 = 1 \end{aligned}$$

  which is the case if and only if $x = 0$ or $x = 1$.

And so, if we consider the **(LEM)** to be a desirable property of our logic, then we must exclude $\mathbb{F}^1$ and $\mathbb{F}^2$.

However, concerning $\mathbb{F}^1$, *if* min and max are desirable functions that we wish to express in our logic, then $\mathbb{F}^3$ is strong enough to express these already as **defined operators**. We may introduce two additional binary operators $\underline{\wedge}$ and $\underline{\vee}$ as **abbreviations** of the following formulas:

$$x \underline{\wedge} y \;=\; (x \to y) \wedge x \qquad \text{and} \qquad x \underline{\vee} y \;=\; (x \to y) \to y$$

which take on the minimum and maximum values of their arguments, respectively. This is to say that Łukasiewicz logic $\mathbb{F}^3$ is an **extension** of $\mathbb{F}^1$, and any reasoning we may like to do in regarding $\mathbb{F}^1$ can just as well be done with $\mathbb{F}^3$[13]

[13] Formally speaking, there is a translation $t : \mathit{Fml} \to \mathit{Fml}$, obtained by iteratively replacing all occurrences of $\wedge$ and $\vee$ with $\underline{\wedge}$ and $\underline{\vee}$, respectively, such that $\models^{\mathbb{F}^1} F$ if and only if $\models^{\mathbb{F}^3} t(F)$.

Exercise

Show that, for all $x, y \in [0,1]$ we have that:

$$(x \underline{\wedge} y) = \min(x,y) \qquad \text{and} \qquad (x \underline{\vee} y) = \max(x,y).$$

- We have that

$$\begin{aligned} x \underline{\wedge} y &= (x \to y) \wedge x \\ &= \max(0, (x \to y) + x - 1) \\ &= \max(0, \min(1, 1 - x + y) + x - 1) \\ &= \max(0, \min(1 + x - 1, 1 - x + y + x - 1)) \\ &= \max(0, \min(x,y)) = \min(x,y) \end{aligned}$$

- We have that

$$\begin{aligned} 1 - \min(1, 1 - x + y) + y &= 1 + \max(-1, -1 + x - y) + y \\ &= \max(1 - 1 + y, 1 - 1 + x - y + y) \\ &= \max(y,x) = \max(x,y) \end{aligned}$$

Using the fact that $-1 \cdot \min(a,b) = \max(-a,-b)$. Hence $(x \to y) \to y = \min(1, \max(x,y)) = \max(x,y)$

Great! So can we do the same for the product $\mathbb{F}^2$ and find some combination of formulas that would behave in $\mathbb{F}^3$ as do the $\wedge$ and $\vee$ in $\mathbb{F}^2$, that we could use as suitable abbreviation for additional defined operators?

After a short (or long) period of trial-and-error, you may convince yourself that this is, indeed, a futile and Sisyphean task, and that no combination of connectives of finite length — no matter how complex and creative — can quite mirror the behaviour of the $\mathbb{F}^2$ t-norm and t-conorm. But how can we be sure that we haven't just missed something? Maybe we just need longer formulas!

To show that this isn't possible, regardless of how creative we are, we need some sort of **characterisation result** that tells us precisely what *kinds* of functions can be expressed by formulas in $\mathbb{F}^3$. This is where McNaughton's Theorem comes in.

5.2 McNaughton's Theorem (Part 1)

When it comes to classical propositional logic, we see that every Boolean function $f : \{0,1\}^n \to \{0,1\}$ is the truth-function of *some* propositional formula. That is to say that there is some $F \in$ *Fml* such that $V(F) = f$ — for instance, we could construct a formula in disjunctive normal form by taking the disjunction of the conjunctions of literals corresponding to inputs for which $f(x_1, \ldots x_n) = 1$.

$$F \;:=\; \bigvee_{f(x_1,\ldots,x_n)=1} \left(\bigwedge_{i=1}^{n} p_i^{x_i} \right)$$

where $p_i^1 = p_i$ and $p_i^0 = \neg p_i$.

However, it should come as no surprise that the same is *not* true for arbitrary real-valued functions $f : [0,1]^n \to [0,1]$ with respect to Łukasiewicz semantics (or indeed any fuzzy semantics)[14] But then, this raises the question: just what kinds of functions *can* be represented by propositional formulas under Łukasiewicz semantics?

Theorem 5.1: McNaughton's Theorem (1 variable) [7]

A one-variable real-valued function $f : [0,1] \to [0,1]$ can be represented by a propositional formula under Łukasiewicz semantics if and only if there is some increasing sequence

$$0 = \gamma_0 < \gamma_1 < \gamma_2 < \cdots < \gamma_n = 1$$

and family of linear function $f_i(x) = a_i x + b_i$ with $a_i, b_i \in \mathbb{Z}$, for $i < n$, such that

$$f(x) = f_i(x) = a_i x + b_i \qquad \text{for all } x \in [\gamma_i, \gamma_{i+1}]$$

In particular, we have that $f_i(\gamma_{i+1}) = f_{i+1}(\gamma_{i+1})$, for all $i < n$.

In other words, McNaughton's Theorem says that we can always partition the truth function of any one-variable propositional formula is a **continuous piecewise-linear function with integer coefficients** [7]. And conversely, that *every* one-variable,

[14]A simple argument is that there there are $2^{2^{\aleph_0}}$ such functions from $[0,1]^n$ to $[0,1]$ (or $2^{\aleph_0}$ if we restrict ourselves to continuous functions), however there are 'only' $\aleph_0$ propositional formulas.

continuous piecewise-linear function with integer coefficients is the truth function of some one-variable propositional formula.

Lemma 5.2

The truth-function of every one-variable propositional formula is a continuous piecewise-linear function with integer coefficients.

Proof: We prove this by induction on the size of the formula.

- Suppose that $n = 1$ then $F = p$ is just a propositional variable, for which we can take $0 = \gamma_0 < \gamma_1 = 1$ with $\overline{F}(x) = x$, as required.

- Suppose for induction that the result holds for all formulas of length $< n$ and let φ have length n. We have four cases to consider

  - If $F = \neg G$ for some G then since G is of length $< n$ we have that there are $0 = \gamma_0 < \gamma_1 \cdots < \gamma_t = 1$ and $g_i(x) = a_i x + b_i$ for $a_i, b_i \in \mathbb{Z}$, for $i < t$, such that $\overline{G}(x) = g_i(x)$ for all $x \in [\gamma_i, \gamma_{i+1}]$.

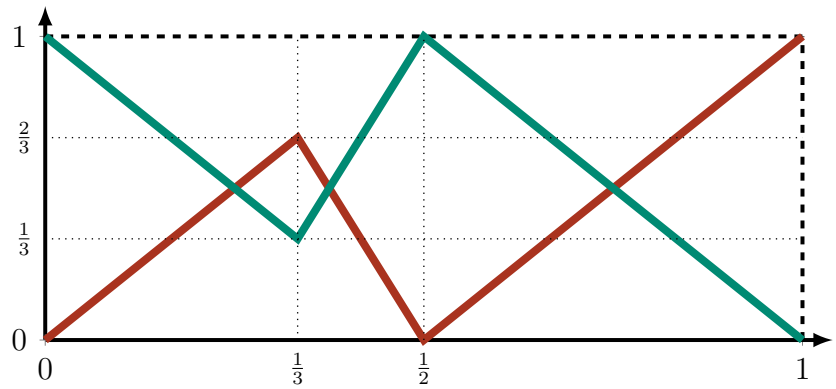
    We define a new family of functions $f_i(x) = a'_i x + b'_i$, for each $i < n$, by taking $a'_i = -a_i \in \mathbb{Z}$ and $b'_i = (1 - b_i) \in \mathbb{Z}$, so that for $x \in [\gamma_i, \gamma_{i+1}]$

$$\begin{aligned} \overline{F}(x) &= 1 - \overline{G}(x) \\ &= 1 - g_i(x) \\ &= 1 - (a_i x + b_i) \\ &= (-a_i)x + (1 - b_i) \;=\; f_i(x) \end{aligned}$$

    as required.

Exercise

Suppose that G is a formula whose truth-function $\overline{G}$ is as shown below. Sketch the truth-function of the formula $F = \neg G$ on the same axes.



- If $F = G \wedge H$ then since the length of G and H are both $< n$, by the Induction Hypothesis:

  (i) there is some sequence $0 = \gamma_0 < \gamma_1 < \cdots < \gamma_t = 0$ and $g_i(x) = a_i x + b_i$ with $a_i, b_i \in \mathbb{Z}$, such that $\overline{G}(x) = g_i(x)$ for all $x \in [\gamma_i, \gamma_{i+1}]$;

  (ii) there is some sequence $0 = \delta_0 < \delta_1 < \cdots < \delta_s = 0$ and $h_i(x) = c_i x + d_i$ with $c_i, d_i \in \mathbb{Z}$, such that $\overline{H}(x) = h_i(x)$ for all $x \in [\delta_i, \delta_{i+1}]$.

  Without loss of generality we may assume that $t = s$ and $\gamma_i = \delta_i$, for all $i < t$, since we do not require F_φ and F_ψ to not *non-linear* at each γ_i and δ_i, respectively.

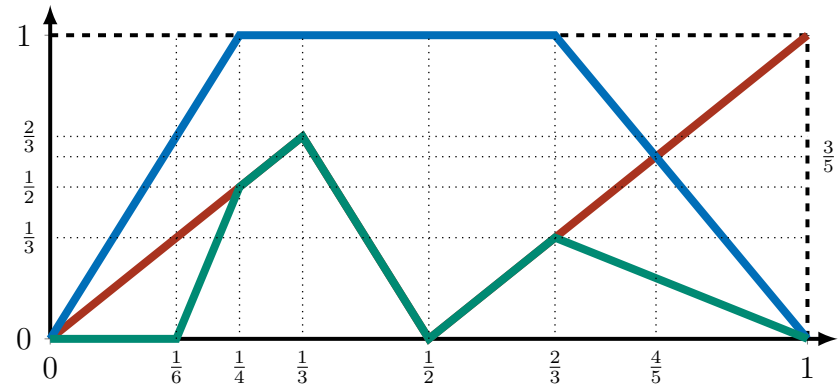
  Hence we can take $f_i(x) = (a_i + c_i)x + (b_i + d_i - 1)$ so that

$$\begin{aligned}
\overline{F}(x) &= \max(0, \overline{G}(x) + \overline{H}(x) - 1) \\
&= \max(0, (a_i x + b_i) + (c_i x + d_i) - 1) \\
&= \max(0, (a_i + c_i)x + (b_i + d_i - 1))
\end{aligned}$$

  for all $x \in [\gamma_i, \gamma_{i+1}]$, as required.

Exercise

Suppose that G and H are two formulas whose truth-function $\overline{G}$ (red) and $\overline{H}$ (blue) are as shown below. Sketch the truth-function of the formula $F = (G \wedge H)$ on the same axes.



- If $F = (G \vee H)$ then we note that, under the Łukasiewicz semantics, $F \equiv \neg(\neg G \wedge \neg H)$, and so it follows immediately from the previous two cases. Likewise, if $F = (G \to H)$, which is equivalent to $(F \vee G) \equiv \neg(F \wedge \neg G)$.

5.3 McNaughton's Theorem (Part 2)

Lemma 5.3

Let $n, m \in \mathbb{Z}$ be integers and let $f : [0,1] \to [0,1]$ be defined as

$$f_{n,m}(x) \;=\; [nx + m]_0^1$$

where $[z]_0^1 = \min(1, \max(0, z))$ denotes the restriction of z to the interval $[0,1]$. Then there exists some one-variable propositional formula $F_{n,m}$ whose truth-function is $f_{n,m}$.

Proof:

Base Case) First suppose $n = 0$, so that

$$f_{0,m}(x) = [m]_0^1 = \begin{cases} 0 & \text{if } m \leq 0 \\ 1 & \text{if } m \geq 1 \end{cases}$$

Hence, we $m \leq 0$ we can take

$$F_{0,m} = \begin{cases} \bot & \text{if } m \leq 0 \\ \top & \text{if } m \geq 1 \end{cases}$$

Where $\top = (p \vee \neg p)$ abbreviates a tautology and $\bot = (p \wedge \neg p)$ abbreviates a contradiction.

Induction Hypothesis) Suppose for all the result is true for all $n, m \in \mathbb{Z}$ where $|n| \leq k$, for some $k \geq 0$. We want to show that the result also holds when $|n| = (k+1)$.

Induction Case for $n = (k+1)$) We want to show that for all $m \in \mathbb{Z}$ there is some function $F_{n,m}$ whose truth-function is $f(x) = [nx + m]_0^1$.

To this end, let

$$f_{k,m}(x) = [kx + m]_0^1 \qquad \text{and} \qquad f_{k,(m+1)}(x) = [kx + (m+1)]_0^1$$

Then by the Induction Hypothesis, there are formulas $F_{k,m}$ and $F_{k,(m+1)}$ whose truth-functions are $f_{k,m}$ and $f_{k,(m+1)}$ respectively.

Hence, we can define a new function $F_{(k+1),m}$ by taking

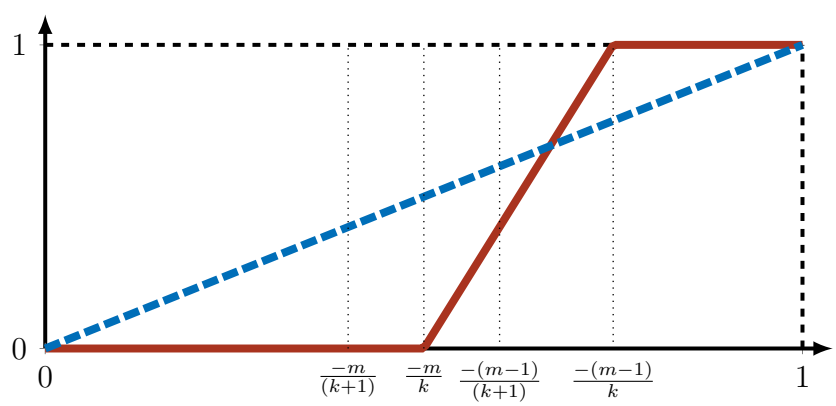
$$F_{(k+1),m} \;=\; (F_{k,m} \vee p) \wedge F_{k,(m+1)}$$

We can then verify that the truth-function of $F_{(k+1),m}$ is $f_{(k+1),m}$, by considering it's behaviour over the intervals $[0, \frac{-m}{(k+1)}]$, $[\frac{-m}{(k+1)}, \frac{-m}{k}]$, $[\frac{-m}{k}, \frac{(1-m)}{(k+1)}]$ and $[\frac{(1-m)}{(k+1)}, 1]$.

- Consider first the truth functions for $F_{k,m}$ and p:

Notes

Sketch on the following axes, the truth functions of $F_{k,m}$ and p.

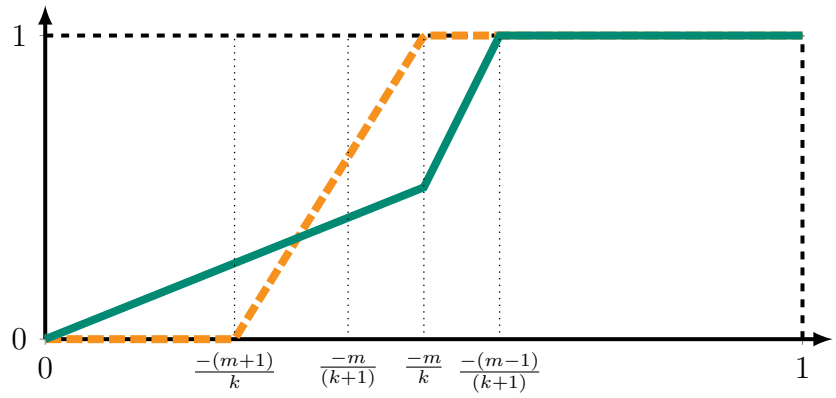


Over the interval $[0, \frac{-m}{k}]$ we have that $f_{k,m}(x) + x = 0 + x = x$, while over the interval $[\frac{-(m-1)}{(k+1)}, 1]$ we have that $f_{k,m}(x) + x \geq 1$. Over the interval $[\frac{-m}{k}, \frac{-(m-1)}{(k+1)}]$ we have that $f_{k,m}(x) + x = (kx + m) + x = (k+1)x + m$.

- We can then consider the truth functions of $(F_{k,m} \vee p)$ and $F_{k,(m-1)}$:

Notes

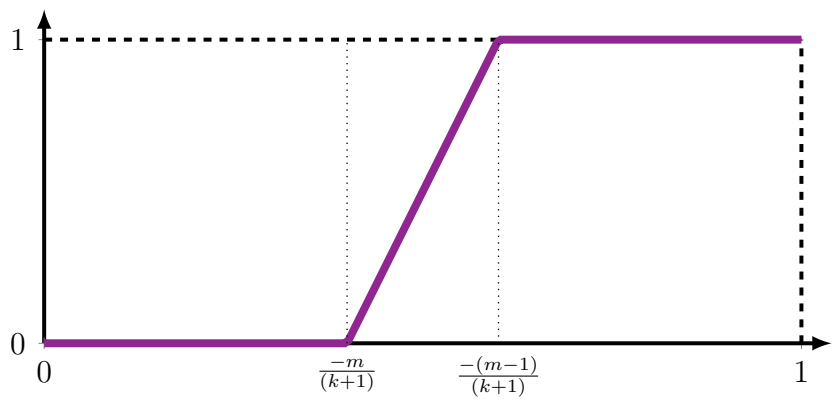
Sketch on the following axes, the truth functions of $(F_{k,m} \vee p)$ and $F_{k,(m+1)}$:



Over the interval $[0, \frac{-m}{(k+1)}]$ we have that $x + f_{k,(m+1)}(x) - 1 \leq 0$, while over the interval $[\frac{-m}{k}, 1]$ we have that $f_{k,(m+1)}(x) = 1$ and so $g(x) + fk, (m+1) - 1 = g(x)$. Over the interval $[\frac{-m}{(k+1)}, \frac{-m}{k}]$ we have that $x + f_{k,(m+1)}(x) - 1 = x + (kx + (m+1)) - 1 = (k+1)x + m$.

Notes

Sketch on the following axes, the truth function of $(F_{k,m} \vee p) \wedge F_{k,(m+1)}$.



Induction Case for $n = -(k+1)$) We want to show that for all $m \in \mathbb{Z}$ there is some function $F_{n,m}$ whose truth-function is $f(x) = [nx + m]_0^1$.

To this end, let

$$f_{-k,m}(x) = [-kx + m]_0^1 \qquad \text{and} \qquad f_{-k,(m-1)}(x) = [-kx + (m-1)]_0^1$$

As above, by the Induction Hypothesis, there are formulas $F_{-k,m}$ and $F_{-k,(m-1)}$ whose truth-functions are $f_{-k,m}$ and $f_{-k,(m-1)}$ respectively.

Hence, we can define a new function $F_{-(k+1),m}$ by taking

$$F_{-(k+1),m} \;=\; (F_{-k,(m-1)} \vee \neg p) \wedge F_{-k,m}$$

We can then verify that the truth-function of $F_{-(k+1),m}$ is $f_{-(k+1),m}$, by considering it's behaviour over the intervals $[0, \frac{-m}{(k+1)}]$, $[\frac{-m}{(k+1)}, \frac{-m}{k}]$, $[\frac{-m}{k}, \frac{(1-m)}{(k+1)}]$ and $[\frac{(1-m)}{(k+1)}, 1]$.

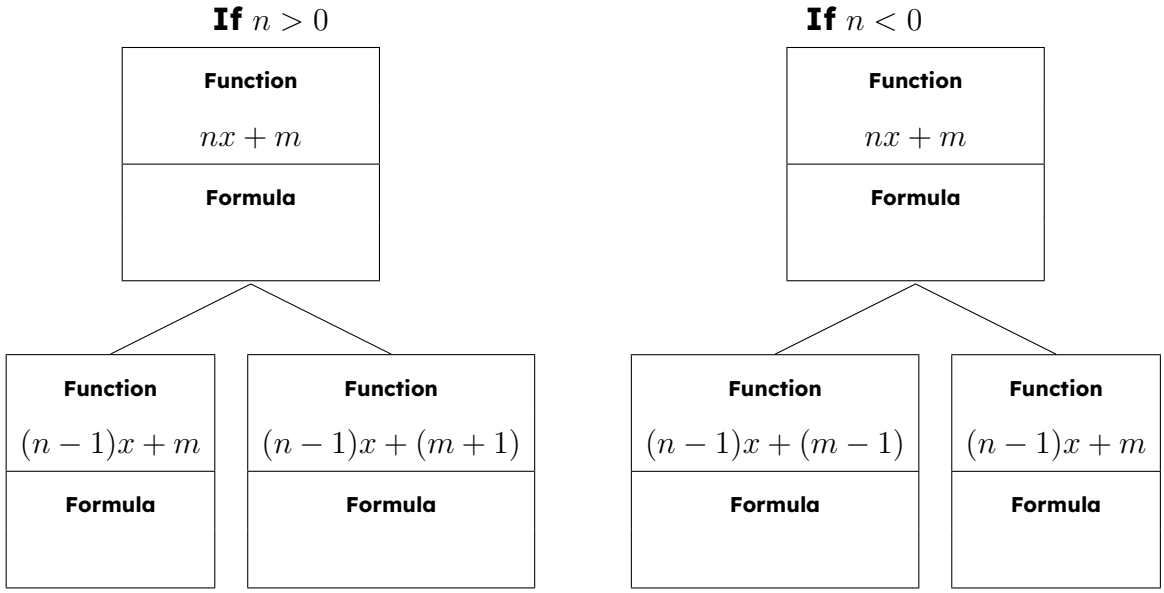
Hence it follows by induction that for all $n, m \in \mathbb{Z}$ there is some formula $F_{n,m}$ whose truth function is $f_{n,m}(x) = [nx + m]_0^1$, as required. ∎

We can turn the induction of Lemma 5.3 into a algorithm for recursively generating formulas whose truth-function is of the form $f(x) = [nx+m]_0^1$ for any integers $n, m \in \mathbb{Z}$, as follows:

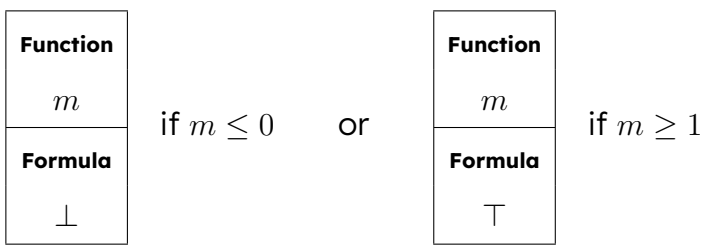
Step 1) Start at the root of a tree with a node labelled with

Function
$nx + m$
Formula

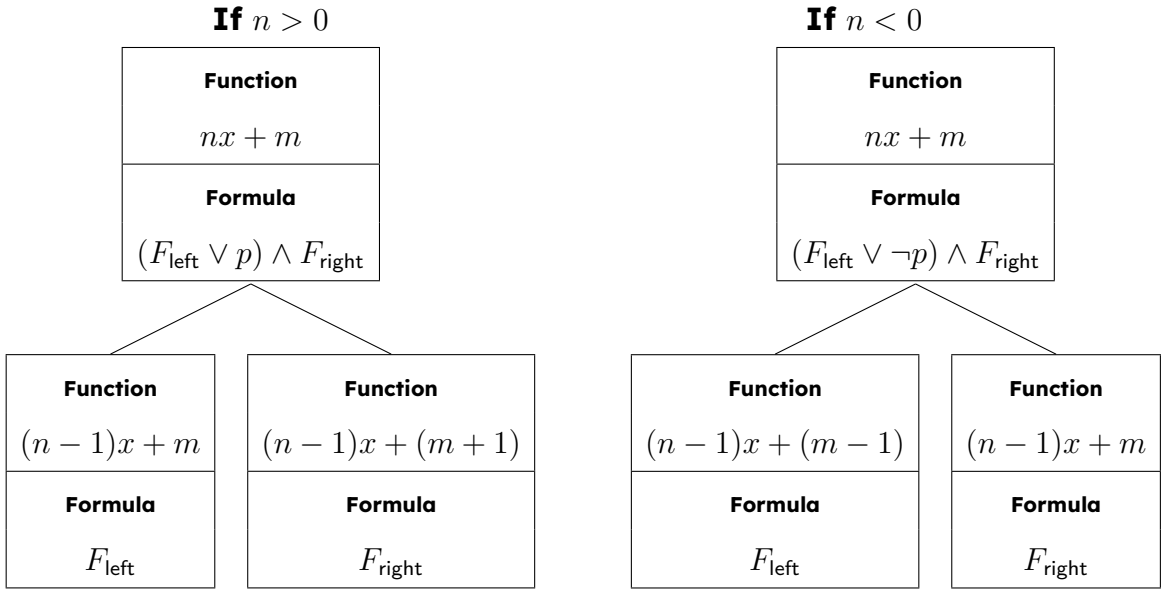
Step 2) From each node of the tree, if $n \neq 0$ then create two child nodes, each labelled as follows, depending on whether n if positive or negative:



Step 3) Once we reach a node with $n = 0$ we can fill in the cell beneath with a corresponding propositional formula



Step 4) Working our way up the tree to the root, we can populate the formula cell based on the formulas contained in the formula cells of the



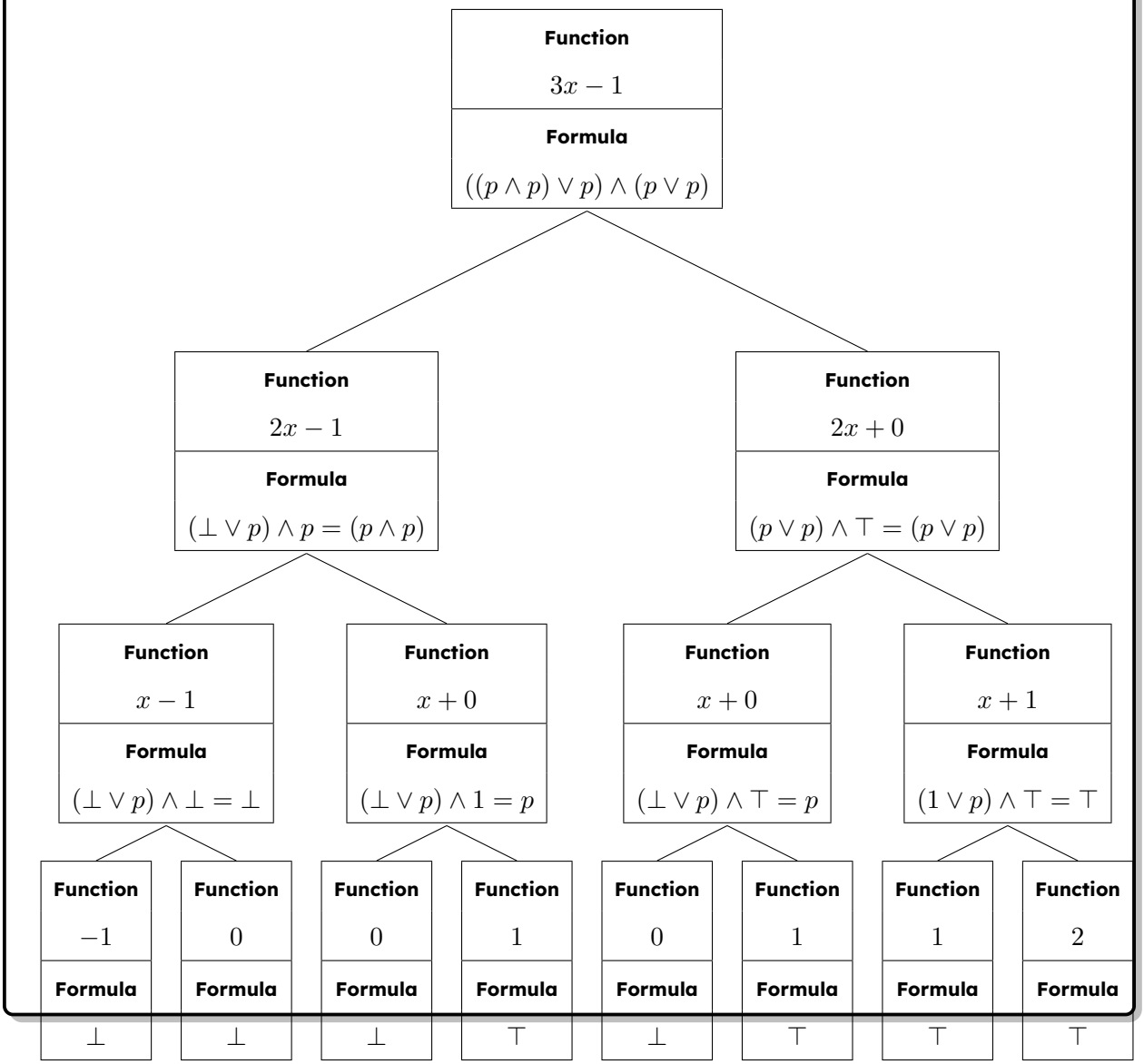
We can **simplify** any formulas as we go by applying the rules that

$$
\begin{aligned}
(F \wedge \bot) &\equiv (\bot \wedge F) &\equiv \bot \\
(F \wedge \top) &\equiv (\top \wedge F) &\equiv F \\
(F \vee \bot) &\equiv (\bot \vee F) &\equiv F \\
(F \vee \top) &\equiv (\top \vee F) &\equiv \top
\end{aligned}
$$

Exercise

Use the tree algorithm to construct a propositional formula whose truth function is

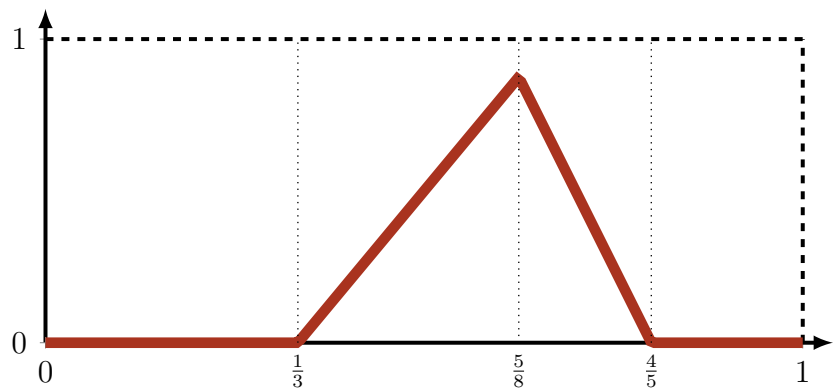
$$f(x) \;=\; [3x-1]_0^1$$



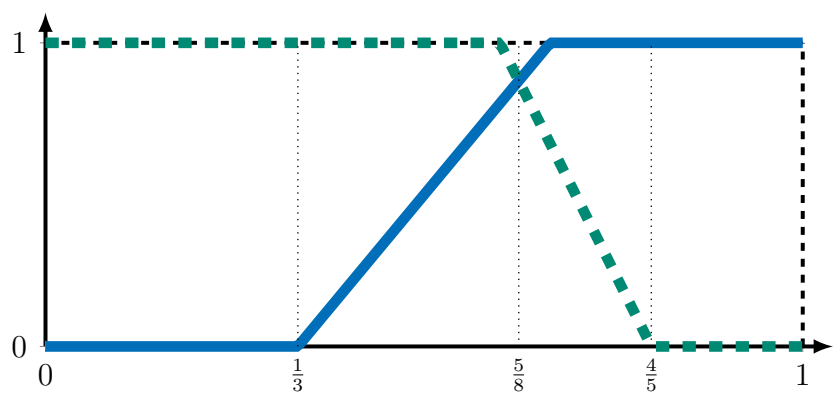
Lemma 5.3 shows that for any (continuous) piecewise linear function f, we can find propositional formulas $F_1, \ldots, F_n$ such that truth-function of F_i aligns with f along the interval $[\gamma_{i-1}, \gamma_i]$. However, we still need to a way to combine these formulas into

a single formula that aligns with f along the whole of its domain $[0, 1]$. We achieve this with some judicious combination of our defined operators $\underline{\wedge}$ (min) and $\underline{\vee}$ (max). For example, consider the following function

$$f(x) = \begin{cases} 0 & \text{if } x \leq \frac{1}{3} \\ 3x - 1 & \text{if } \frac{1}{2} < x \leq \frac{5}{8} \\ 4 - 5x & \text{if } \frac{5}{8} < x \leq \frac{4}{5} \\ 0 & \text{if } x > \frac{4}{5} \end{cases}$$



Then it is clear that f is simply takes on the minimum of $f_{3,-1} = [3x - 1]_0^1$ (blue) and $f_{-5,4} = [-5x + 4]_0^1$ (green/dashed), shown below:

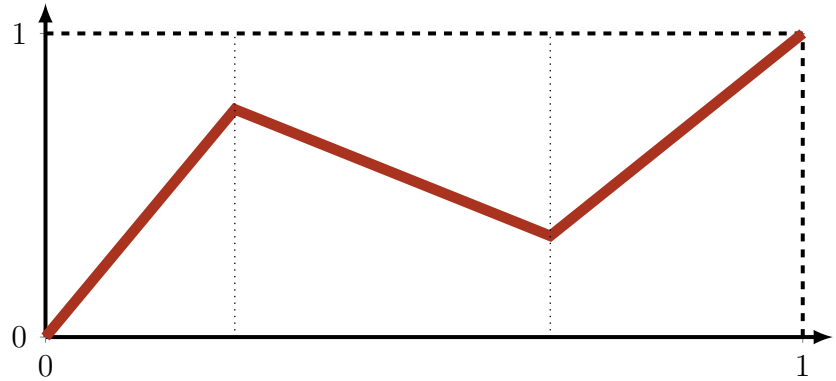


Hence we would be able to take F to be

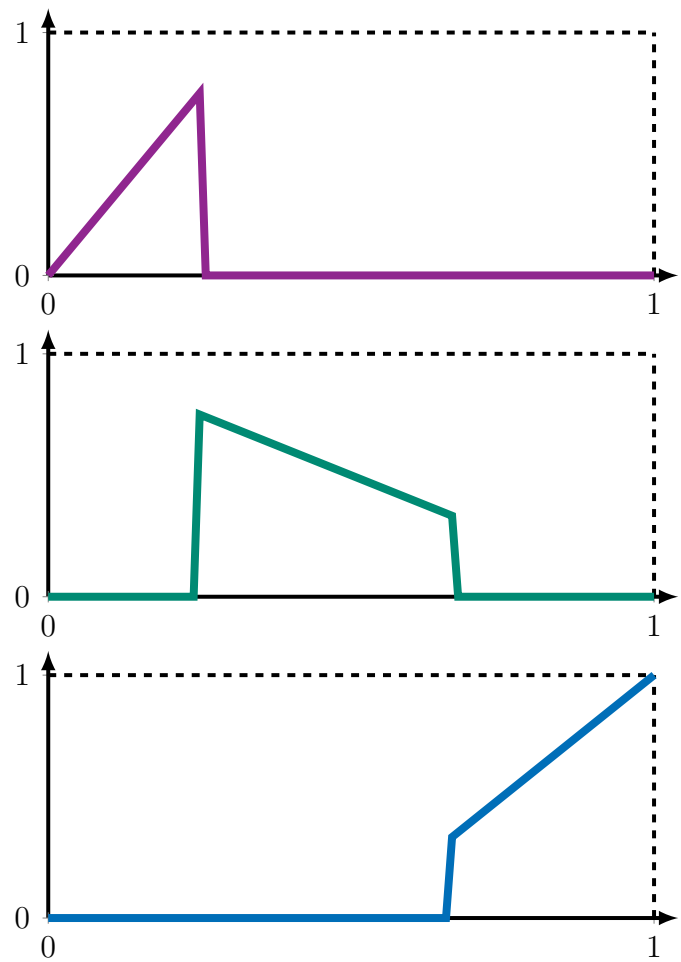
$$F = F_{3,-1} \underline{\wedge} F_{-5,4}$$

where $F_{3,-1}$ and $F_{-5,4}$ are as defined in the proof of Lemma 5.3.

For more complex functions such as the following



we cannot always simply take a minimum or maximum of functions of the form $f_{n,m}$. Instead, what we must first do, is decompose f into 'wedges-shaped' pieces like so:



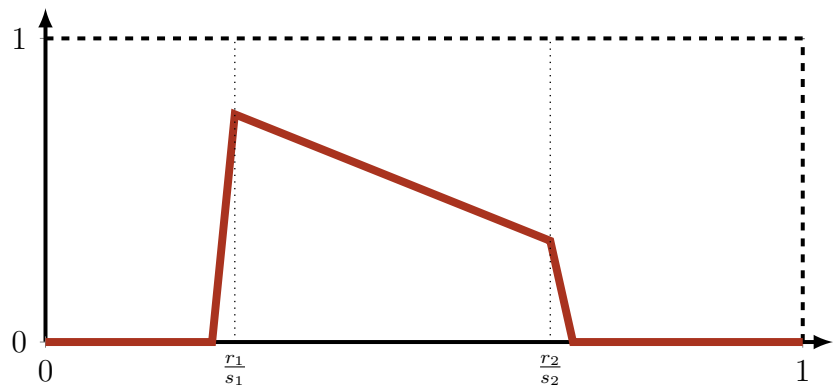
each of which can be described by a piecewise-linear function with integers coefficients, and whose maximum is the desired function f. (Note that the 'vertical' portions of

these functions are not actually vertical, as that would not be a well-defined function, but rather linear functions with a large positive or negative gradient.)

5.4 McNaughton's Theorem (Part 3) — Optional

[This section is for the sake of completeness and is not examinable]

We want to show that for any linear function $f_{n,m}(x) = [nx + m]_0^1$ with $n, m \in \mathbb{Z}$, and rational points $0 \leq \frac{r_1}{s_1} < \frac{r_2}{s_2} \leq 1$, we can construct a 'wedge-shaped' function:



in which the slope slope of the linear components immediately before r_1/s_2 and immediately after r_2/s_2 can be made arbitrarily steep.

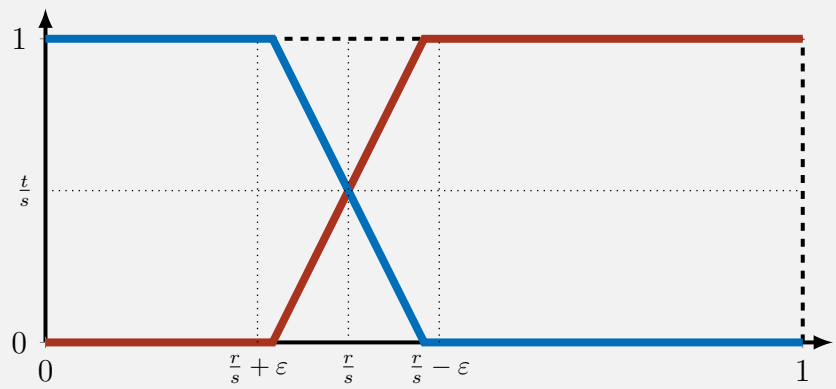
Lemma 5.4

Let $r, s, t \in \mathbb{N}$ be such that gcd$(r, s) = 1$. Then for all $\varepsilon > 0$, there exists formulas G_1 and G_2, with truth-functions g_1 and g_2, respectively, such that:

(i) $g_i(\frac{r}{s}) = \frac{t}{s}$, for $i = 1, 2$,

(ii) $g_1(x) = 0$ and $g_2(x) = 1$, for all $x < \frac{r}{s} - \varepsilon$,

(iii) $g_1(x) = 1$ and $g_2(x) = 0$, for all $x > \frac{r}{s} + \varepsilon$,



Proof: Let r, s, t be as described, and let $\varepsilon > 0$ be fixed. By **Bézout's Identity**, we have that there are integers $a, b \in \mathbb{Z}$ such that $ar + bs = \gcd(r, s) = 1$, and hence $art + bst = t$ which is to say that

$$\begin{aligned} ar + bs = 1 \quad &\Longrightarrow \quad art + bst = t \\ &\Longrightarrow \quad art + bst + (krs - krs) = t \qquad \text{for any } k \in \mathbb{Z} \\ &\Longrightarrow \quad (at + ks)r + (bt - kr)s = t \end{aligned}$$

Hence, we can take $n = (at + ks)$ and $m = (bt - kr)$ so that

$$f_{n,m}\left(\frac{r}{s}\right) = n\left(\frac{r}{s}\right) + m = (at + ks)\left(\frac{r}{s}\right) + (bt - kr) = \left(\frac{t}{s}\right)$$

By choosing k to be sufficiently large and positive, we can ensure that $f_{n,m}(x) = [nx + m]_0^1 = 0$ for all $x < \frac{r}{s} - \varepsilon$, as required for g_1. And by choosing k to be sufficiently large and negative, we can ensure that $f_{n,m}(x) = [nx + m]_0^1 = 0$ for all $x > \frac{r}{s} + \varepsilon$, as required for g_2. ∎

With Lemma 5.4, we are able to construct a formula corresponding to any 'wedge-shape' function as shown above, which then allows us to complete the proof of McNaughton's Theorem in one-variable (Theorem 5.2).

Proof:

- Let piecewise linear function $f : [0,1] \to [0,1]$ be a continuous piecewise linear function with $0 = \gamma_0 < \gamma_1 < \cdots < \gamma_n = 1$, which that $f(x) = a_i(x) + b_i$, for all $x \in [\gamma_i, \gamma_i + 1]$.

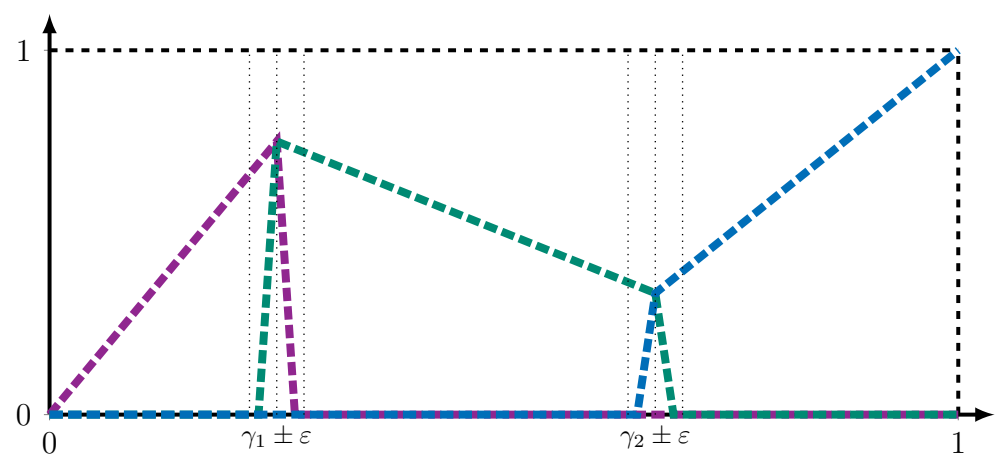
- For each $i < n$, let

$$F_i = (G_1^i \mathbin{\underline{\wedge}} F_{a_i,b_i} \mathbin{\underline{\wedge}} G_2^i)$$

  where G_1^i and G_2^i denote the formulas corresponding to the functions g_1 and g_2 of Lemma 5.4 such that (i) $g_1(\gamma_i) = a_i\gamma_i + b_{i+1}$ and $g_2(\gamma_{i+1}) = a_{i+1}\gamma_{i+1} + b_{i+1}$, and (ii) ε is chosen to be sufficiently small so that g_1 and g_2 not not intersect f at any other point $x \in [0,1]$.

- We then define F to be the maximum of each of the formulas F_i, for $i < n$, corresponding to each of the wedge shaped sections:

$$F \;=\; F_0 \mathbin{\underline{\vee}} F_1 \mathbin{\underline{\vee}} F_2 \mathbin{\underline{\vee}} \cdots \mathbin{\underline{\vee}} F_n$$



6 Introduction to Modal Logics

Learning Objectives

By the end of this section you should be able to:

- ☐ Define what it means for a modal formula to be **satisfiable** or **valid** in a given **Kripke model**;
- ☐ Identify all worlds in which a given formula is satisfied.

6.1 Motivation

Consider the following premises

- *"It is not possible that Alice and Bob are both telling the truth"*;
- *"It is possible that Alice is telling the truth"*.

From this we would like to conclude, for example, that *"it is not necessarily the case that Bob is telling the truth"*, that is to say *"it is possible that Bob is lying"*. However, if we were to try any capture these premises with classical propositional logic, we would quickly encounter an issue: how do we deal with the 'possibility'?

Grammatically, words such as 'possibly' and 'necessarily' fall under a category of words called **modalities** that change the **mode of truth** away from objective, here-and-now static world of propositional logic[15]. It is not immediately obvious how to represent these in terms of propositional logic, and so one approach that is available to us is to simply introduce some new connectives!

[15] Other modalities of interest include: *'it should/ought to be true'*, *'it will be true'*, *'I know it to be true'*, *'I believe it to be true'*

Let $\Box$ and $\Diamond$ be to fresh unary connectives, which we call **Modal operators**, with the intended interpretations that:

$$\Diamond F \quad : \quad \text{“}F \textit{ is possibly true}\text{”}$$

$$\Box F \quad : \quad \text{“}F \textit{ is necessarily true}\text{”}$$

We can then extend our set of well-formed formulas by allowing us to apply modal operators to any subformulas in the way that we would with negations — our only other unary connective.

Definition 6.1: Well-formed Modal Formulas

Given a set of **propositional variables** Prop $= \{p_1, \ldots, p_n \ldots\}$ we define the set *Fml*$^{\Box}$ of all **(well-formed) modal formulas** inductively as follows:

- If $p \in$ Prop then $p \in$ *Fml*$^{\Box}$ is a well-formed formula;
- If $F, G \in$ *Fml*$^{\Box}$ are well-formed formulas then so are:

$$\neg F, \qquad (F \wedge G), \qquad (F \vee G), \qquad (F \to G), \qquad \Box F, \qquad \Diamond F$$

Exercise

Write down some examples of modal formulas from *Fml*$^{\Box}$.

- $(\Diamond p \wedge \Diamond \neg p)$
- $\Box p \to \Diamond p$
- $\Box(p \to q) \to (\Box p \to \Box q)$

Now, we may wonder, did we even need to introduce these additional connectively? Couldn't we have just worked out a truth table that captured our intended semantics for 'possibility' and 'necessity', and then find an equivalent formula using just $\{\neg, \wedge, \vee, \to$

} (or some other adequate set of connectives)? Unfortunately, any attempt to pin down the semantics of our newly introduced modal operators is doomed to fail.

Each of our Boolean connectives are **truth-functional**, which is to say that their truth tables are fully determined by the truth values that are assigned to their subformulas. However, consider the following formula:

$$\Diamond\mathsf{FlyingCat} \quad : \quad \textit{“It is possible that Mr Whiskers can fly”}$$

If it is the case that Mr Whiskers can fly, that is, if FlyingCat is assigned to be true then it is reasonable that $\Diamond$FlyingCat is also assigned to be true[16]. But if Mr Whiskers can't fly, that is, if FlyingCat is assigned to be false, then it does not say anything about whether or not it is *possible* for Mr Whiskers to fly. Indeed, there are only four possible truth tables for unary connectives, none of which seem to capture our intended semantics for 'possibly' or 'necessarily':

F	Op_1F
1	0
0	0

F	Op_2F
1	0
0	1

F	Op_3F
1	1
0	0

F	Op_4F
1	1
0	1

The first truth table is for the 'false' operator that simply returns false, regardless of the formula F. If we choose this for our interpretation of $\Diamond F$ then it would say that **nothing is possible**. The second truth table is just the standard truth table for negation, while the third table is just the 'identity' operator that simply returns the truth value of F. If this were our interpretation of $\Diamond F$ then it would say that **only that which is true is possible** — this is certainly one philosophically defensible position to take, although it may not quite capture some of the nuances that we would like to capture. The final table is the 'true' operator that simply returns true regardless of the formula F. If we were to choose that as our interpretation for $\Diamond F$ then it would say that **nothing is impossible**, which — while an admirably motivational outlook on life — doesn't quite capture our intended semantics either.

[16] we will revisit this assumption again shortly

The unavoidable conclusion is, therefore, whatever semantics we come up with for our modal operations $\Box$ and $\Diamond$, it can't be based in truth tables — 'possibility' and 'necessity' are **non-truth-functional** operators.

6.2 Kripke Semantics

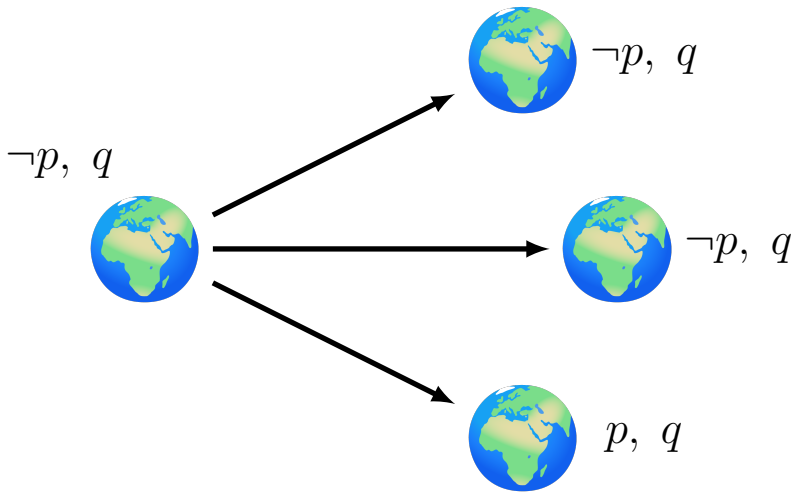
One approach to defining a semantics for Modal logic(s) was proposed in 1963 by American philosopher and logician Saul Kripke [5], and which now bears his name: **Kripke semantics**.

The idea is this: suppose that you inhabit a world that is governed by some propositional valuation — a model of propositional logic / a row of a truth table. In your local world you can reason, as you would about any formulas of propositional logic (note that *Fml* $\subseteq$ *Fml*$^{\Box}$). For example, if we inhabited a world in which *"Mr whiskers cannot fly"* ($\neg p$) and *"1 + 1 is 2"* (q) then we can conclude that *"either Mr whiskers can fly or 1 + 1 is 2"* ($p \vee q$) and *"If Mr Whiskers can fly then 1+1 is 2"* ($p \to q$) are true statements, based simply off evaluating the appropriate row of the truth table.

$$\neg p, \ q$$

However, if we wish to decide whether is is *possible* that Mr Whiskers could *hypothetically* fly[17] then we must look **outside of our local world** to all the **possible worlds** that we could imagine might exists with other possible propositional models.

[17] Having, for example, undergone sufficient training and obtained his pilot's licence.



If we can imagine some world (here depicted with directed edge) in which Mr Whiskers *can* fly, that is where p is true, then it must be the case that *"It is possible that Mr Whiskers can fly"* ($\Diamond p$). On the other hand, if in all the worlds that we can imagine, we can't imagine *any* wold in which *"1 + 1 is <u>not</u> 2"* then it must be the case that *"1 + 1 is necessarily 2"* ($\Box q$).

But what about more complex modal formulas such as *"It is necessarily the case that Alice is possibly telling the truth"*? How do we make sense of such **nested modalities**? Well, in exactly the same way — to determine whether it is necessarily the case that *"Alice is possibly telling the truth"*, we must check that *"Alice is possibly telling the truth"* holds in **every** world that we can imagine. However, in order to evaluate whether Alice is possibly telling the truth in each of *these* worlds, we must explore all of the worlds that we could imagine **from** those worlds!

That is to say, in order to define our semantics for modal logics, we need not only a description of the propositional variables that are true in our local world, we also need to know about what other worlds are possible, and what variables are true in each of those possible worlds. And, to deal with nested modalities, we need not only a set of possible worlds all accessible from the 'local world' but a **directed graph of possible worlds** connected is some in any arbitrary configuration. The resulting structure is what is called a **Kripke model** [5].

Definition 6.2: Kripke models

We define a **Kripke model** to be a triple $\mathfrak{M} = \langle W, R, V \rangle$ where

- W is a set of vertices referred to as **(possible) worlds**;
- $R \subseteq W \times W$ is a binary **accessibility relation** on W describing the (directed) edges between possible worlds;
- $V : W \times \text{Prop} \to \{0, 1\}$ is a propositional that maps each pair (w, p), with $w \in W$ and $p \in \text{Prop}$ to some Boolean value, 0 ('false') or 1 ('true').

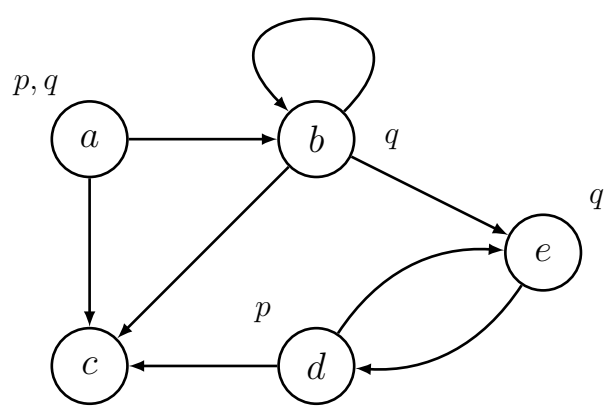
We refer to the pair $\mathfrak{F} = \langle W, R \rangle$ as the **Kripke frame** underlying $\mathfrak{M}$.

Notes

Sketch a diagram illustrating the following Kripke model $\mathfrak{M} = \langle W, R, V \rangle$ where

$$W = \{a, b, c, d, e\} \qquad \text{and} \qquad R = \{(a, b), (b, c), (a, c), (d, c), (b, b), (b, e), (e, d), (d, e)\}$$

and $V(w, p) = 1$ if and only if $w \in \{a, d\}$ and $V(w, q) = 1$ if and only if $w \in \{a, b, e\}$



Definition 6.3: Satisfiability and Validity

Given a Kripke model $\mathfrak{M} = \langle W, R, V \rangle$ and a modal formula $F \in \mathit{Fml}^{\Box}$ we can extend the valuation V to the domain $W \times \mathit{Fml}^{\Box}$ of all pairs (w, F) where F is a modal formula as follows:

$$\begin{aligned}
V(w, \neg F) &= 1 - V(w, F) \\
V(w, F \wedge G) &= \min\big(V(w, F),\ V(w, G)\big) \\
V(w, F \vee G) &= \max\big(V(w, F),\ V(w, G)\big) \\
V(w, F \to G) &= \max\big(1 - V(w, F),\ V(w, G)\big) \\
V(w, \Diamond F) &= \max\big\{V(u, F) : (w, u) \in R\big\} \\
V(w, \Box F) &= \min\big\{V(u, F) : (w, u) \in R\big\}
\end{aligned}$$

We say that F is **satisfiable** in $\mathfrak{M}$ at world $w \in W$ if $V(w, F) = 1$; in which case we write

$$\mathfrak{M}, w \models F \qquad \leftrightsquigarrow \qquad \text{“}F \textit{ is satisfied in } \mathfrak{M} \textit{ at world } w\text{”}$$

That is to say that we evaluate the Boolean connectives in exactly the same as we do for classical propositional logic, with the only difference being that we must evaluate that at every world $w \in W$ of our model. Whereas our new box and diamond operators take on the minimum and maximum values of their successors with respect to the accessibility relation R. This is equivalent to the following, perhaps more intuitive definition[18]

$$\begin{aligned}
\mathfrak{M}, w \models \Diamond F \quad &\iff \quad \exists u \in W;\ (w, u) \in R \text{ and } \mathfrak{M}, u \models F \\
\mathfrak{M}, w \models \Box F \quad &\iff \quad \forall u \in W;\ (w, u) \in R \text{ implies } \mathfrak{M}, u \models F
\end{aligned}$$

[18] Here we are using $\exists$ and $\forall$ as a convenient notational shorthand, rather than as a symbol of a formal logical language; however, we will consider what formal logical language this *could* be represented in later in the course.

Definition 6.4: Validity

Given a Kripke frame $\mathfrak{F} = (W, R)$, a Kripke model $\mathfrak{M} = \langle W, R, V \rangle$ with underlying frame $\mathfrak{F}$, and a modal formula $F \in \mathit{Fml}^{\Box}$.

- We say that F is **valid over the model** $\mathfrak{M}$ if $\mathfrak{M}, w \models F$ for all $w \in W$; in which case we write:

$$\mathfrak{M} \models F \qquad \leftrightsquigarrow \qquad \text{“}M \textit{ is valid over the model } \mathfrak{M}\text{”}$$

- We say that F is **valid over the frame** $\mathfrak{F} = (W, R)$ if $\mathfrak{M} \models F$ for **every** Kripke model $\mathfrak{M} = \langle W, R, V \rangle$ with underlying frame $\mathfrak{F}$; in which case we write:

$$\mathfrak{F} \models F \qquad \leftrightsquigarrow \qquad \text{“}F \textit{ is valid over the frame } \mathfrak{F}\text{”}$$

We can think of each of these three (ab)uses of $\models$ turnstile as fixing some components of the model that are stated on the left-hand-side, with any components that are not stated being implicitly universally quantified: for example, in the first case $\mathfrak{M}, w \models F$ of satisfiability in a model $\mathfrak{M}$ at a world $w \in W$, we fix the model, *i.e.* the frame and the valuation, and some world w. In the case $\mathfrak{M} \models F$ of validity over the model, we fix only the frame and valuation, and consider all possible worlds. In the case $\mathfrak{F} \models F$ of validity over a frame, we fix only the frame and consider all possible worlds under all possible valuations.

We can go one step further, and consider formulas that are valid over **all frames** (or all frames belonging to some given **class of frames** $\mathcal{C}$, *e.g.* reflexive frames, transitive frames, *etc.*).

Exercise

Show that the formula

$$F \;=\; \Box(p \to q) \to (\Box p \to \Box q)$$

is valid in **all frames**.

- Suppose for contradiction that $\mathfrak{M}, w \not\models \Box(p \to q) \to \Box(p \to q)$ for some Kripke model $\mathfrak{M} = \langle W, R, V \rangle$ and some $w \in W$.
- It follows that

$$\mathfrak{M}, w \models \Box(p \to q) \tag{12}$$

$$\mathfrak{M}, w \not\models \Box p \to \Box q \tag{13}$$

  from (13) we then also find that we must have

$$\mathfrak{M}, w \models \Box p \tag{14}$$

$$\mathfrak{M}, w \not\models \Box q \tag{15}$$

- By (15) we must have that there is some $v \in W$ such that $(w, v) \in R$ and $\mathfrak{M}, v \not\models q$.
- By (14), we must have that $\mathfrak{M}, v \models p$, since $(w, v) \in R$.
- By (12), we must have that $\mathfrak{M}, v \models p \to q$, again since $(w, v) \in R$.
- However, it then follows that $\mathfrak{M}, v \models q$ since $p \wedge (p \to q) \models q$.

Given this Kripke semantics, we can now define the (minimal) modal logic $\models^{\mathbf{K}}$, in an analogous way to how we define logical consequence $\models$ in classical logic, by considering all the models that make the left-hand-side true, and then checking that the right-hand-side is also true in these models. It is just the case that here our models are more complex, and we must also consider all of the possible worlds of our models.

Definition 6.5

Given a set of modal formulas $\Gamma \subseteq \mathit{Fml}^{\Box}$ and $F \in \mathit{Fml}^{\Box}$, we write

$$\Gamma \models^{\mathbf{K}} F \qquad \Longleftrightarrow \qquad \mathfrak{M}, w \models \Gamma \quad \text{implies} \quad \mathfrak{M}, w \models F$$

for all Kripke models $\mathfrak{M} = \langle W, R, V \rangle$ and all $w \in W$.

In particular, the set of all **validities / tautologies** with respect to $\models^{\mathbf{K}}$ is the set $\mathbf{K} \subseteq \mathit{Fml}^{\Box}$ of all formulas that are valid in **all Kripke frames**:

$$F \in \mathbf{K} \quad \Longleftrightarrow \quad \top \models^{\mathbf{K}} F \quad \Longleftrightarrow \quad \mathfrak{F} \models F \text{ for all } \mathfrak{F}$$

where $\top$ denotes a propositional tautology.

It is commonplace in the modal logic literature to refer to the set of validities **K** a the **(minimal) modal logic** rather than the consequence relation. However, for our purposes the two notions are interchangeable since for **finite** sets of formulas $\Gamma \subseteq \mathit{Fml}^{\Box}$ we have that

$$\Gamma \models^{\mathbf{K}} F \qquad \Longleftrightarrow \qquad (\bigwedge \Gamma \to F) \in \mathbf{K}.$$

There is, of course, a syntactic proof theory, as well, for the modal logic **K**, from which we may derive a provability relation $\vdash^{\mathbf{K}}$, for which we have an appropriate soundness and completeness theorem. However, this will take us beyond the scope for this course; and so we state it here for the sake of completeness without proof.

Theorem 6.6: Completeness Theorem for K [5]

Let $\Gamma \subseteq \mathit{Fml}^{\Box}$ and $F \in \mathit{Fml}^{\Box}$. Then we have that

$$\Gamma \models^{\mathbf{K}} F \qquad \Longleftrightarrow \qquad \Gamma \vdash^{\mathbf{K}} F.$$

6.3 A Labelling Algorithm for Model-checking

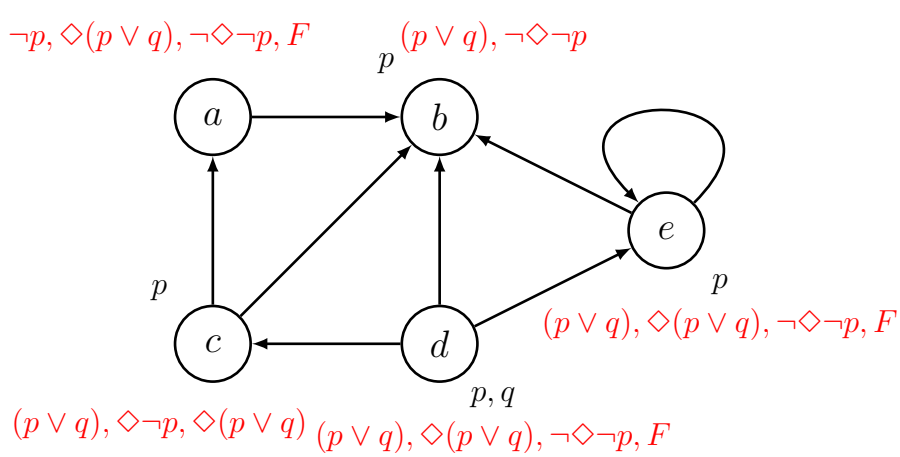
Suppose we are given some (finite) Kripke model $\mathfrak{M} = \langle W, R, V \rangle$. If we wish to not only determine whether a given formula $F \in \mathit{Fml}^{\Box}$ is satisfiable in M at a given world $w \in W$ (so-called **model-checking**), but instead wish to determine whether F is **valid** over $\mathfrak{M}$, we would need to evaluate $V(w, F)$ for all $w \in W$ — this may result in a lot of necessary duplication of work. Fortunately, we can eliminate some of this duplication by answering a more general question: compute the set of all worlds $W(F) \subseteq W$ in which F is satisfied. We could then check whether F is valid by simply checking whether or not $W(F) = W$. The benefit of this slight change in perspective, is that we now invite the opportunity to solve this new problem with **divide-and-conquer**.

Notes

Consider the following subformulas of $F = (\neg\Diamond\neg p \wedge \Diamond(p \vee q))$:

$$p, \quad q, \quad \neg p, \quad (p \vee q), \quad \Diamond\neg p, \quad \Diamond(p \vee q), \quad \neg\Diamond\neg p, \quad (\neg\Diamond\neg p \wedge \Diamond(p \vee q))$$

In order of size, annotate the following Kripke model, by labelling all worlds in which each subformula is satisfied. What do you notice?



Labelling Algorithm for Model-checking

Input) Given a Kripke model $\mathfrak{M} = \langle W, R, V \rangle$ and a modal formula $H \in \mathit{Fml}^{\Box}$;

Pre-processing) Re-write H using the following successive equivalences to replace all occurrences of $\rightarrow$ and $\Box$

$$F \rightarrow G \;\; \mapsto \;\; (\neg F \vee G)$$

$$\Box F \;\; \mapsto \;\; \neg \Diamond \neg F$$

Case 1) If H is a propositional variable, then we already have the worlds in which p is satisfied encoded in the valuation V.

$$W(p) \;=\; \{\, w \in W : V(w, p) = 1 \,\}$$

Case 2) If H is a formula of the form $\neg F$, then based on the semantics for V, the set of worlds where H is true is precisely those there F is false. In other words

$$W(H) \;=\; W \setminus W(F)$$

where $A \setminus B$ denotes the set of elements in A that do not appear in B.

Case 3) If H is a formula of the form $(F \wedge G)$, then based on the semantics for V, the set of worlds where H is true is precisely those there F is true and where G is true. In other words

$$W(H) \;=\; W(F) \cap W(G)$$

Case 4) Similarly, if H is a formula of the form $(F \vee G)$, then the set of worlds where H is true is precisely those there F is true or where G is true. In other words

$$W(H) \;=\; W(F) \cup W(G)$$

Case 6) If H is a formula of the form $\Diamond F$, then the set of worlds where H is true is precisely those that can 'see' a world in which F is true. In other words

$$W(H) \;=\; \{w \in W : (w, v) \in R \text{ for some } v \in W(F)\}$$

In truth, there is no inherent need for the preprocessing step to first eliminate all occurrences of $\rightarrow$ and $\Box$ from our formulas, as we could just have easily designed additional cases for these two operators; it just helps when performing the algorithm to have fewer cases[19]

[19] In this spirit, we could have also argued for the removal of the case for either one of $\wedge$ or $\vee$, as De Morgan's laws would have allowed us to express one in terms of the other, however the cost of doing so does not appear to pay off in this case, given the simplicity and symmetry of the intersection and union set operations.

7 The Tableau Algorithm

Learning Objectives

By the end of this section you should be able to:

- ☐ Apply the **Tableau algorithm** to determine whether a given modal formula is satisfiable with respect to a given class of frames;
- ☐ Apply the **Tableau algorithm** to determine whether a given modal formula is valid with respect to a given class of frames.

7.1 A Tableau Algorithm for Propositional Logic

Unlike for classical propositional logic where to determine whether a formula is satisfiable or valid, it is sufficient to draw out a truth table and check whether there is a row that evaluates to true (satisfiable) or whether all rows evaluate to true (valid). This can always be done in a finite amount of time since there are at most 2^n rows to our truth table for a formula with n propositional variables.

However, for modal logics where our Kripke models comprise not only a valuation but a description of the structure of the worlds/states of our system and how they are connected, it is no longer possible to simply exhaust all possible models, since there is no defined limit on the size of the underlying frames.

Indeed, we do not even have a restriction that our Kripke models must be **finite**! In fact, there are often cases where we may want to model situations where we have an **infinite number** of possible worlds, such as in temporal logics where each world corresponds to some future instance of time and we would

If we are to be able to answer modal satisfiability and validity questions in a finite and reasonable amount of time, we need a more careful and systematic approach. One such approach is the **Tableau Algorithm**.

Definition 7.1: Tableaux

A **(propositional) tableau** /ˈtæbləʊ/ (plural: **tableaux** /ˈtæbləʊz/) is a two-column table

True	**False**
⋮	⋮

with

- a **'True' column** where we list all (sub)formulas that we want to be true;
- a **'False' column** where we list all (sub)formulas that we want to be false.

Formally, we can describe a tableau as a pair

$$(\textbf{True}, \textbf{False}) \in \mathscr{P}(\mathit{Fml}) \times \mathscr{P}(\mathit{Fml})$$

where **True** and **False** are (finite) subsets of formulas.

We say that a tableau is **closed** if **True** $\cap$ **False** $\neq \varnothing$ and **open** otherwise

In order to decide whether a propositional $F \in \mathit{Fml}$ is satisfiable we create an **initial tableau** $(\{F\}, \varnothing)$ with F in the 'true' column and nothing in the 'false' column:

	True	**False**
1.	F	

We then successively *'break down'* our formula into its subformulas and move these component parts into either the true column or false column in accordance with the rules outlined below, that emulate the truth table semantics that we are familiar with.

Rules for the True Column

Negation If we have a formula of the form $\neg F$ that appears in the **True** column of a tableau — intuitively, we want $\neg F$ to be true — then we can move F across to the

False — that is to say that we want F to be false. We then mark the formula $\neg F$ as having been expanded so that we can better keep track of which rows of our tableau have yet to be processed:

	True	False
1.	$\cancel{\neg F}$	
2.		F

($\neg$ true)

Conjunction The **($\wedge$ true)** rule says that if we have a formula of the form $(F \wedge G)$ appearing in the **True**, then we can split this conjunction and add each of its conjuncts F and G to be the **True** as well. We can then cross out $(F \wedge G)$ as having been expanded:

	True	False
1.	$\cancel{F \wedge G}$	
2.	F	
3.	G	

($\neg$ true)

Disjunction By contrast the **($\vee$ true)** rule is a **branching rule**: If we have that formula of the form $(F \vee G)$ appearing in the **True**, then we *either* want F to be true *or* for G to be true, but not necessarily both. In order to deal with these two alternatives we **branch** into two tableaux (each continuing the history of the parent tableau), with F in the **True** of the first branch and G in the **True** of the second branch:

	True	False
1.	$\cancel{(F \vee G)}$	
	$\vdots$	$\vdots$

	True	False
2.	F	

	True	False
2.	G	

($\vee$ true)

Implication Like **($\vee_{\textbf{true}}$)**, the **($\rightarrow_{\textbf{true}}$)** rule is also a branching rule, since for a formula of the form $(F \rightarrow G)$ to be true, it could **either** be because F is false **or** because G is true, so we must consider both cases in separate branches:

($\vee_{\textbf{true}}$)

	True	False
1.	$\cancel{(F \rightarrow G)}$	
	$\vdots$	$\vdots$

	True	False
2.		F

	True	False
2.	G	

Taking note that for this rule, we move F into the **False** of the first branch, whereas G remains in the **True** of the second branch.

Alternatively, we can think of the **($\rightarrow_{\textbf{true}}$)** rule as a combination of the rules for negation and disjunction since we have that $F \rightarrow G \equiv \neg F \vee G$.

Rules for the False Column

We are half-way there, having defined each of the rules for the **True**; we now just need to define the rules for the **False**. Fortunately, these closely mirror the rules we have already seen:

Negation If we have a formula of the form $\neg F$ that appears in the **False** column then we can move F across to the **True**. After which we mark the formula $\neg F$ as having been expanded:

($\neg_{\textbf{false}}$)

	True	False
1.		$\cancel{\neg F}$
2.	F	

Conjunction The **($\wedge_{\textbf{false}}$)** rule is a **branching rule** that behaves similar to the **($\vee_{\textbf{true}}$)** rule, since a formula of the form $(F \wedge G)$ is false if and only if *either* F is false *or* G is false:

	True	False
1.		$\cancel{(F \wedge G)}$
	$\vdots$	$\vdots$

	True	False
2.		F

	True	False
2.		G

($\wedge_{\textbf{false}}$)

Disjunction The **($\vee_{\textbf{false}}$)** rule, on the other hand, behaves more like the **($\wedge_{\textbf{true}}$)** rule, since a formula of the form $(F \vee G)$ is false if and only if *both* F and G are false:

	True	False
1.		$\cancel{F \vee G}$
2.		F
3.		G

($\vee_{\textbf{false}}$)

Implication Lastly, the **($\rightarrow_{\textbf{true}}$)** rule also behaves like the **($\wedge_{\textbf{true}}$)** rule, since a formula of the form $(F \rightarrow G)$ is false if and only if F is true *and* G is false:

	True	False
1.		$\cancel{F \rightarrow G}$
2.		F
3.	G	

($\vee_{\textbf{false}}$)

Again, we can think of the **($\rightarrow_{\textbf{false}}$)** rule as a combination of the rules for negation and disjunction, as above.

We continue applying the rules until all formulas in both the **True** and **False** have been fully expanded leaving only propositional variables, that can no further be expanded by any of our rules.

- If we encounter a fully expanded tableau that is **open** along any of our branches then we may conclude that our initial formula F is **satisfiable**. Indeed, we can 'read off' a satisfying assignment directly from the fully expanded open tableau by setting:

$$V(p) \;=\; 1 \qquad \Longleftrightarrow \qquad p \in \textbf{True}$$

- If, on the other hand, every branch of our algorithm results in a closed tableau then we can conclude that the formula is **unsatisfiable**.

Exercise

Apply the Tableaux Algorithm to determine whether or not the following formula is satisfiable:

$$F \;=\; \neg(p \to \neg(r \vee q)) \wedge (p \to \neg r)$$

	True	False
1.	$\neg(p \to \neg(r \vee q)) \wedge (p \to \neg r)$	
2.	$\neg(p \to \neg(r \vee q))$	
3.	$(p \to \neg r)$	
4.		$(p \to \neg(r \vee q))$
	$\vdots$	$\vdots$

	True	False
5.		p
6.		$\neg(r \vee q))$
7.	p	
	$\times$ **Closed** (5 & 7)	

	True	False
5.	$\neg r$	
6.		$\neg(r \vee q)$
7.	p	
8.		r
9.	$r \vee q$	
	$\vdots$	$\vdots$

	True	False
10.	r	
	$\times$ **Closed** (8 & 10)	

	True	False
10.	q	
	✓ **Open**	

The final tableau is open and fully expanded and so the formula is **satisfiable** with $V(p) = V(q) = 1$ and $V(r) = 0$.

7.2 Deciding Validity with Tableaux

We can also utilise the Tableau Algorithm to determine, not only whether a formula is satisfiable, but whether or not it is **valid** *i.e.* a **tautology**. We do this by utilising the property that

$$F \text{ is valid/a tautology} \quad \Longleftrightarrow \quad \neg F \text{ is } \textbf{NOT} \text{ satisfiable}$$

Since we can use our tableau algorithm to determine whether or not a formula is satisfiable, in order to determine whether a given formula F is a tautology, we can simply run our algorithm on its negation $\neg F$, and say that it is a tautology if **every** fully expanded tableau branch is **closed**, since any fully expanded open tableau branch would correspond to a counter-model in which F was false.

Equivalently, in order to decide whether a propositional $F \in$ *Fml* is satisfiable we create an **initial tableau** $(\varnothing, \{F\})$ with F in the 'false' column and nothing in the 'true' column:

	True	**False**
1.		F

7.3 The Modal Tableau Algorithm

In order to extend our Tableau algorithm to modal logics, we need to make a few small modifications: First, since modal satisfiability may vary from world to world in our Kripke models, we need to note not only whether or not a subformula should be true or false, but in which worlds it should be true or false.

Instead of storing just the subformulas in our **True** and **False** of our tableaux, we can instead store a 'tagged formula' that ties the truth or falsity of a formula to a specific world.

Definition 7.2: Tagged Formulas

A **tagged formula** is a pair of the form $(w : F)$ where w is a label used to designate a given world of our to-be-determined Kripke model W, and $F \in \mathit{Fml}^{\Box}$ is a modal formula.

Next, in addition of keeping track of which (tagged) subformulas we need to be true and which we need to be false, we also need to keep track of any relations between worlds that we need to include in our Kripke model. Hence we must extend our definition of a tableau to capture this information as well[20].

Definition 7.3: Modal Tableaux

A **(modal) tableau** is a three-column table

True	False	Relation
⋮	⋮	⋮

with

- a **'True' column** where we list all *tagged* (sub)formulas that we want to be true;

[20] Some sources will choose to place this information in the **True**, but I find it clearer to record this in a separate column as it makes it easier to read off a satisfying model from a fully expanded open tableau.

- a **'False' column** where we list all *tagged* (sub)formulas that we want to be false;
- a **'Relation' column** where we list all the *relations* that we want to include in our Kripke model.

As with propositional tableaux, we say that a modal tableau is **closed** if **True** $\cap$ **False** $\neq \varnothing$ and **open** otherwise

All of our previous propositional tableau rules remain effectively the same, with the one modification that we now work with tagged formula. For example, the **($\rightarrow_{\textbf{true}}$)** rule for the modal tableaux algorithm would be the following:

($\vee_{\textbf{true}}$)

	True	False	Relation
1.	$\cancel{w : (F \rightarrow G)}$		
	$\vdots$	$\vdots$	$\vdots$

	True	False	Relation
2.		$w : F$	

	True	False	Relation
2.	$w : G$		

Finally, we need only to introduce the additional rules for formulas of the form $\Box F$ and $\Diamond F$ appearing in either the **True** or **False**, and it is here where we need to be most careful

Diamond True If we have a tagged formula of the form $w : \Diamond F$ that appears in the **True** column of a tableau, then we want for $\Diamond F$ to be true at world w in our Kripke model. But in order for this formula to be true there must be some $u \in W$ such that $(w, u) \in R$, where F is true at u.

	True	False	Relation
1.	$\cancel{w : \Diamond F}$		
2.	$u : F$		(w, u)

($\Diamond$ true)

It is important that u is a **fresh label** that does not appear elsewhere in the current branch of the tableau. This is most easily maintained by simply enumerating the world labels as $w_0, w_1, \ldots$, and simply using the next available index whenever the **($\Diamond$ true)** rule is used.

Box False The **($\Box$ false)** rule behaves similarly to **($\Diamond$ true)** rule, since for a formula of the form $\Box F$ to be false at world w, there must be some $u \in W$, with $(w, u) \in R$, in which F is false.

	True	False	Relation
1.		$\cancel{w : \Box F}$	
2.		$u : F$	(w, u)

($\Box$ false)

Box True The **($\Box$ true)** rule says that whenever we have a tagged formula of the form $(w : \Box F)$ appearing in the **True** column, we must look through our **Relation** for any relations of the form (w, u), and add $(u : F)$ to the **True** column.

	True	False	Relation
1.	$w : \Box F$		
2.	$u_1 : F$		
	$\vdots$	$\vdots$	$\vdots$
3.	$u_n : F$		

where $(w, u_1), \ldots, (w, u_n) \in$ **Relation** **($\Box$ true)**

IMPORTANT NOTE: Moreover, in contrast to all of the other rules seen so far, we **do not** mark off the formula $(w : \Box F)$ as having been expanded, as we may still need to **re-apply** this rule in the event that additional relations of the form (w, u) are added to the **Relation**.

Diamond False The **($\Diamond$ false)** rule behaves similarly to the **($\Box$ true)** rule and says that whenever we have a tagged formula of the form $(w : \Diamond F)$ appearing in the **False** column, we must look through our **Relation** for any relations of the form (w, u), and add $(u : F)$ also to the **False** column.

	True	False	Relation
1.		$w : \Diamond F$	
2.		$u_1 : F$	
	$\vdots$	$\vdots$	$\vdots$
3.		$u_n : F$	

where $(w, u_1), \ldots, (w, u_n) \in$ **Relation** **($\Diamond$ false)**

As with the **($\Box$ true)** rule, we **do not** mark off the tagged formula $(w : \Diamond F)$ as having been expanded, as we still may need to reapply the rule if we later add additional relations of the form (w, u) are added to the **Relation**.

We continue applying the rules until all formulas in both the **True** and **False** have been fully expanded, and every boxed formula appearing in the **True** column, and every diamond formula appearing in the **False** column, have been applied to all relations appearing in **Relation**.

- If we encounter a fully expanded tableau that is **open** along any of our branches then we may conclude that our initial formula F is **satisfiable**. Indeed, we can

'read off' a satisfying Kripke model $\mathfrak{M} = \langle W, R, V \rangle$ from the fully expanded open tableau by taking:

$$W = \{w_i : (w_i : p) \in \textbf{True} \cup \textbf{False}\}$$

to be the set of all world labels that belong to any tagged formulas appearing in either the **True** column or **False** of our open tableau;

$$(w_i, w_j) \in R \quad \Longleftrightarrow \quad (w_i, w_j) \in \textbf{Relation}$$

to be the set of all relations that are added to our **Relation** column of our open tableau; and taking

$$V(w_i, p) \;=\; 1 \quad \Longleftrightarrow \quad (w_i : p) \in \textbf{True}$$

for all $w \in W$ and $p \in$ Prop.

- If, on the other hand, every branch of our algorithm results in a closed tableau then we can conclude that the formula is **unsatisfiable**.

Exercise

Apply the (Modal) Tableaux Algorithm to determine whether or not the following formula is satisfiable (with respect to the class of all Kripke frames):

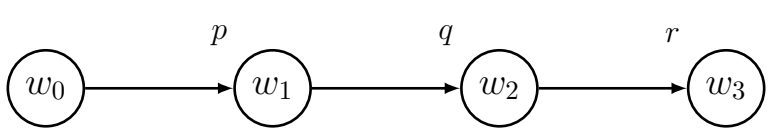
$$F \;=\; \Diamond(p \wedge \Box q) \wedge \Box(p \to \Diamond\Diamond r)$$

	True	False	Relation
1.	$w_0 : \Diamond(p \wedge \Box q) \wedge \Box(p \to \Diamond\Diamond r)$		
2.	$w_0 : \Diamond(p \wedge \Box q)$		
3.	$w_0 : \Box(p \to \Diamond\Diamond r)$		
4.	$w_1 : p \wedge \Box q$		(w_0, w_1)
5.	$w_1 : p \to \Diamond\Diamond r$		
6.	$w_1 : p$		
7.	$w_1 : \Box q$		
	$\vdots$	$\vdots$	$\vdots$

	True	False	Relation
16.		$w_1 : p$	
	$\times$ **Closed** (14 & 16)		

	True	False	Relation
16.	$w_1 : \Diamond\Diamond r$		
17.	$w_2 : \Diamond r$		(w_1, w_2)
18.	$w_2 : q$		
19.	$w_3 : r$		(w_2, w_3)
	$\checkmark$ **Open**		

It follows that F is satisfiable at w_0 in the following Kripke model



7.4 Deciding Modal Satisfiability with respect to $\mathcal{C}$

By examining the construction of the satisfying Kripke models generated by our tableau algorithm, we note that they all take on a very specific form: every frame is a **tree** — this is often referred to as the **tree model property**: every satisfiable formula is satisfiable in a model whose underlying Kripke frame is a tree.

However, there may be occasions where we want to determine whether or not a formula $F \in \mathit{Fml}^{\Box}$ is satisfiable, not just with respect to the class of all frames, or even with respect to the class of all trees, but with respect to some other arbitrary class of Kripke frames $\mathcal{C}$. For example, with respect to **reflexive frames**, or with respect to **transitive frames**.

In these cases we "simply" need to ensure that after each application of the **($\Diamond_{\textbf{true}}$)** rule and **($\Box_{\textbf{false}}$)** rule — the two rules which require us to add additional relations to the **Relation** column — we 'close' the set of pairs appearing in **Relation** by adding any additional edges that may be required to ensure that the resulting frame has the intended structure.

For example, when deciding whether a given formula F is satisfiable **with respect to transitive frames**, we "simply" need to check that whenever we add edges (x, y) and (y, z) to the **Relation** column, we also add the edge (x, z) to **Relation** so that we ensure that the resulting frame (in any fully-expanded open tableau) is transitive.

Exercise

Apply the (Modal) Tableaux Algorithm to determine whether or not the following formula is satisfiable, with respect to the class of all **transitive** Kripke frames:

$$F \;=\; \Diamond\big(p \wedge \neg\Box(q \wedge \Box r)\big) \wedge \Box(q \wedge r)$$

	True	False	Relation
1.	$w_0 : \Diamond\big(p \wedge \neg\Box(q \wedge \Box r)\big) \wedge \Box(q \wedge r)$		
2.	$w_0 : \Diamond\big(p \wedge \neg\Box(q \wedge \Box r)\big)$		
3.	$w_0 : \Box(q \wedge r)$		
4.	$w_1 : p \wedge \neg\Box(q \wedge \Box r)$		(w_0, w_1)
5.	$w_1 : q \wedge r$		
6.	$w_1 : p$		
7.	$w_1 : \neg\Box(q \wedge \Box r)$		
8.	$w_1 : q$		
9.	$w_1 : r$		
10.		$w_1 : \Box(q \wedge \Box r)$	
11.		$w_2 : q \wedge \Box r$	(w_1, w_2)
12.			$(w_0, w_2)^*$
13.	$w_2 : q \wedge r$		
14.	$w_2 : q$		
15.	$w_2 : r$		
	$\vdots$	$\vdots$	$\vdots$

	True	False	Relation
16.		$w_2 : q$	
	$\times$ **Closed** (14 & 16)		

	True	False	Relation
16.		$w_2 : \Box r$	
17.		$w_3 : r$	(w_2, w_3)
18.			$(w_1, w_3)^*, (w_0, w_3)^*$
19.	$w_3 : q \wedge r$		
20.	$w_3 : q$		
21.	$w_3 : r$		
	$\times$ **Closed** (17 & 21)		

7.5 Deciding Modal Validity with Tableaux

As with propositional logic we can utilise the Tableau Algorithm to determine whether or not it is **valid** with respect to a given class of frames $\mathcal{C}$ using the property that

$$F \text{ is valid with respect to } \mathcal{C} \quad \Longleftrightarrow \quad \neg F \text{ is } \textbf{NOT} \text{ satisfiable with respect to } \mathcal{C}$$

And so in order to decide whether a modal formula $F \in \mathit{Fml}^{\Box}$ is satisfiable with respect to a given class of frames $\mathcal{C}$, we create an **initial tableau** with F in the 'false' column and nothing in the 'true' column:

	True	**False**	**Relation**
1.		F	

And conclude that F is **valid with respect to** $\mathcal{C}$ if and only if every fully-expanded branching tableaux is closed.

8 Correspondence Theory

Learning Objectives

By the end of this section you should be able to:

- ☐ Define what it means for a formula to be a **Salqvist formula**;
- ☐ Apply the **Salqvist–van Benthem algorithm** to compute first-order correspondents of arbitrary Salqvist formulas

8.1 Applications of Modal Logic

To justify the US invasion of Iraq in 2003, then-Defence Secretary Donald Rumsfeld famously said in relation to the existence of Weapons of Mass Destruction (WMDs):

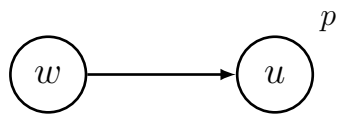
> *"[...] there are* **known knowns***; there are things we know we know. We also know there are* **known unknowns***; that is to say we know there are some things we do not know. But there are also* **unknown unknowns** *— the ones we don't know we don't know."*
>
> Donald Rumsfeld, February 12, 2002

Setting aside the question of whether or not this makes for sound and sensible defence policy, we may be interested to see how we can model Rumsfeld's epistemology[21] with modal logic and determine whether or not it is internally consistent?

So far we have been using the interpretation, or **'reading'**, of the modal operators $\Box$ and $\Diamond$ as *'necessarily true'* and *'possibly true'*. However, even with this particular interpretation, you may have been somewhat uneasy that some formulas that we would expect to be valid tautologies, are not valid with respect to our defined Kripke semantics. For example, the formula $p \to \Diamond p$ that says *"If p is true then p is possible"*.

[21] Epistemology (/əˌpɪstəˈmɒlə͡ʤi/): A particular theory of knowledge and understanding.

It is easy to see that this formula is not valid in all Kripke models; for example, we could take $\mathfrak{M} = \langle W, R, V \rangle$ to be the following Kripke model with $W = \{w, u\}$, $R = \{(w, u)\}$, and $V(w, p) = 0$, $V(u, p) = 1$.



But while this formula may be not valid over the class of *all* Kripke frames, it *is* valid over the class of all **reflexive** Kripke frames.

And so, if we insisted that in order for our semantics to be an accurate representation of our intuitive reading of what it means for a formula to be "possibly true", then we could simply restrict our attention to the class of all reflexive frames.

Likewise, if we wanted to model knowledge in some single-agent or multi-agent system then we may decide choose to read $\Box p$ as *"I know p to be true"* (or in a multi-agent system we may decide to introduce many modal operators $\Box_1, \Box_2, \ldots, \Box_n$ with the interpretation that $\Box_i p$ means *"Agent i knows that p is true"*, for $i = 1, \ldots, n$). We might then introduce some properties (or axioms) that we expect should be valid tautologies for this particular reading. For example:

- **True Knowledge** *"If I know p to be true then p must be true (I can't <u>know</u> things that are false)"*

$$\Box p \to p$$

- **Positive Introspection** *"If I know p then I know that I know p (I am aware of what I know)"*

$$\Box p \to \Box\Box p$$

- **Negative Introspection** *"If I don't know p then I know that I don't know p (I am aware of what I don't know)"*

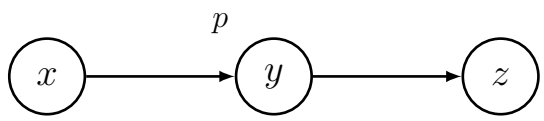
$$\neg\Box p \to \Box\neg\Box p$$

Exercise

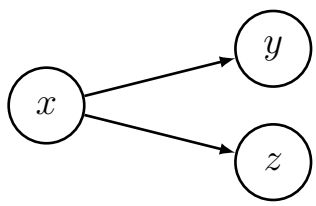
Show that True Knowledge, Positive Introspection, and Negative Introspection are *not* valid with respect to the class of all Kripke frames.

- We have that $\mathfrak{M}, x \not\models \Box p \to p$ where

- We have that $\mathfrak{M}, x \not\models \Box p \to \Box\Box p$ where



- We have that $\mathfrak{M}, x \not\models \neg\Box p \to \Box\neg\Box p$ where



So, if we want to be able to utilise modal logics to be able to model a variety of real-world modal phenomenon such as knowledge (*epistemic logic*), beliefs (*doxastic logic*), obligations and permissions (*deontic logic*), time (*temporal logics*), as well as various transition systems (*dynamic logics*), then we need some way to restrict the semantics of modal logic to an appropriate class of frames that validate properties inherent to those phenomena.

8.2 First-Order Correspondence

It may not have escaped your noticed that the semantics of our modal operators is eerily reminiscent of the semantics for quantifiers in first order predicate logic.

$$\mathfrak{M}, w \models \Diamond F \iff \exists u \in W; (w, u) \in R \text{ and } \mathfrak{M}, u \models F$$

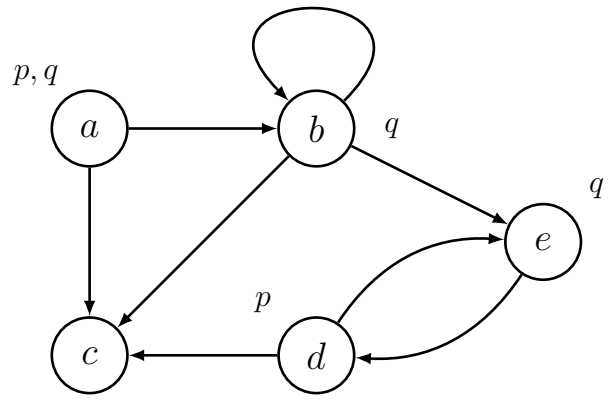
$$\mathfrak{M}, w \models \Box F \iff \forall u \in W; (w, u) \in R \text{ implies } \mathfrak{M}, u \models F$$

Indeed, we might be tempted to think that we don't need modal logic at all — can we not simply translate all of our modal logic formulas into predicate logic with some suitably choice of predicate symbols?

For example, for a given Kripke model $\mathfrak{M} = \langle W, R, V \rangle$, we could consider a first-order predicate language that contains a unary predicate symbol $P_i \in$ Pred, for each propositional variable $p_i \in$ Prop, and a single binary predicate $R \in$ Pred, to describe the accessibility relation. For each $w_i \in W$, we could introduce a constant symbol $c_i \in$ Cons, which would allow us to describe $\mathfrak{M}$ with a predicate formula. For example, the Kripke model $\mathfrak{M}$

Notes

Write down a first-order predicate formula, over the language with Pred $= \{P, Q, R\}$, that describes the following Kripke model:



$$\big(R(a,b) \wedge R(b,c) \wedge R(a,c) \wedge R(d,c) \wedge R(b,b) \wedge R(b,e) \wedge R(e,d) \wedge R(d,e)\big)$$
$$\wedge \big(P(a) \wedge P(d) \wedge Q(a) \wedge Q(b) \wedge Q(e)\big)$$

Moreover, not only can we describe our Kripke models in the language of predicate logic, but we can also specify what it means for a modal formula F to be satisfiable at a given world $x \in W$ in our model in predicate logic, by first translating F into are equisatisfiable predicate formula $\mathsf{ST}_x(F)$ that we call its **standard translation**.

Definition 8.1: Standard Translation

Given a modal formula $F \in \mathit{Fml}^{\Box}$, we define the **standard translation of** F (relative to some variable $x \in \mathsf{Vars}$) to be $\mathsf{ST}_x(F)$, defined inductively as follows:

$$\begin{aligned} \mathsf{ST}_x(p_i) &= P_i(x) \\ \mathsf{ST}_x(\neg G) &= \neg \mathsf{ST}_x(G) \\ \mathsf{ST}_x(G \odot H) &= \mathsf{ST}_x(G) \odot \mathsf{ST}_x(H) \end{aligned}$$

for $\odot \in \{\wedge, \vee, \rightarrow\}$, and

$$\begin{aligned} \mathsf{ST}_x(\Diamond G) &= \exists y\big(R(x,y) \wedge \mathsf{ST}_y(G)\big) \\ \mathsf{ST}_x(\Box G) &= \forall y\big(R(x,y) \rightarrow \mathsf{ST}_y(G)\big) \end{aligned}$$

where $y \in \mathsf{Vars}$ is a fresh first-order variable.

In other words, the standard translation maps each propositional variable $p_i \in \mathsf{Prop}$ to some unary first-order predicate symbols $P_i \in \mathsf{Pred}$, and maps each diamond formula $\Diamond G$ to $\exists y\big(R(x,y) \wedge \mathsf{ST}_y(G)\big)$ and each box formula $\Box G$ to $\forall y\big(R(x,y) \rightarrow \mathsf{ST}_y(G)\big)$, where $\mathsf{ST}_y(G)$ is defined inductively, with all Boolean operators being unchanged by the translation. We can interpret each Kripke model $\mathfrak{M} = \langle W, R, V\rangle$ as a predicate model $\mathfrak{M}* = \langle W, \mathcal{I}\rangle$ over this predicate language, by taking the domain to be the set of worlds W, and interpreting

$$R^{\mathcal{I}} = R \qquad \text{and} \qquad P_i^{\mathcal{I}} = V(p_i) \quad \text{for all } p_i \in \mathsf{Prop}$$

We then have the following correspondence between modal satisfiability and first-order predicate satisfiability:

Theorem 8.2

Let $\mathfrak{M} = \langle W, R, V \rangle$ be an arbitrary Kripke model and let $F \in \mathit{Fml}^{\Box}$ be a modal formula. Then

$$\mathfrak{M}, x \models F \qquad \Longleftrightarrow \qquad \mathfrak{M}^* \models^h \mathsf{ST}_x(F) \tag{16}$$

where $\mathfrak{M}^* = \langle W, \mathcal{I} \rangle$ with $R^{\mathcal{I}} = R$ and $P_i^{\mathcal{I}} = V(p_i)$, for all $p_i \in$ Prop, and $h(x) = x$.

Proof: The proof is by induction on the size of the formula F. ∎

In other words, F is satisfiable in a some Kripke model $\mathfrak{M}$ at world x if and only if $\mathsf{ST}_x(F)$ is satisfiable in the predicate model $\mathfrak{M}^*$ associated with $\mathfrak{M}$[22]. If we wished to check whether F were **valid/globally satisfied** over model $\mathfrak{M}$, *i.e.* true at *all* worlds in W then we can accomplish this by universally quantifying over our variable stand-in for the world x:

$$F \text{ is globally satisfied in } \mathfrak{M} \qquad \Longleftrightarrow \qquad \forall x\, \mathsf{ST}_x(F) \text{ is satisfiable in } \mathfrak{M}^*.$$

However, for F to be **valid** over the underlying *frame* $\mathfrak{F} = (W, R)$ — what we really want — then we would need for this formula to be true, not just at all worlds, which we can achieve with the first-order universal quantifier $\forall x$, but under all possible valuations V of the propositional variables $p_1, \ldots, p_n \in Prop$. In other words, we want

[22] Incidentally, this would give us an alternative approach to deciding modal satisfiability, were it not for the fact that satisfiability in first-order predicate logic is undecidable in general!

to quantify over all possible interpretations of the predicate symbols $P_1, \ldots, P_n$ in our standard translation!

$$F \text{ is valid over } \mathfrak{F} \quad \iff \quad \forall P_1 \ldots \forall P_n \, \forall x \, \mathsf{ST}_x(F) \text{ is satisfiable in } \mathfrak{M}^*.$$

But wait! $\forall P_1 \ldots \forall P_n \, \forall x \, \mathsf{ST}_x(F)$ is <u>not a well-formed formula</u> of first-order predicate logic! We can only quantify over variables that stand-in for individual objects from our domain, not over predicates which stand-in for sets of objects (or tuples of objects).

We have ventured beyond the realm of first-order logic into what is known as **second-order predicate logic**, which allows from quantification over the evaluation of first-order predicate symbols. In order for us to derive some first-order condition on the class of frames that validate F, we must find a way to **eliminate** all occurrences of the quantified predicates symbols $P_1, \ldots, P_n$ from $\mathsf{ST}_x(F)$, in a way that respects satisfiability; leaving us with a first-order formula involving only the binary predicate symbols R describing the accessibility relation.

8.3 Sahlqvist Formulas

We might well ask whether we can find some algorithm that when given an arbitrary modal formula (or set of formulas), would allow us to compute the class of frames on which that formula or formulas are valid. For example, by providing the formula $p \to \Diamond p$ as input, would return the first-order correspondent $\forall x R(x, x)$ describing reflexivity.

Unfortunately, no such algorithm exists that will work in general even in those cases where there is a first-order correspondent [1].

Theorem 8.3: Chagrova's Theorem [1]

It is undecidable whether a given modal axiom has a first-order correspondent.

Proof: Beyond the scope of the course. ∎

Fortunately, there is are large classes of modal formulas for which we can guarantee the existence of a first-order correspondent *and* have algorithms to compute these correspondents; the most widely studied such class are known as **Sahlqvist formulas**. To define this class, we first need a few supplementary definitions.

Definition 8.4: Positive Formulas

A formula F is said to be **positive** (resp. **negative**) with respect to some variable $p \in \mathsf{Prop}$ if every occurrence of p in the **conjunctive normal form** or F appears under the scope of an even (resp. odd) number of negation symbols.
We say that F is **positive** (resp. **negative**) if it is positive (resp. negative) with respect to <u>all</u> propositional variables appearing in F.

Positive (resp. negative) formulas have an interesting property in that they are **monotonically increasing (resp. decreasing)** with respect to the valuation of the propositional variables.

Lemma 8.5

Suppose that $\mathfrak{M} = \langle W, R, V \rangle$ and $\mathfrak{M}' = \langle W, R, V' \rangle$ are Kripke models that differ only in the propositional valuation, and that $V(p) \subseteq V'(p)$ for every propositional variable $p \in \mathsf{Prop}$.

(i) If F is a **positive formula** then

$$\mathfrak{M}, w \models F \quad \Longrightarrow \quad \mathfrak{M}', w \models F$$

for all $w \in W$.

(ii) If F is a **negative formula** then

$$\mathfrak{M}, w \models F \quad \Longleftarrow \quad \mathfrak{M}', w \models F$$

for all $w \in W$.

The next ingredients we need are generalisations of atomic propositional variables, which we call **boxed atoms**.

Definition 8.6: Atoms and Boxed Atoms

We refer to any propositional variable $p \in \mathsf{Prop}$ as an **atom** and define a **boxed atom** to be any formula of the form $\Box^n p$, for some $p \in \mathsf{Prop}$ and $n \geq 1$; that is to say, a propositional variable p in the scope of one or more box operators.

Together, atoms, boxed atoms, positive, and negative formulas form the building blocks that we may assemble in countless ways, subject to some permitted construction rules, to form Sahlqvist formulas.

Definition 8.7: Sahlqvist Formulas

A **Sahlqvist antecedent** is a model formula that is built up from $\top$, $\bot$, atoms, boxed atoms and negative formulas using only conjunction $\wedge$ and disjunction $\vee$.

A **Sahlqvist implication** is a modal formula of the form $F \to G$ where

- F is a **Sahlqvist antecedent**, and
- G is a **positive formula**

A **Salqvist formula** is a formula that is built up from Salqvist implications by freely applying boxes $\Box$ and conjunctions, and disjunctions only between formulas that do not share are propositional variables in common.

Intuitively, Sahlqvist antecedents allow the nesting of boxes within the scope of diamonds, such as $\Diamond\Box p$, but do not allow the nesting of diamonds within boxes, such as $\Box\Diamond p$. This means that a formula such as $\Diamond\Box p \to \Box\Diamond p$ is a Sahlqvist formula, while a formula such as $\Box\Diamond p \to \Diamond\Box p$ is not Sahlqvist[23].

Theorem 8.8: Sahlqvist's Theorem [9]

(i) Every Sahlqvist formula has a first-order correspondent;

(ii) There is an algorithm to compute the first-order correspondent of any Sahlqvist formula.

For the purposes of this course, we will focus only on the special case of Sahlqvist implications, as the algorithms for computing first-order correspondents for the more general class of Sahlqvist formulas involves simply identifying correspondents for each of the constituent Sahlqvist implications and then combining these in an obvious way.

[23] Nor, in this case, is it equivalent to any Sahlqvist formula. This is not obvious, but follows from Sahlqvist's theorem since it is known that $\Box\Diamond p \to \Diamond\Box p$ does *not* have any first-order correspondent!

8.4 Interlude: Lambda Expressions

In what follows it will be convenient to use λ**-notation** as a shorthand to describe predicates/subsets in terms of some formulas[24]. Given an arbitrary predicate formula with (at least one) free variable $F(x)$, we write $(\lambda x.F(x))$ to denote the **unary predicate**

"The set of all those x who satisfy $F(x)$".

In particular, if we evaluate $(\lambda x.F(x))$ at some term t we simply get back the result of uniformly substituting t in place of x in $F(x)$.

$$(\lambda x.F(x))(t) \;=\; F(x/t)$$

This is the most we will need from lambda expressions, for our purposes, but in general such lambda expression can be used to form an entire model of computation, known as the **Lambda calculus** which is known to be as computationally expression as Turing Machines.

8.5 The Sahlqvist–van Benthem Algorithm

The Sahlqvist–van Benthem algorithm, proposed independently by Henrik Sahlqvist ([9]) and Johan van Benthem ([11]), takes an input a **Sahlqvist implication** and computes its **first-order correspondent**. The correctness and successful termination of the algorithm depends on the restricted format of the input — it only works on formulas that are Sahlqvist formulas!

Before we state the algorithm rigorously, let's take a look at a few examples to help motivate the algorithm!

[24]The same *lambda expressions* are used in some programming languages such as Python to define **anonymous functions**, and behave in a similar way

Example 1 Let's start by fixing an arbitrary model $\mathfrak{M}$ and considering the following formula:

$$F \;=\; p \to \Diamond p$$

that says *"If p is true, then p is possibly true"*. We can compute the standard translation of F to get obtain a formula that describes a conditions of what it would mean for F to be satisfied at a model $\mathfrak{M}$ at world x:

$$\mathsf{ST}_x(F) \;=\; P(x) \to \big(\exists y\big(R(x,y) \wedge P(y)\big)$$

Now lets consider what would be required for this formula to be valid in $\mathfrak{M}$?

Well, for any x such that $P(w)$ is false, the formula $\mathsf{ST}_x(F)$ would be automatically true. Hence, we can restrict our attention to those x where $P(x)$ is true.

However, since the consequent of $\mathsf{ST}_x(F)$ is **positive**, by Lemma 8.3, we need only consider the **minimal** case(s) where the antecedent $P(x)$ is true. So let's ask the question:

"If the only thing I told you about P is that $P(x)$ is true, what would the minimum that you could conclude about the valuation of P?"

The answer is surely that $P^{\mathcal{I}} = \{x\}$. In other would the **minimal valuation for** P is given by the lambda expression $\lambda u.\, u = x$, that says *"the set of all u such that $u = x$"*

Hence, if we apply the substitution $\sigma(P) = (\lambda u.\, u = x)$ that substitutes all occurrences of P in $\mathsf{ST}_x(F)$ with the minimal valuation $(\lambda u.\, u = x)$, we will obtain a simplified formula that does not depend on P:

$$\begin{aligned}
\sigma(\mathsf{ST}_x(F)) \;&=\; \sigma(P)(x) \;\to\; \exists y\big(R(x,y) \wedge \sigma(P)(y)\big) \\
&=\; (\lambda u.\, u = x)(x) \;\to\; \exists y\big(R(x,y) \wedge (\lambda u.\, u = x)(y)\big) \\
&=\; (x = x) \to \exists y\big(R(x,y) \wedge (y = x)\big)
\end{aligned}$$

Note that the antecedent is now trivially satisfied by this substitution, as expected, and the resulting formula is free of all occurrences of the quantified variable P. This

formula now says that for F to be valid in a model $\mathfrak{M} = \langle W, R, V \rangle$ then it must because the $(x = x) \to \big(\exists y\big(R(x,y) \wedge (y = x)\big)$ is true for all $x \in W$, irrespective of the valuation V.

Hence, this formula describes a first-order condition on the accessibility relation R, that should be true for all x. That is to say, where the accessibility relation satisfies $\forall x\, \sigma(\mathsf{ST}_x(F))$. If we tidy this up a bit using successive equivalences we get the following:

$$\begin{aligned}
\forall x\, \sigma(\mathsf{ST}_x(F)) &= \forall x\Big((x = x) \to \exists y\big(R(x,y) \wedge (y = x)\big)\Big) \\
&\equiv \forall x\Big(\top \to \exists y\big(R(x,y) \wedge (y = x)\big)\Big) \\
&\equiv \forall x\Big(\exists y\big(R(x,y) \wedge (y = x)\big)\Big) \\
&\equiv \forall x\Big(\exists y\big(R(x,x)\big)\Big) \\
&\equiv \forall x\, R(x,x)
\end{aligned}$$

Hence, we can conclude that F is valid precisely over the class of **reflexive** frames, satisfying the first-order condition $\forall x R(x,x)$, as required.

Example 2 Let's now consider a slightly involved formula

$$F \;=\; (p \wedge \Diamond\Box p) \to \Box p$$

The standard translation of which is given by

$$\mathsf{ST}_x(F) \;=\; P(x) \wedge \exists y\big(R(x,y) \wedge \forall z\big(R(y,z) \to P(z)\big)\big) \;\to\; \forall u\big(R(x,u) \to P(u)\big)$$

As in the previous example, the antecedent here is **positive**, but the antecedent is a little messy. We can tidy things up a bit by applying successive equivalences to **'pull out'** the existential quantifier $\exists y$ to the front.

$$\begin{aligned}
\mathsf{ST}_x(F) \;&\equiv\; \exists y\big(P(x) \wedge R(x,y) \wedge \forall z\big(R(y,z) \to P(z)\big)\big) \;\to\; \forall u\big(R(x,u) \to P(u)\big) \\
&\equiv\; \forall y\Big[\big(P(x) \wedge R(x,y) \wedge \forall z\big(R(y,z) \to P(z)\big)\big) \;\to\; \forall u\big(R(x,u) \to P(u)\big)\Big]
\end{aligned}$$

For simplicity, let's 'hide' the $\forall y$ quantifier so that we can better examine the anatomy of the inner subformula

$$\mathsf{ST}_x(F) \;\equiv\; \underbrace{P(x)}_{\text{AT}} \wedge \underbrace{R(x,y)}_{\text{REL}} \wedge \underbrace{\forall z\big(R(y,z) \to P(z)\big)}_{\text{BOX}} \;\to\; \underbrace{\forall u\big(R(x,u) \to P(u)\big)}_{\text{POS}}$$

Here we see that our antecedent is composed of three components: a conjunction of (translations of) atoms AT, a conjunction of relational formulas REL, and a conjunction of (translations of) boxed atoms BOX, while the consequent is a positive formula.

As in the first example, we now look for a **minimal valuation** for P that would make the antecedent true. So we ask the question:

"If the only thing I told you about P is that $P(x)$ is true and $P(z)$ is true whenever $R(y,z)$ is true, and I told you that you are located at a world where P is true, what might you conclude about your location?"

In this case the answer is *either* that you are located at x *or* that you are located at some world that is a successor of y,

$$P^{\mathcal{I}} \;=\; \{x\} \cup \{y : R(x,y)\}$$

Since these are the only worlds that you know satisfy P, given the information provided by the antecedent. In other would the **minimal valuation for** P is given by the lambda expression $\lambda u.\,\big(u = x \vee R(y,u)\big)$, that says *"the set of all u such that $u = x$ or u is a successor of y"*.

We now apply the substitution $\sigma(P) = \lambda u.\,\big(u = x \vee R(y,u)\big)$ that substitutes all occurrences of P in $\mathsf{ST}_x(F)$ with the minimal valuation to get

$$\begin{aligned}
\sigma(\mathsf{ST}_x(F)) \;&=\; \big(\sigma(P)(x) \wedge R(x,y) \wedge \forall z\big(R(y,z) \to \sigma(P)(z)\big)\big) \\
&\qquad \to \forall u\big(R(x,u) \to \sigma(P)(u)\big) \\
&=\; \big(\top \wedge R(x,y) \wedge \top\big) \\
&\qquad \to \forall u\big(R(x,u) \to \lambda u.\,\big(u = x \vee R(y,u)\big)(u)\big) \\
&=\; R(x,y) \to \forall u\big(R(x,u) \to (u = x \vee R(y,u))\big)
\end{aligned}$$

Note that both the AT and BOX components of the antecedent become trivially satisfied by this substitution, as expected, and the resulting formula is free of all occurrences of the quantified variable P. Applying some logical equivalences and variable relabelling to tidy up this resulting formula (not forgetting the $\forall y$ that we hid earlier) we get the following **first-order correspondent** for F:

$$\forall x \sigma(\mathsf{ST}_x(F)) \;\equiv\; \forall x \forall y \forall z \Big(R(x,y) \wedge R(y,z) \;\to\; (x = z \vee R(y,z)) \Big)$$

From these examples we can generalise the procedure into the following algorithm that is guaranteed to return a first-order correspondent for all Sahlqvist implications.

The Sahlqvist–van Benthem Algorithm

Input) Given a modal formula $F \in \mathit{Fml}^{\Box}$;

Step 1) Compute the Standard Translation

The first step of the algorithm is to compute the standard translation $\mathsf{ST}_x(F)$ of F

Step 2) Pull out 'Diamonds'

Wherever we see an existential quantifier in the **antecedent** of $\mathsf{ST}_x(F)$, we can apply the following logical equivalences to 'pull out' the existential quantifier to the front, in doing so converting them into universal quantifiers

$$\begin{aligned} \big(\exists x A(x) \wedge B\big) &\equiv \exists x\big(A(x) \wedge B\big) \\ \big(\exists x A(x)\big) \to B &\equiv \forall x\big(A(x) \to B\big) \end{aligned}$$

Step 3) Group Terms

Identify a predicate symbol P_i and apply successive equivalences to reorganise the antecedent into the form

$$\forall x_1 \ldots \forall x_n \Big[\big(\mathsf{REL} \ \wedge \ \mathsf{AT} \ \wedge \ \mathsf{BOX} \ \wedge \ \mathsf{NEG}\big) \ \to \mathsf{POS}\Big]$$

where

- REL is a conjunction of **relational** formulas of the form $R(x, y)$ for some variables $x, y \in$ Vars;

- AT is a conjunction of (translations of) **atoms** of the form $\mathsf{ST}_x(p_i) = P_i(x)$ for some variable $x \in \mathsf{Vars}$,
- BOX is a conjunction of (translations of) **boxed atoms** of the form $\mathsf{ST}_x(\Box^n p_i)$, form some variable $x \in \mathsf{Vars}$;
- NEG is a (translations of a) **negative formula** with respect to P_i;
- POS is (translations of a) **positive formula** with respect to P_i.

Step 4) Read off Minimal Instances

Identify a minimum valuation $\sigma(P_i)$ that makes makes the conjunction $(\mathsf{AT} \wedge \mathsf{BOX})$ true.

$$\sigma(\mathsf{AT} \wedge \mathsf{BOX}) \equiv \top$$

Step 5) Apply the Minimal Substitution

Apply the minimal substitution σ to $\mathsf{ST}_x(F)$ to eliminate all occurrences of P_i, and simplify the resulting formula.

Step 7) Repeat (if applicable)

If any propositional variables remain in the formula, repeat from Step 3.

Exercise

Apply the Sahlqvist-van Benthem algorithm to compute a first-order correspondent to the following formula

$$F \;=\; \Diamond\Box p \to \Box\Diamond p$$

that says *"If it is possible that p is necessary, then it is necessarily the case that p is possible"*.

- Taking the standard translation we get

$$\begin{aligned}
\mathsf{ST}_x(F) &= \exists y\big(R(x,y) \wedge \forall z\big(R(y,z) \to P(z)\big)\big) \to \forall z\big(R(x,z) \to \exists u\big(R(z,u) \wedge P(u)\big)\big) \\
&\equiv \forall y\Big[\big(R(x,y) \wedge \forall z\big(R(y,z) \to P(z)\big)\big) \to \forall z\big(R(x,z) \to \exists u\big(R(z,u) \wedge P(u)\big)\big)\Big]
\end{aligned}$$

- The minimal valuation is $\sigma(P) = \lambda u.\ R(y,u)$,

$$\begin{aligned}
\sigma(\mathsf{ST}_x(F)) &= \forall y\Big[\big(R(x,y) \wedge \forall z\big(R(y,z) \to R(y,z)\big)\big) \to \forall z\big(R(x,z) \to \exists u\big(R(z,u) \wedge R(y,u)\big)\big)\Big] \\
&\equiv \forall y \forall z\Big[\big(R(x,y) \wedge \top\big) \to \big(R(x,z) \to \exists u\big(R(z,u) \wedge R(y,u)\big)\big)\Big] \\
&\equiv \forall y \forall z\Big[R(x,y) \wedge R(x,z) \to \exists u\big(R(z,u) \wedge R(y,u)\big)\Big]
\end{aligned}$$

- Hence, the class of frames on which F is valid is the class of frames that satisfy the first-order condition

$$\forall x \forall y \forall z\big[R(x,y) \wedge R(x,z) \to \exists u\big(R(y,u) \wedge R(z,u)\big)\big]$$

which is a property sometimes referred to as *confluence*.

Exercise

Rumsfeld's epistemology asserts that there are **unknown unknowns**, which is a direct refutation of **negative introspection** (that we are aware of what we don't know). On what class of frames would we guarantee the existence of unknown unknowns.

For the existence of **unknown unknowns** we would need to consider frames on which **negative introspection** was **not** valid, so let's identify the class of frames on which it is valid and take its complement.

- We start with the formula for the complement, and (to make life easier) make a substitution $q = \neg p$ (this does not affect frame validity, since frame validity does not depend on the choice of valuation)

$$\begin{aligned} F &= \neg\Box p \to \Box\neg\Box p \\ &\equiv \Diamond q \to \Box\Diamond q \end{aligned}$$

- Taking the standard translation we get

$$\begin{aligned} \mathsf{ST}_x(F) &= \exists y\big(R(x,y) \land Q(y)\big) \to \forall z\big(R(x,z) \to \exists u\big(R(z,u) \to Q(u)\big) \\ &\equiv \forall y\Big[\big(R(x,y) \land Q(y)\big) \to \forall z\big(R(x,z) \to \exists u\big(R(z,u) \land Q(u)\big)\Big] \end{aligned}$$

- The minimal valuation is $\sigma(Q) = \lambda u.\ u = y$,

$$\begin{aligned} \sigma(\mathsf{ST}_x(F)) &= \forall y\Big[\big(R(x,y) \land (\lambda u.\ u = y)(y)\big) \to \forall z\big(R(x,z) \to \exists u\big(R(z,u) \land (\lambda u.\ u = y)(u)\big)\Big] \\ &= \forall y\Big[\big(R(x,y) \land y = y\big) \to \forall z\big(R(x,z) \to \exists u\big(R(z,u) \land u = y\big)\Big] \\ &= \forall y \forall z\Big[R(x,y) \land R(x,z) \to R(z,y)\Big] \end{aligned}$$

- Hence, the class of frames that refute negative introspection is the class of frames that satisfy the complementary first-order condition

$$\exists x \exists y \exists z\big[R(x,y) \land R(x,z) \land \neg R(z,y)\big]$$

References

[1] L. A. Chagrova. An undecidable problem in correspondence theory. *The Journal of Symbolic Logic*, 56(4):1261–1272, 1991.

[2] D. M. Gabbay. Theoretical foundations for non-monotonic reasoning in expert systems. In Krzysztof R. Apt, editor, *Logics and Models of Concurrent Systems*, pages 439–457, Berlin, Heidelberg, 1985. Springer Berlin Heidelberg.

[3] Gerhard Gentzen. Untersuchungen über das logische schließen. i. *Mathematische zeitschrift*, 39(1):176–210, 1935.

[4] Sarit Kraus, Daniel Lehmann, and Menachem Magidor. Nonmonotonic reasoning, preferential models and cumulative logics. *Artificial Intelligence*, 44(1):167–207, 1990.

[5] Saul A. Kripke. Semantical analysis of modal logic i normal modal propositional calculi. *Mathematical Logic Quarterly*, 9(5-6):67–96, 1963.

[6] David Makinson. General theory of cumulative inference. In M. Reinfrank, J. de Kleer, M. L. Ginsberg, and E. Sandewall, editors, *Non-Monotonic Reasoning*, pages 1–18, Berlin, Heidelberg, 1989. Springer Berlin Heidelberg.

[7] Robert McNaughton. A theorem about infinite-valued sentential logic. *The Journal of Symbolic Logic*, 16(1):1–13, 1951.

[8] Paul S Mostert and Allen L Shields. On the structure of semigroups on a compact manifold with boundary. *Annals of Mathematics*, 65(1):117–143, 1957.

[9] Henrik Sahlqvist. Completeness and correspondence in the first and second order semantics for modal logic. In Stig Kanger, editor, *Proceedings of the Third Scandinavian Logic Symposium*, volume 82 of *Studies in Logic and the Foundations of Mathematics*, pages 110–143. Elsevier, 1975.

[10] Enric Trillas. Sobre funciones de negación en la teoría de conjuntos difusos. *Stochastica*, 3(1):47–60, 1979.

[11] Johan Van Benthem. *Modal correspondence theory*. PhD thesis, University of Amsterdam, 1976.